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Leutheuser et al.

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(54) **HOUSING AND ANTENNA ARCHITECTURE
FOR MOBILE DEVICE**

(58) **Field of Classification Search**

CPC H01Q 13/10; H01Q 5/30; H01Q 1/243
See application file for complete search history.

(71) Applicant: **Apple Inc.**, Cupertino, CA (US)

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(73) Assignee: **APPLE INC.**, Cupertino, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 711 days.

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Primary Examiner — Graham P Smith

(22) Filed: **Sep. 21, 2021**

(74) *Attorney, Agent, or Firm* — Brownstein Hyatt Farber Schreck, LLP

(65) **Prior Publication Data**

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Related U.S. Application Data

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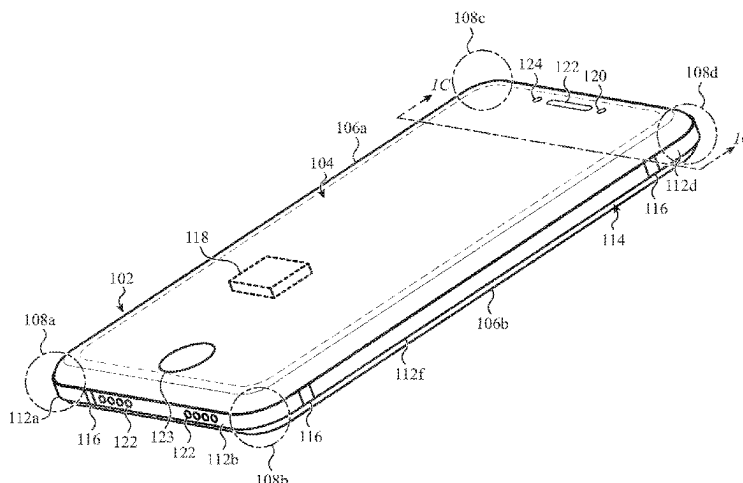
(51) **Int. Cl.**
H01Q 1/24 (2006.01)
H01Q 5/30 (2015.01)
(Continued)

(52) **U.S. Cl.**
CPC **H01Q 1/243** (2013.01); **H01Q 5/30** (2015.01); **H01Q 13/10** (2013.01); **H04M 1/0249** (2013.01); **H04M 1/0283** (2013.01)

(57) **ABSTRACT**

A device includes a display and a housing. The housing at least partially surrounds the display. The housing includes a first housing segment defining at least a first portion of an exterior surface of the device and a first interlock feature having an interlock surface that is offset with respect to an end surface of the first housing segment. The first interlock feature has a first opening formed in the interlock surface. The housing further includes a second housing segment defining at least a second portion of the exterior surface of the device and a second interlock feature having a second opening aligned with the first opening, and a non-conductive housing component defining a third portion of the exterior surface of the device and extending into the first opening and the second opening.

20 Claims, 46 Drawing Sheets



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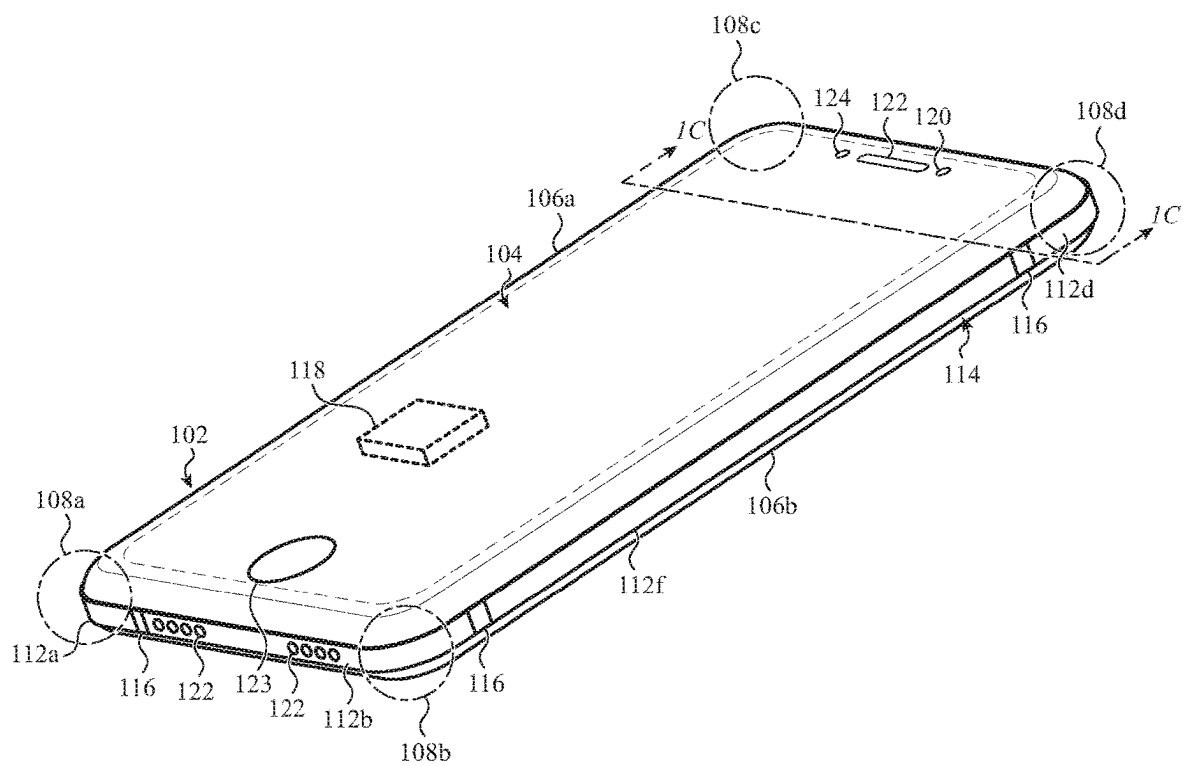


FIG. 1A

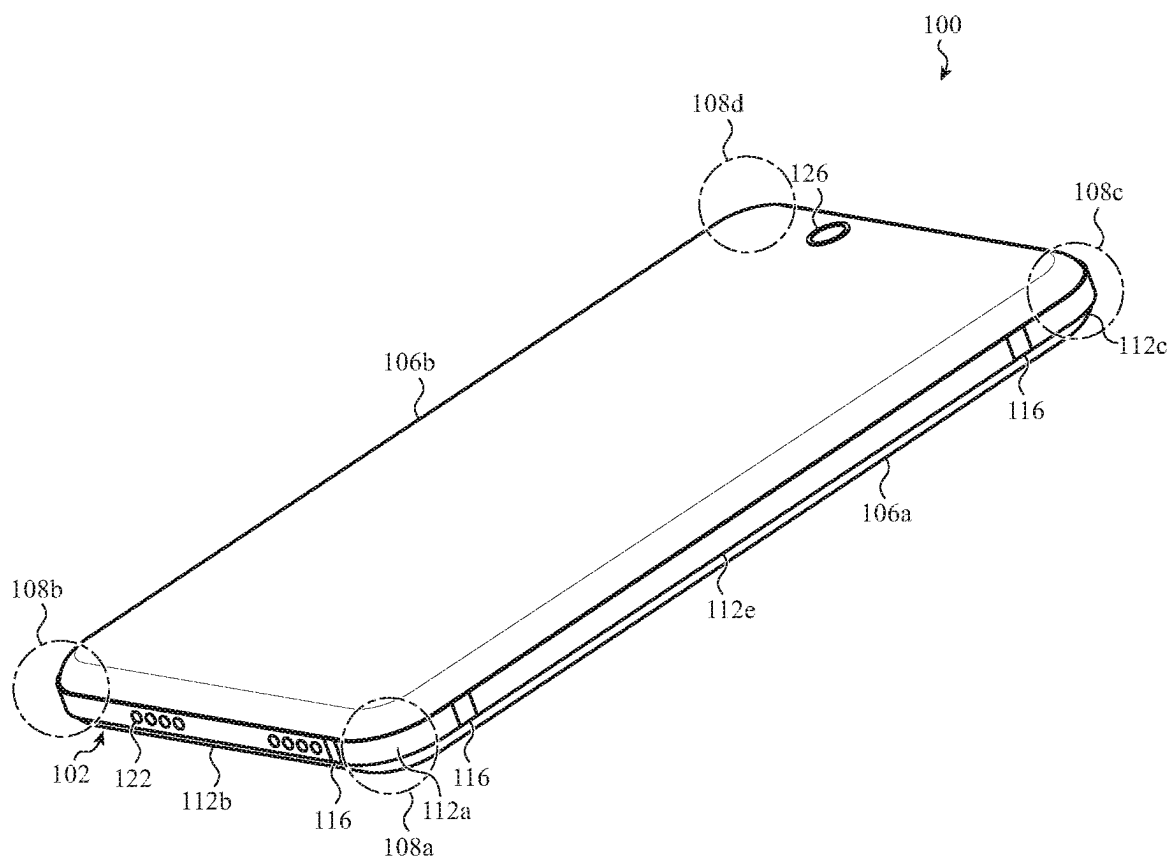


FIG. 1B

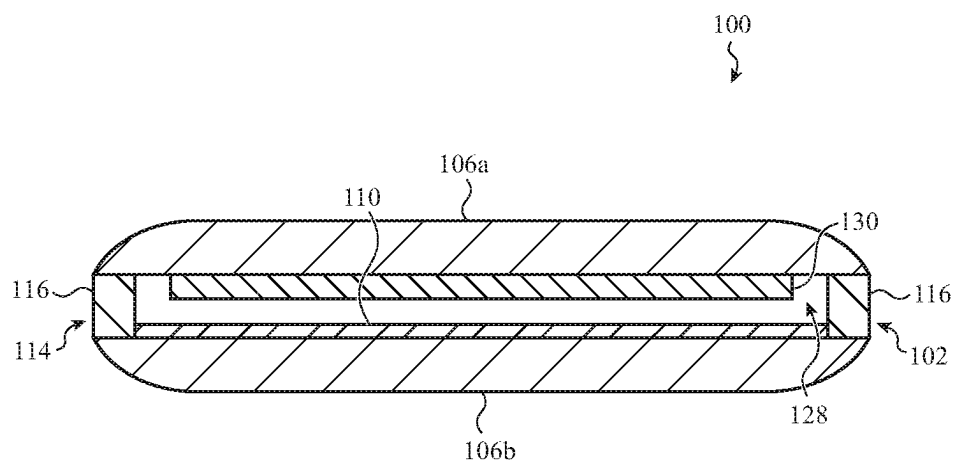


FIG. 1C

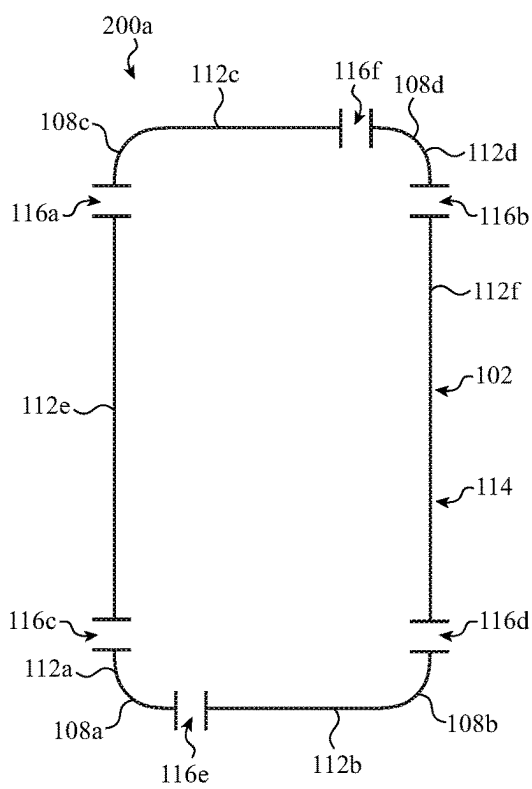


FIG. 2A

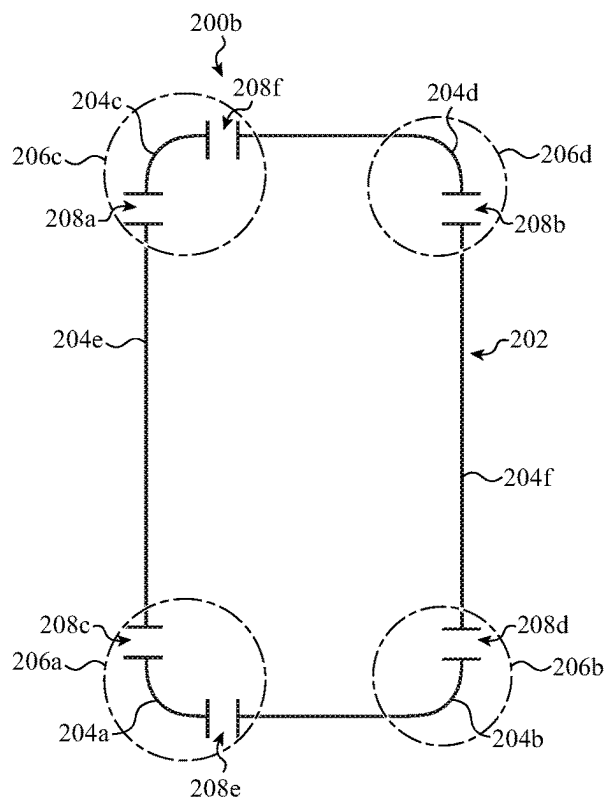


FIG. 2B

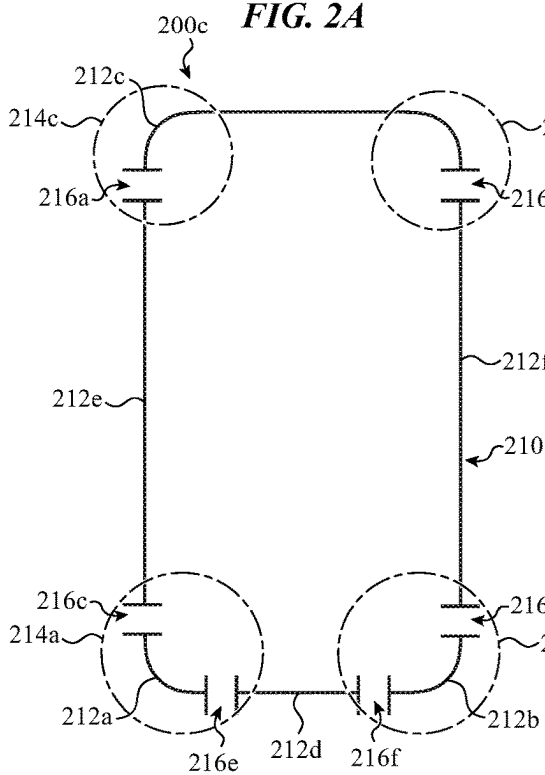


FIG. 2C

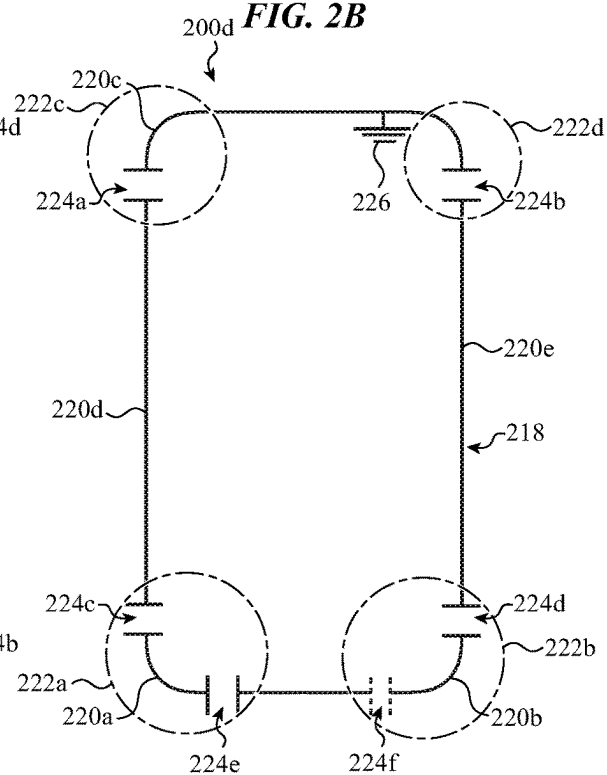


FIG. 2D

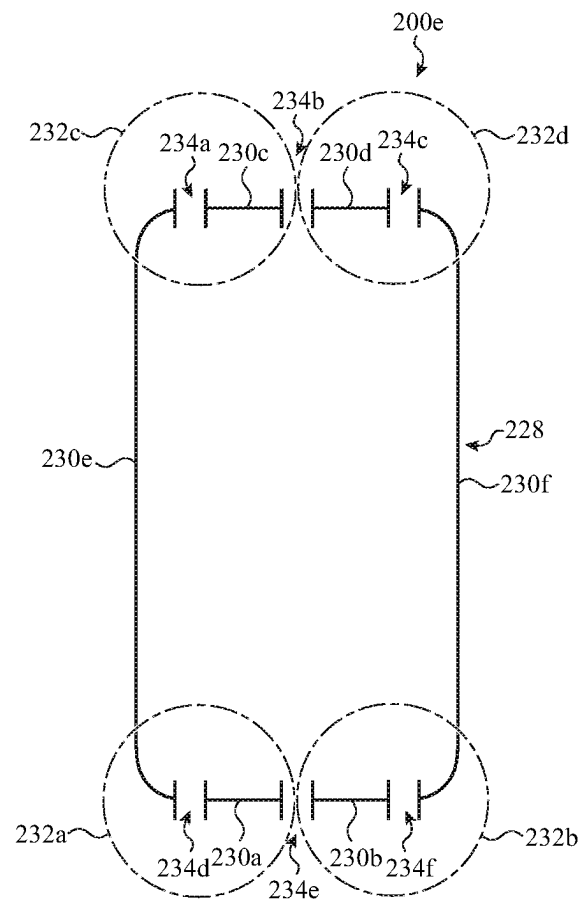


FIG. 2E

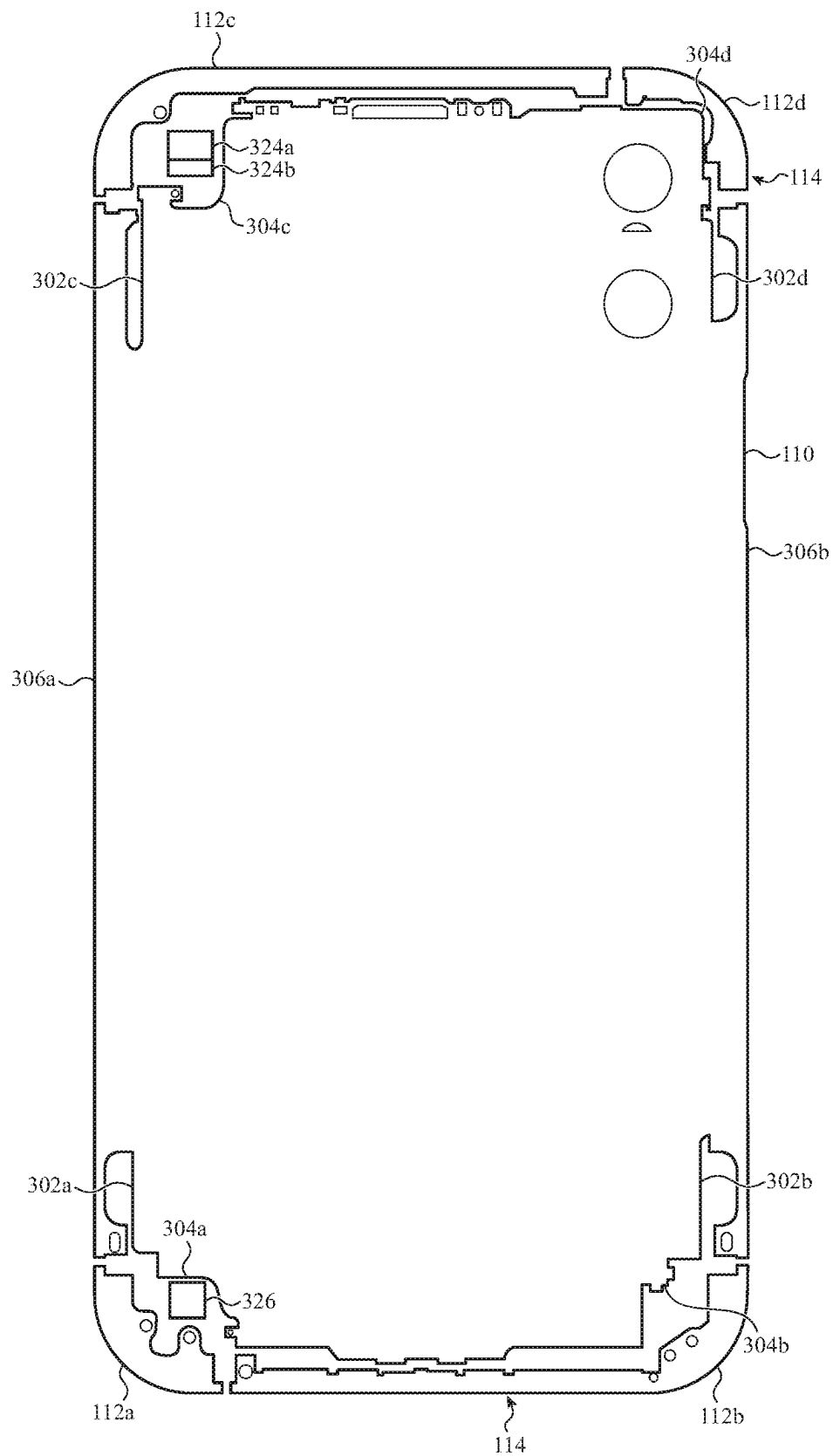


FIG. 3A

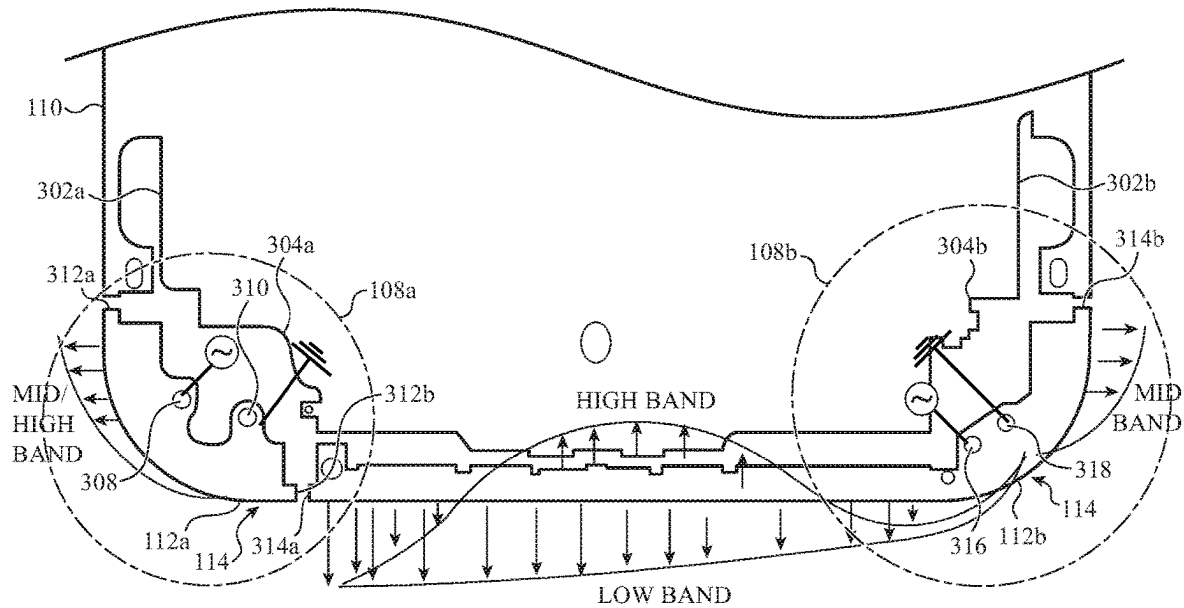


FIG. 3B

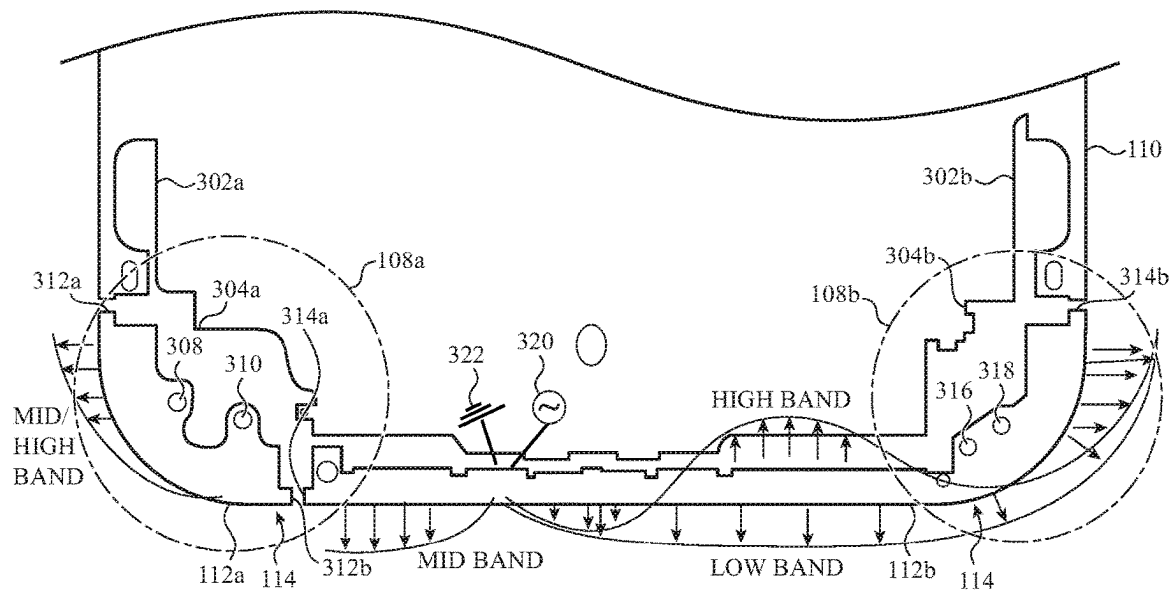


FIG. 3C

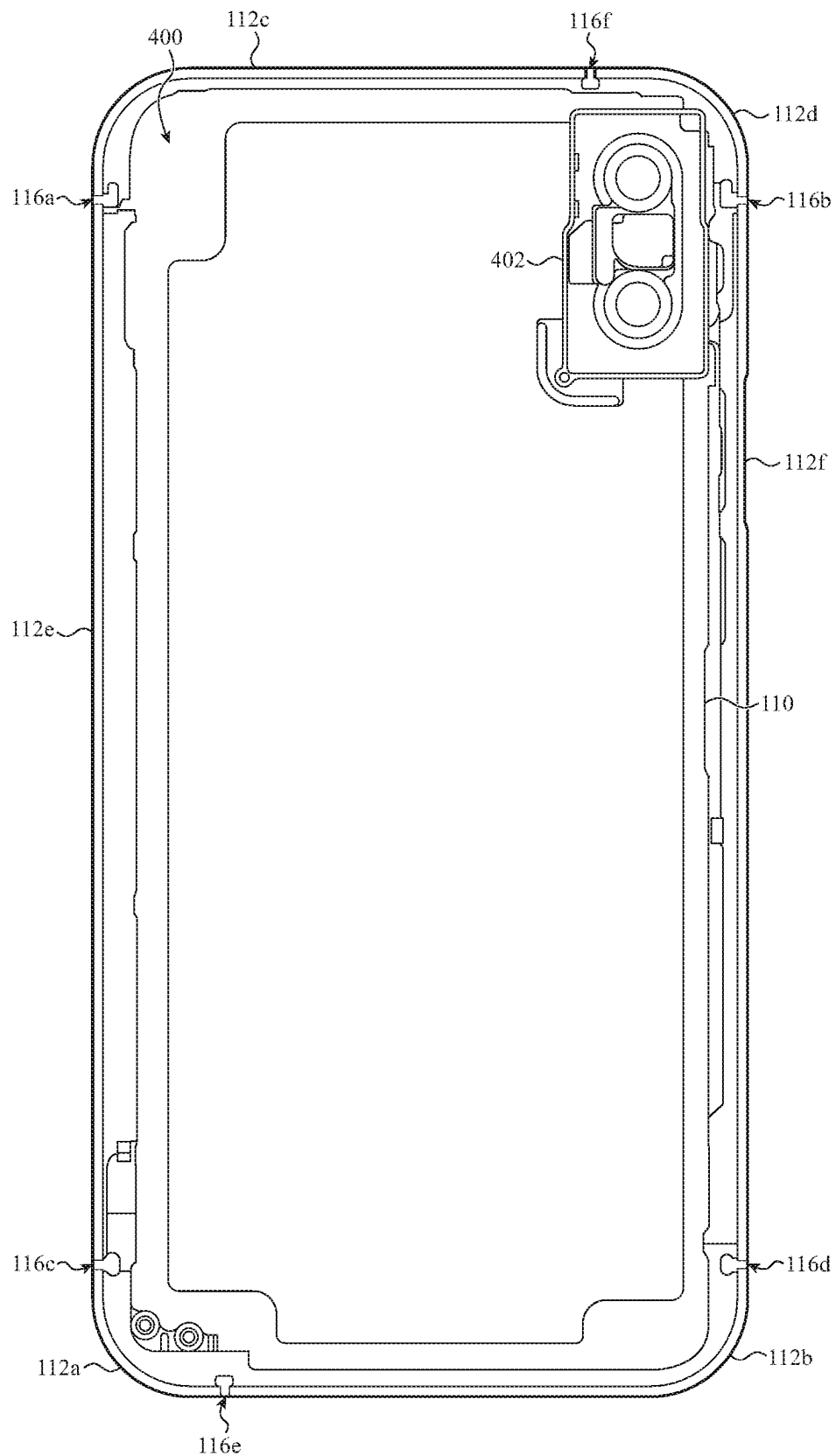


FIG. 4

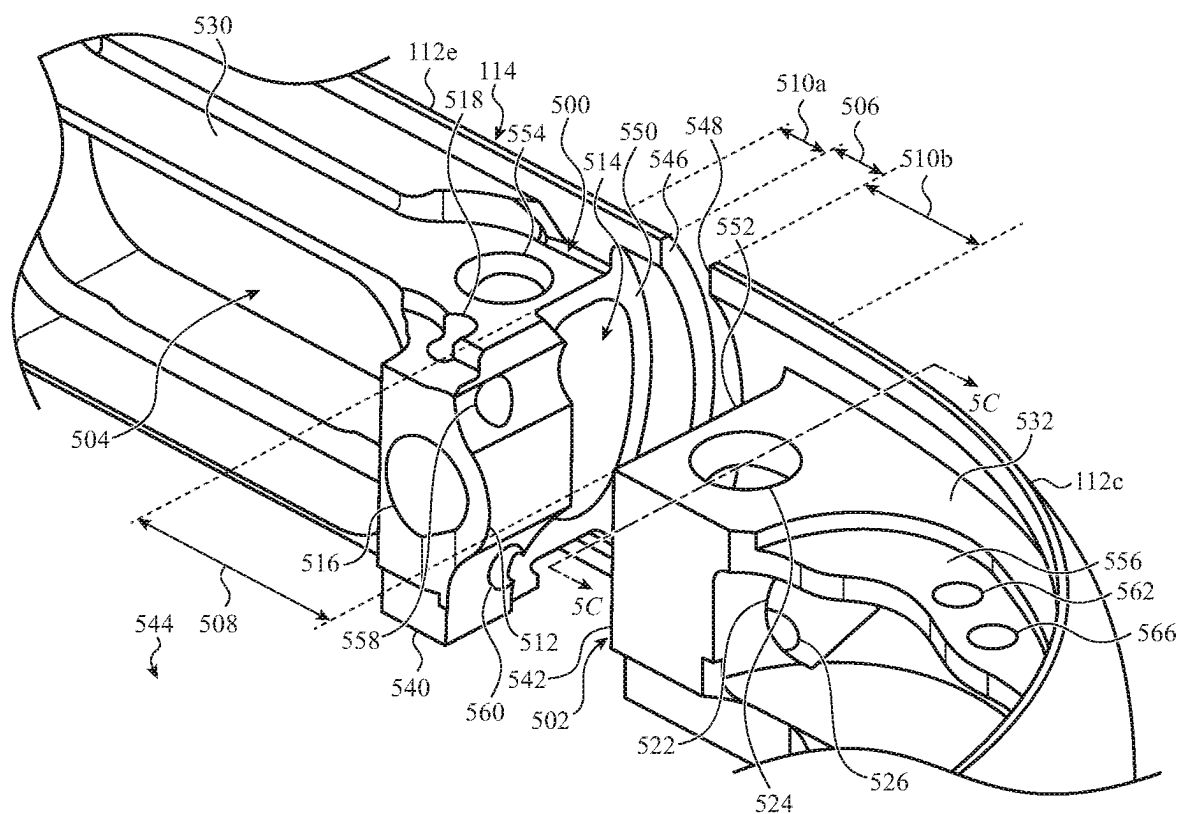


FIG. 5A

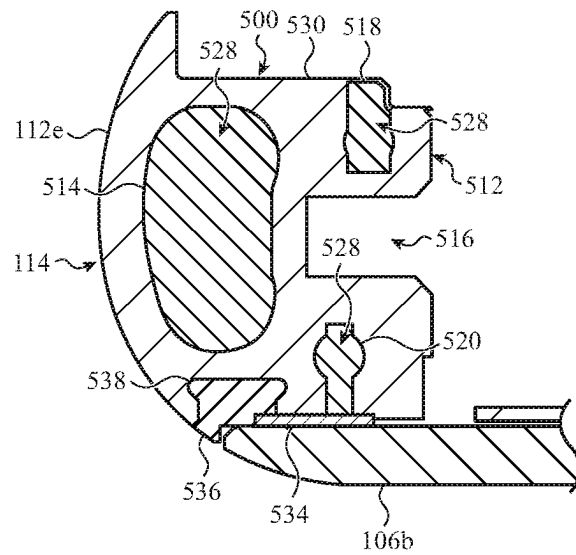


FIG. 5B

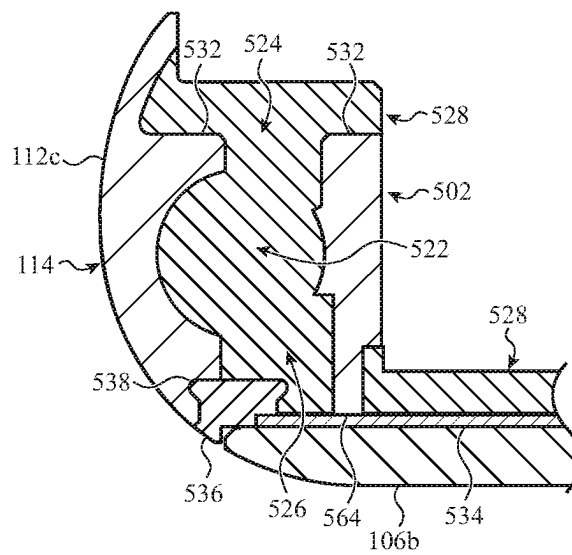


FIG. 5C

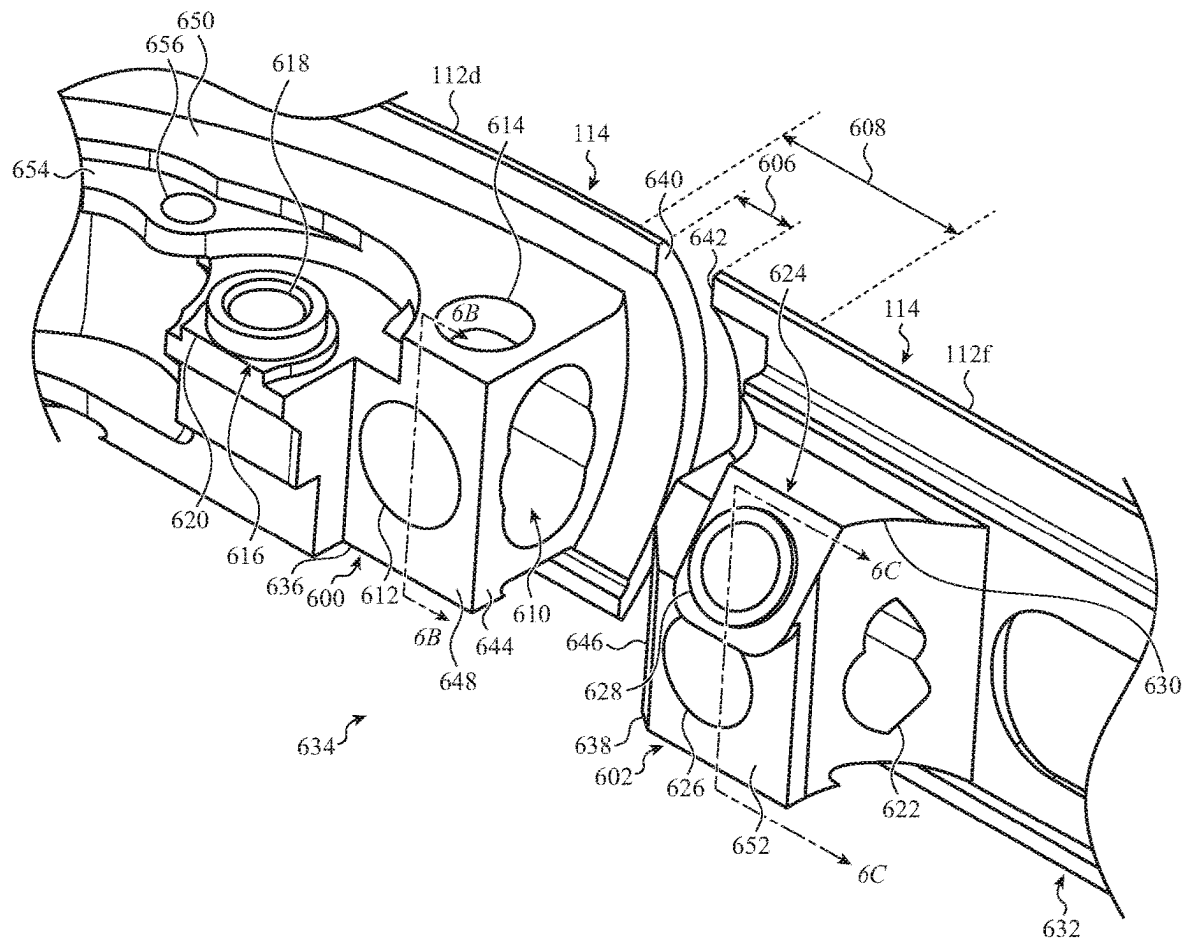


FIG. 6A

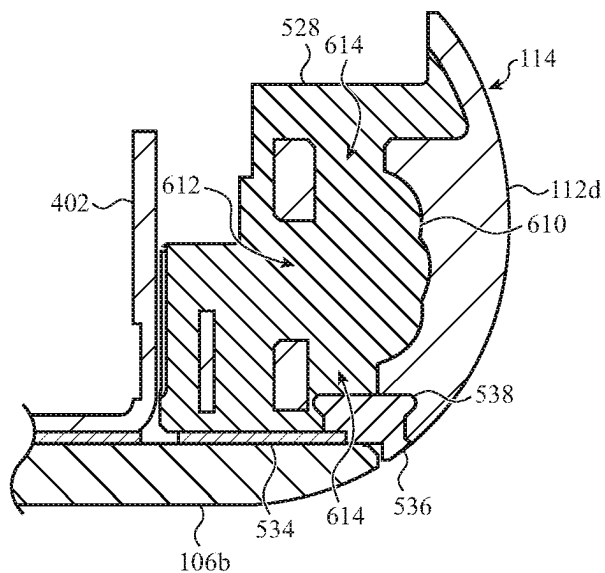


FIG. 6B

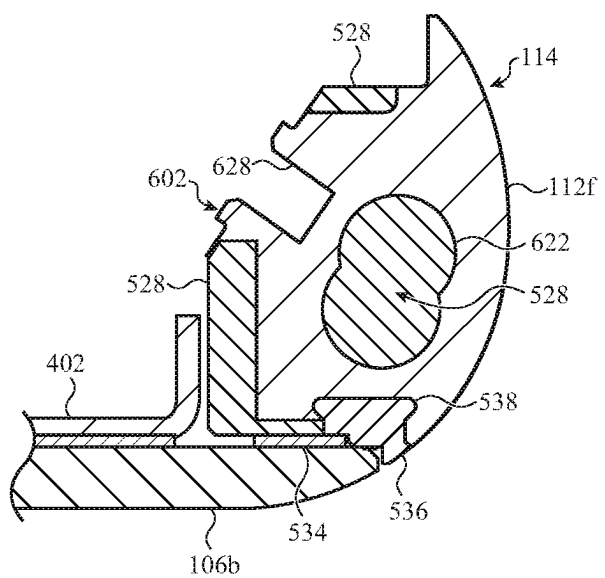


FIG. 6C

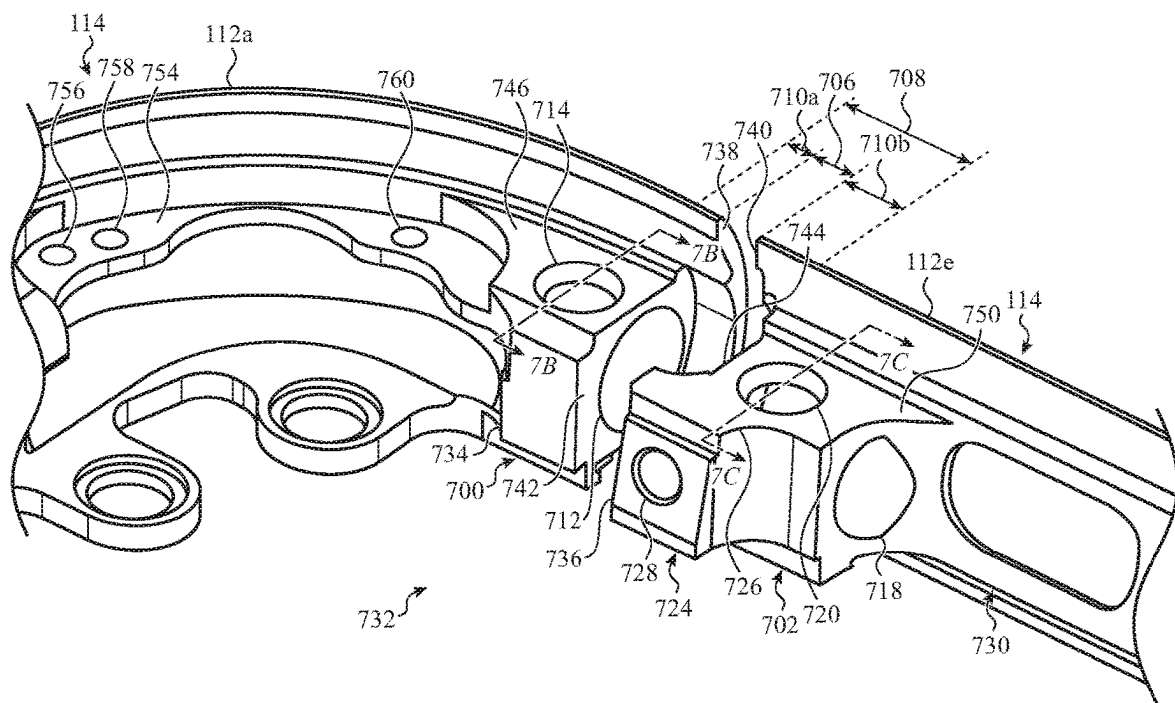


FIG. 7A

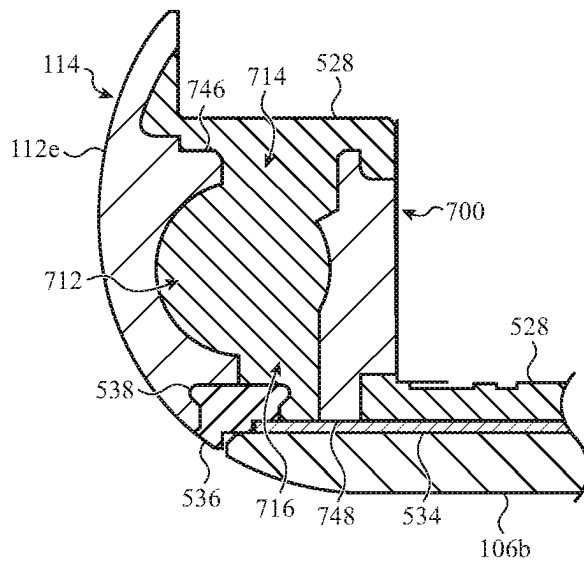


FIG. 7B

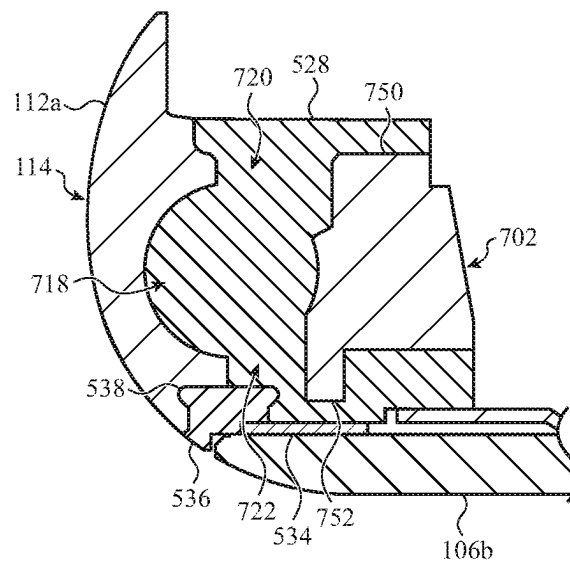


FIG. 7C

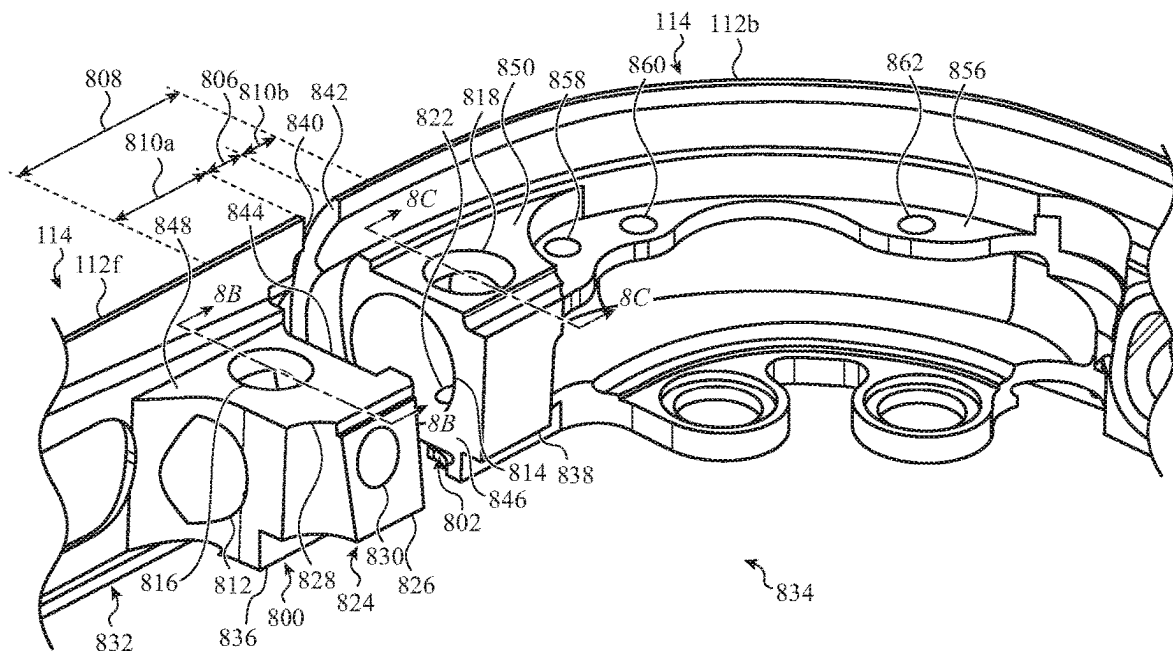


FIG. 8A

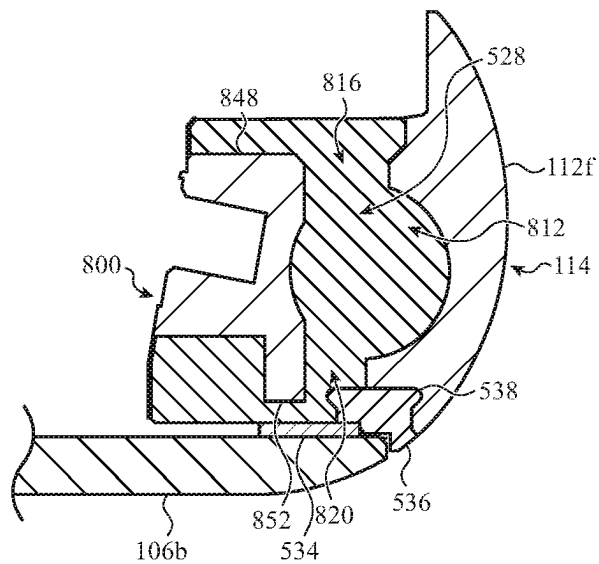


FIG. 8B

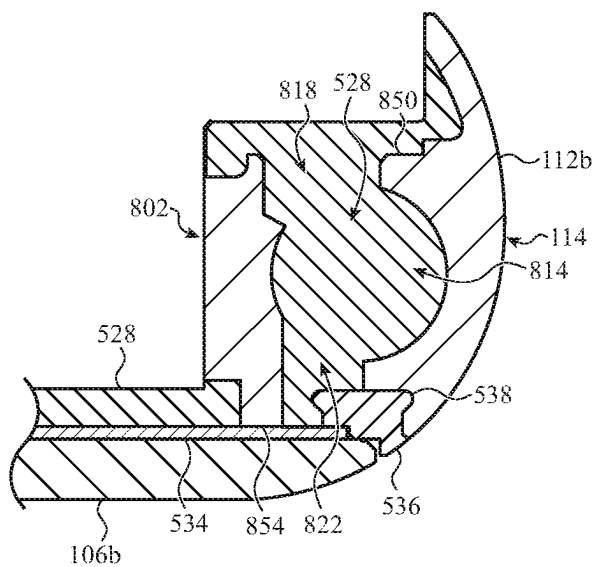


FIG. 8C

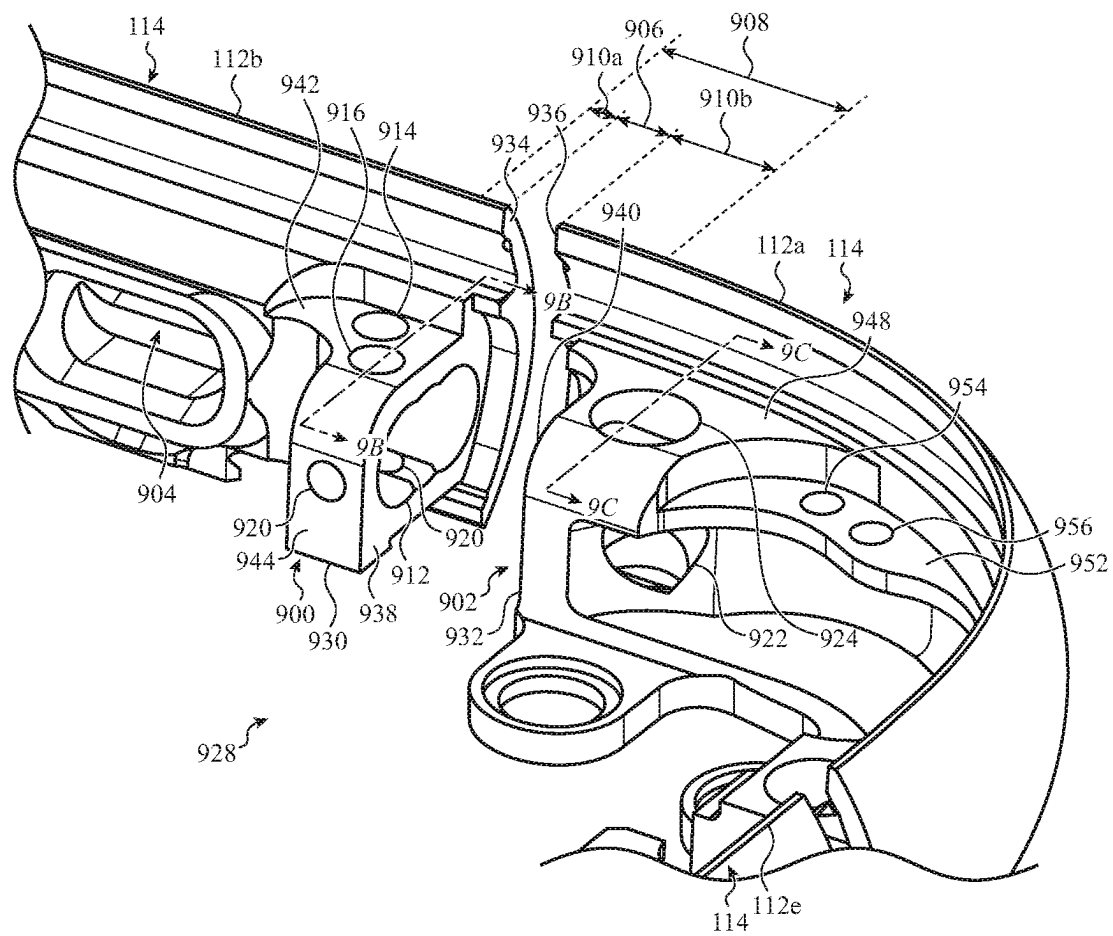


FIG. 9A

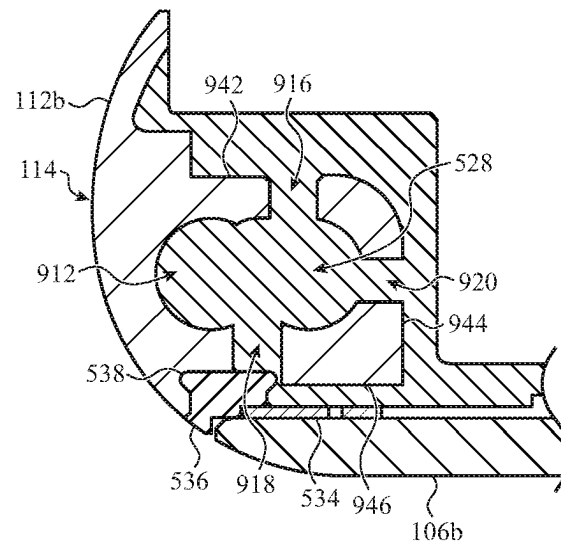


FIG. 9B

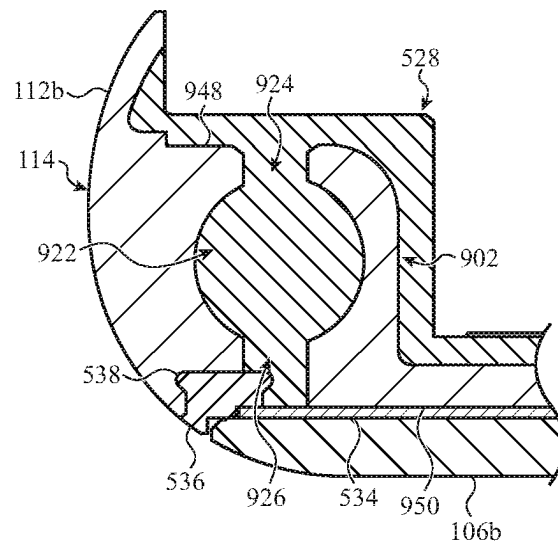


FIG. 9C

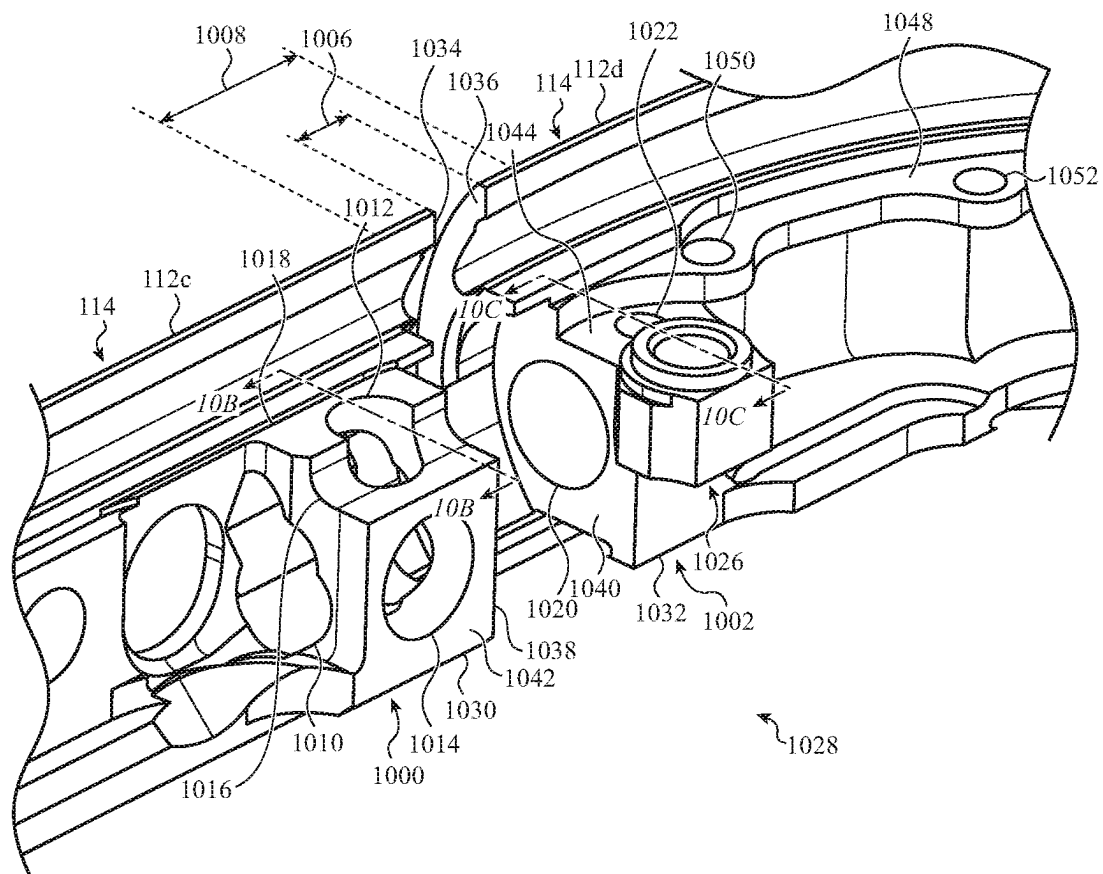


FIG. 10A

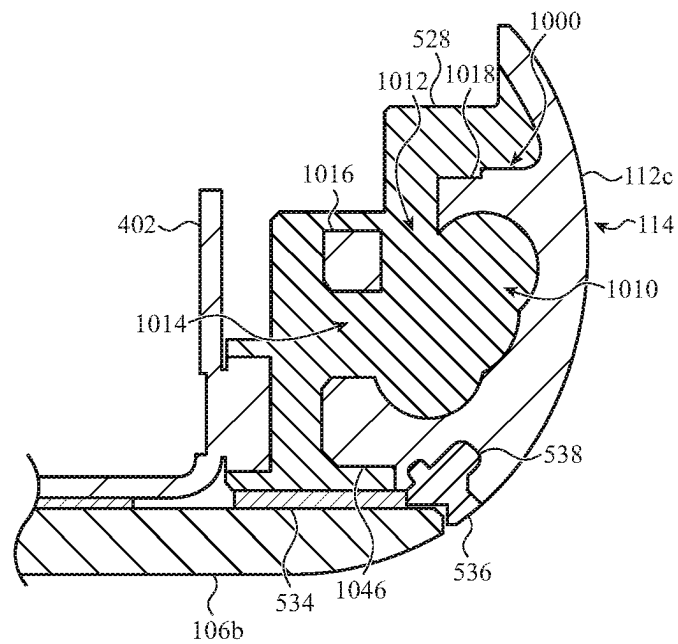


FIG. 10B

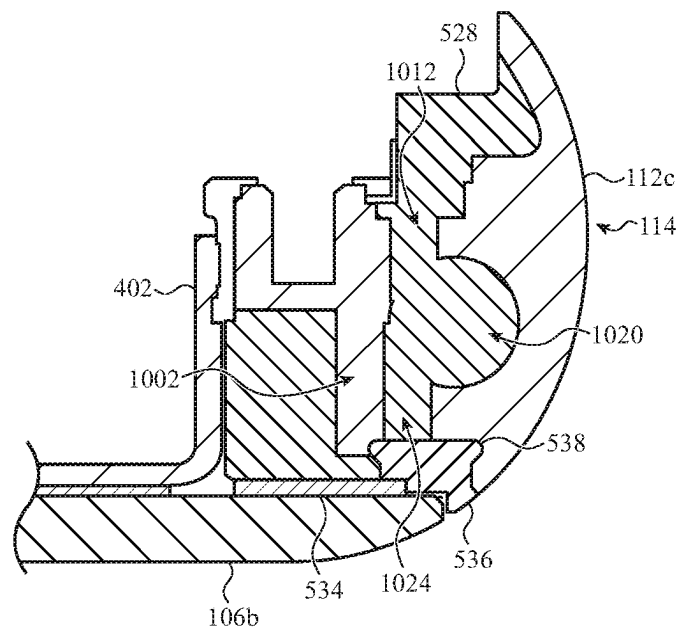


FIG. 10C

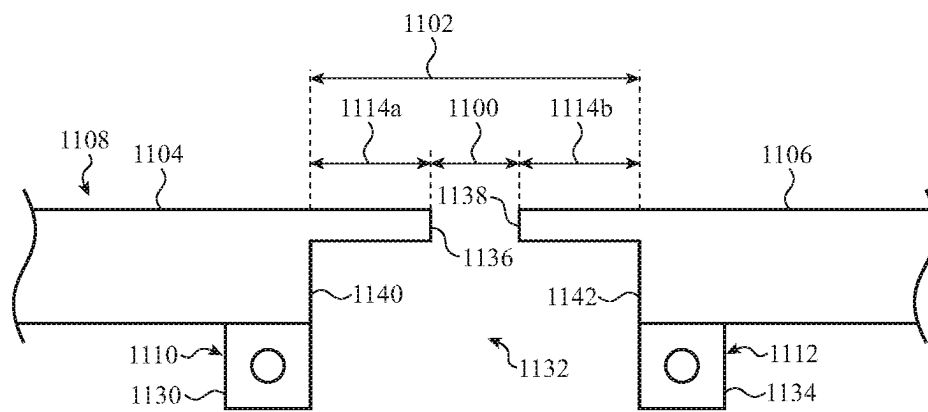


FIG. 11A

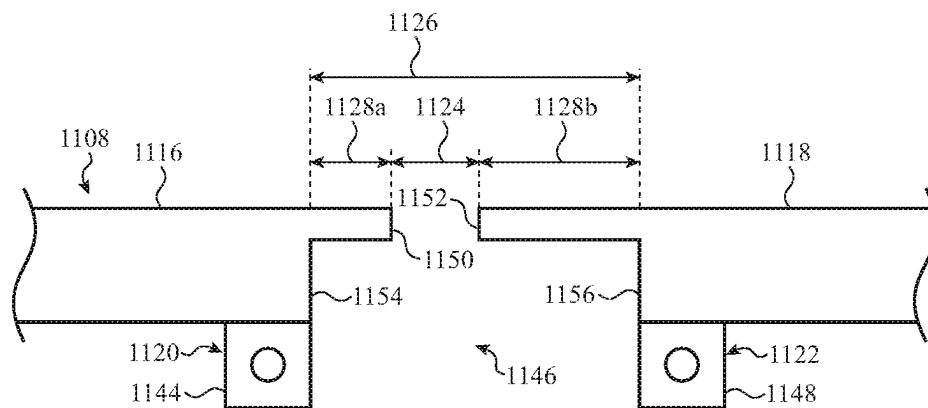


FIG. 11B

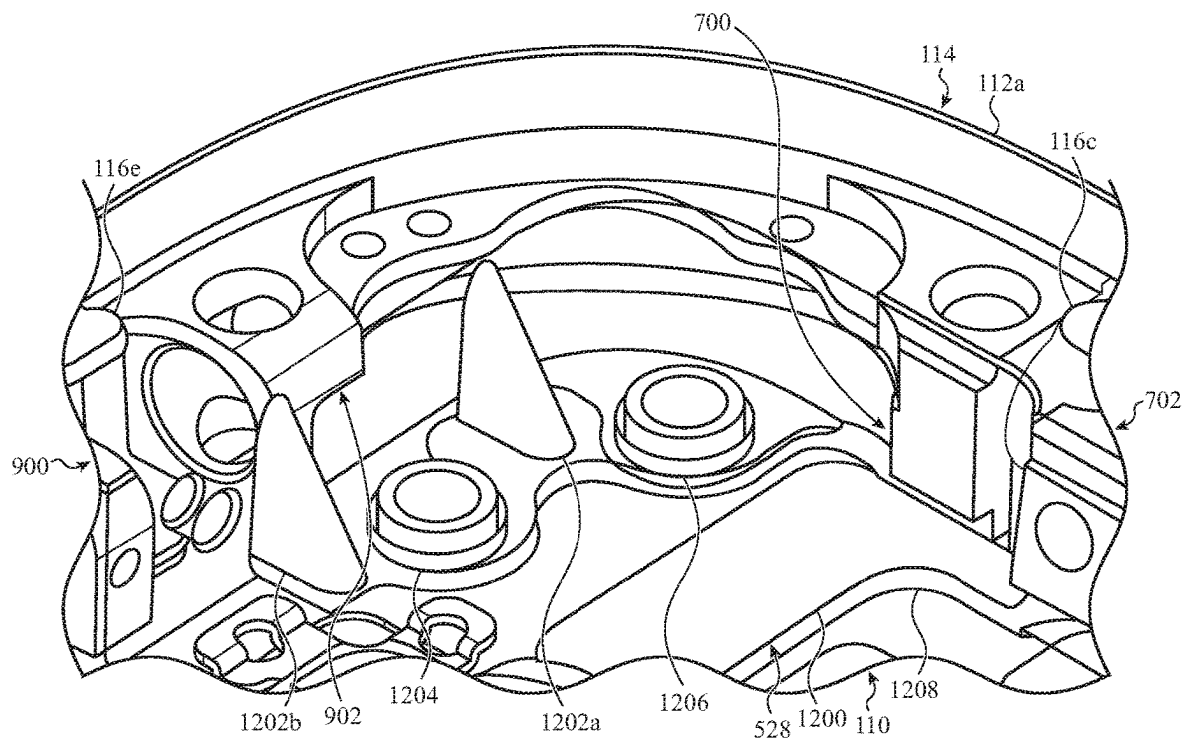


FIG. 12

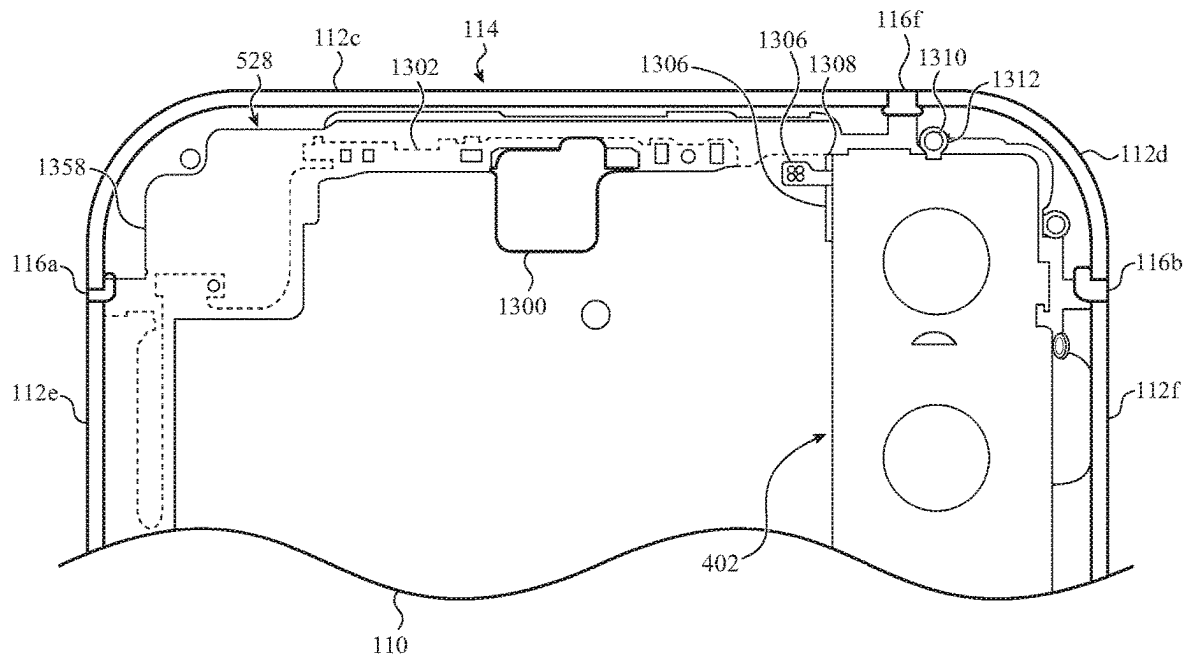


FIG. 13A

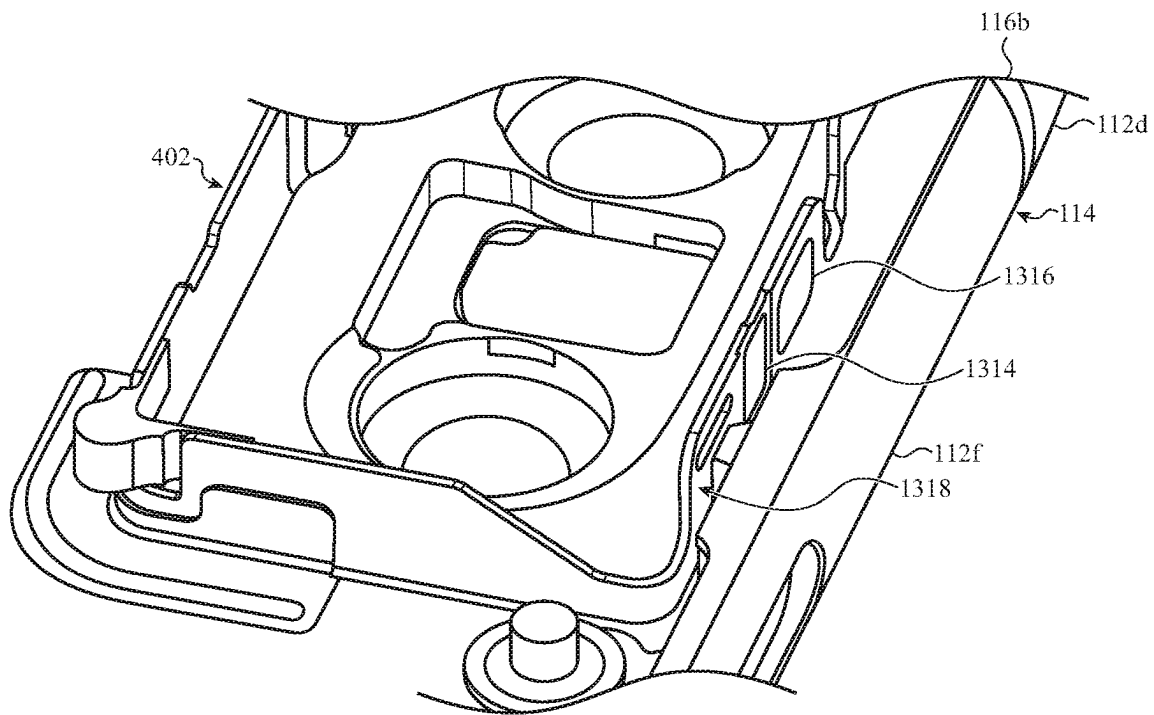


FIG. 13B

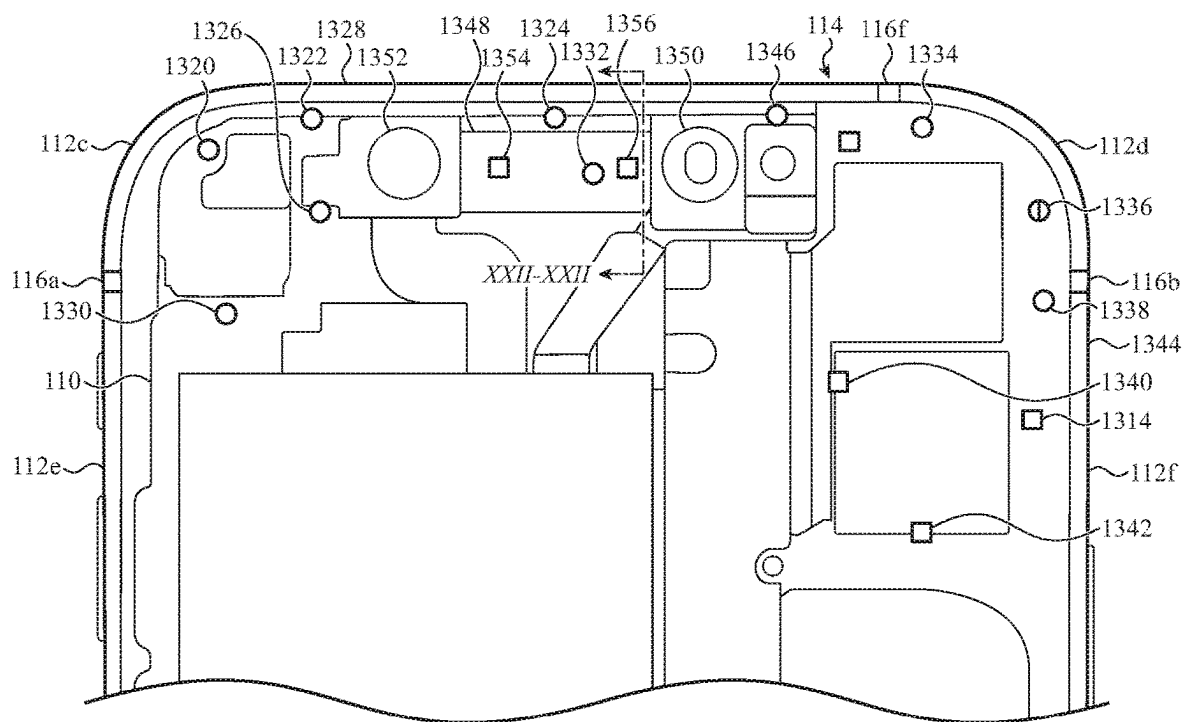


FIG. 13C

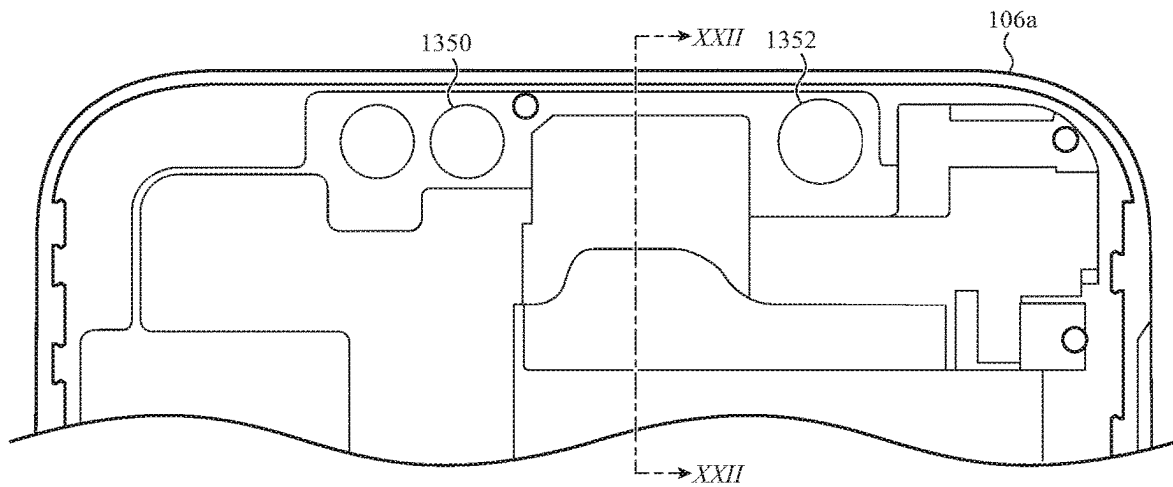


FIG. 13D

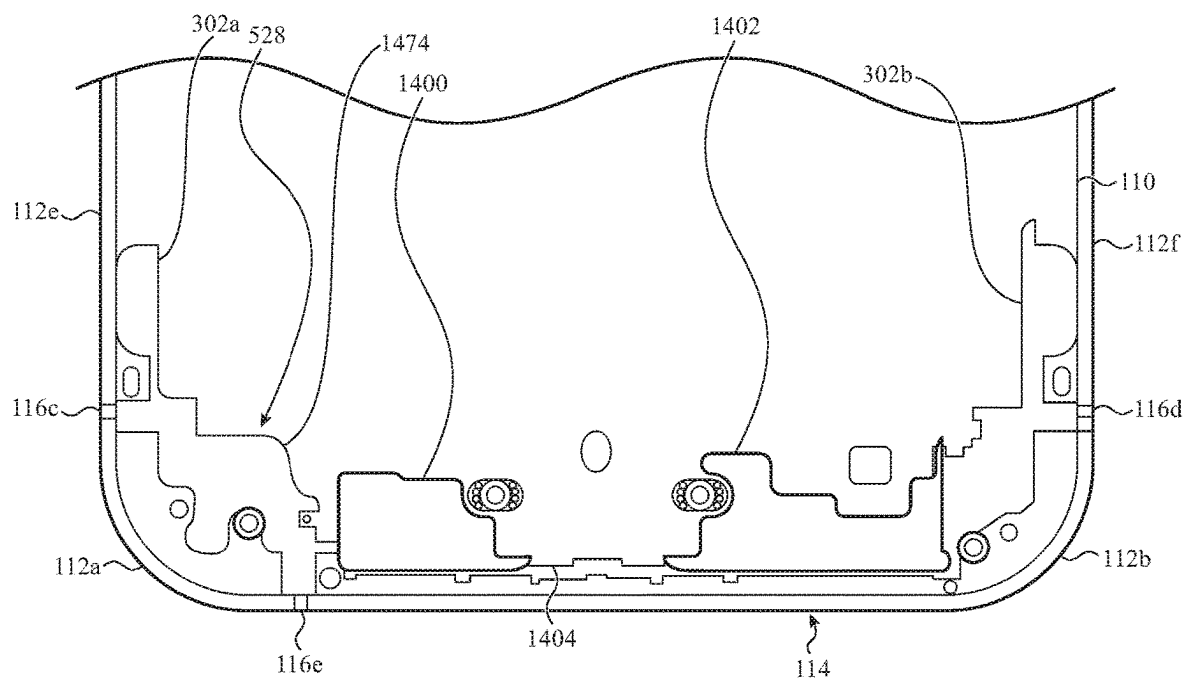


FIG. 14A

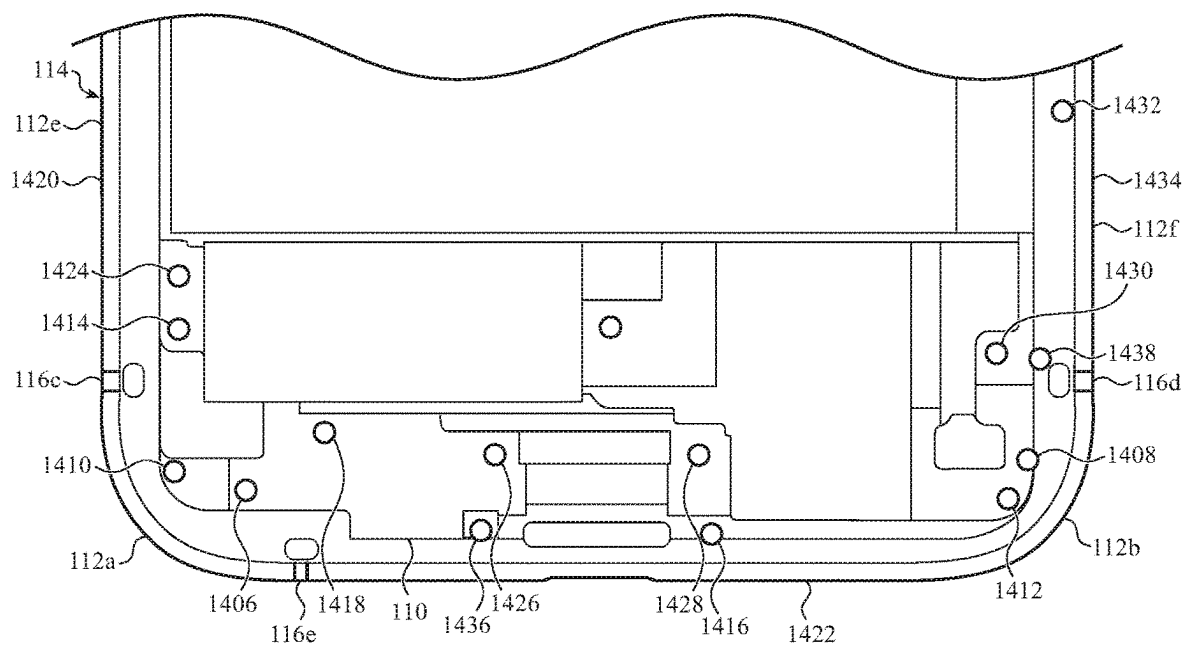


FIG. 14B

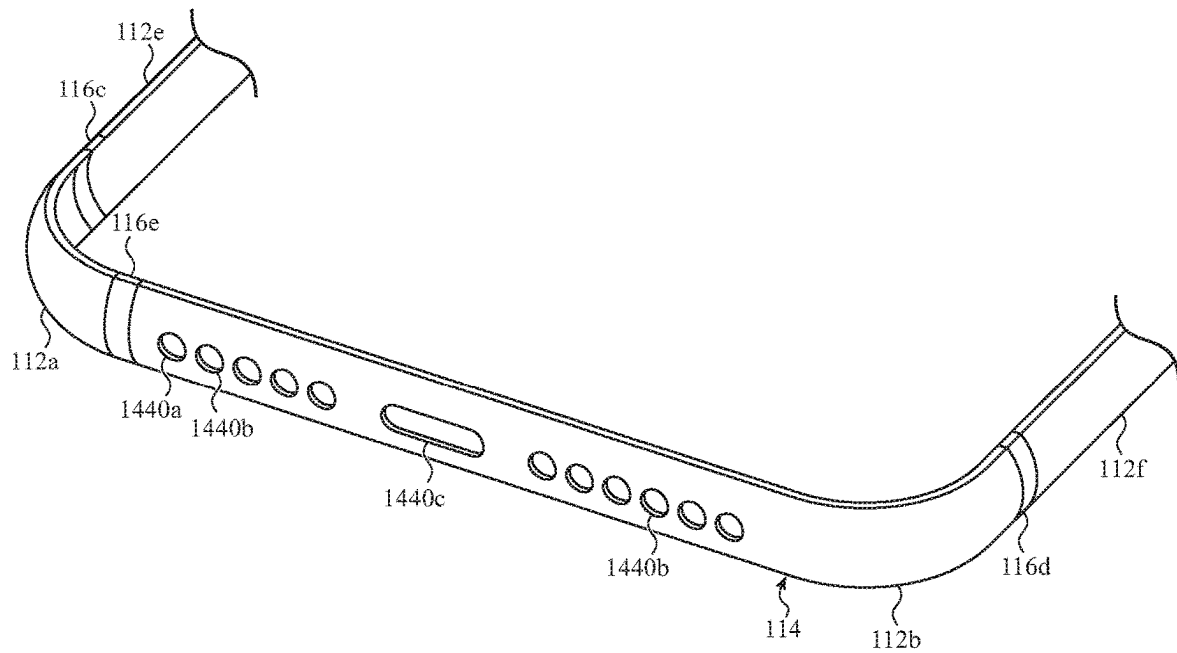


FIG. 14C

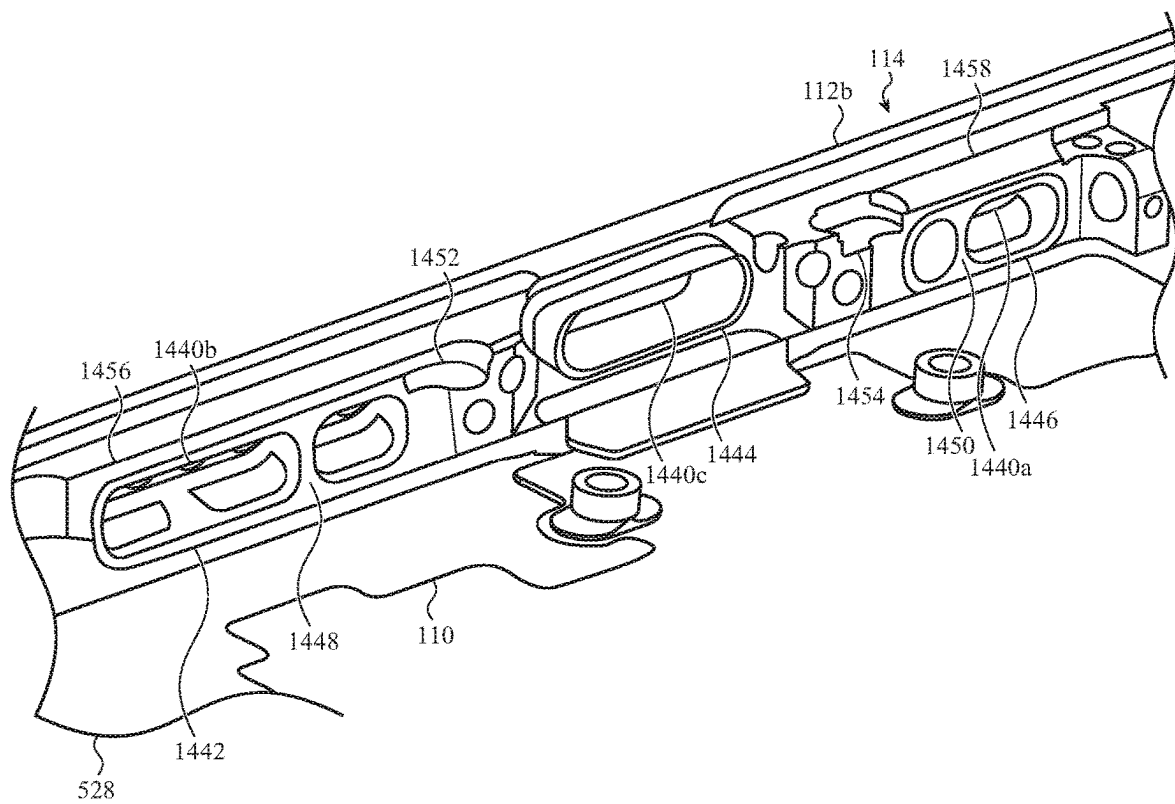


FIG. 14D

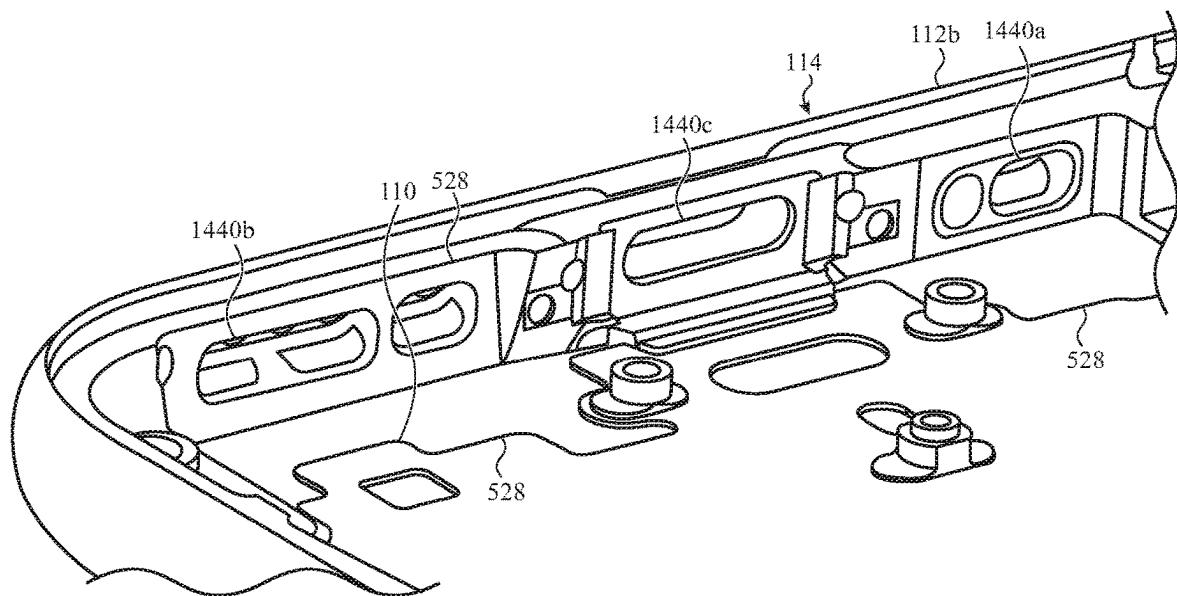


FIG. 14E

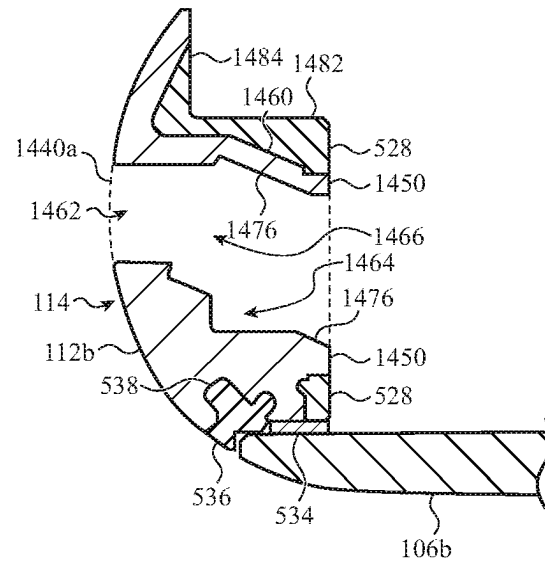


FIG. 14F

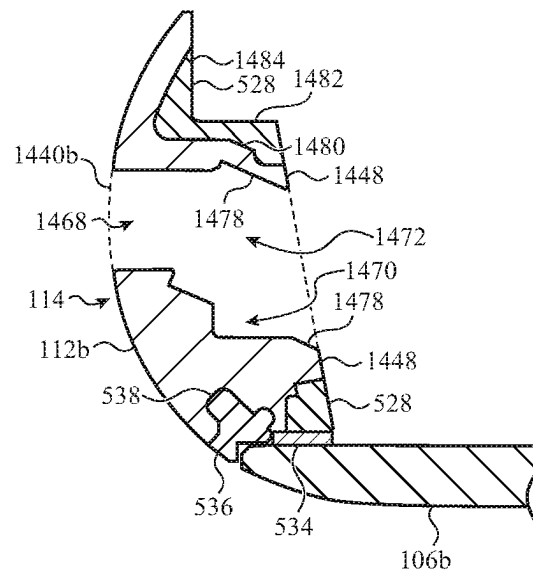


FIG. 14G

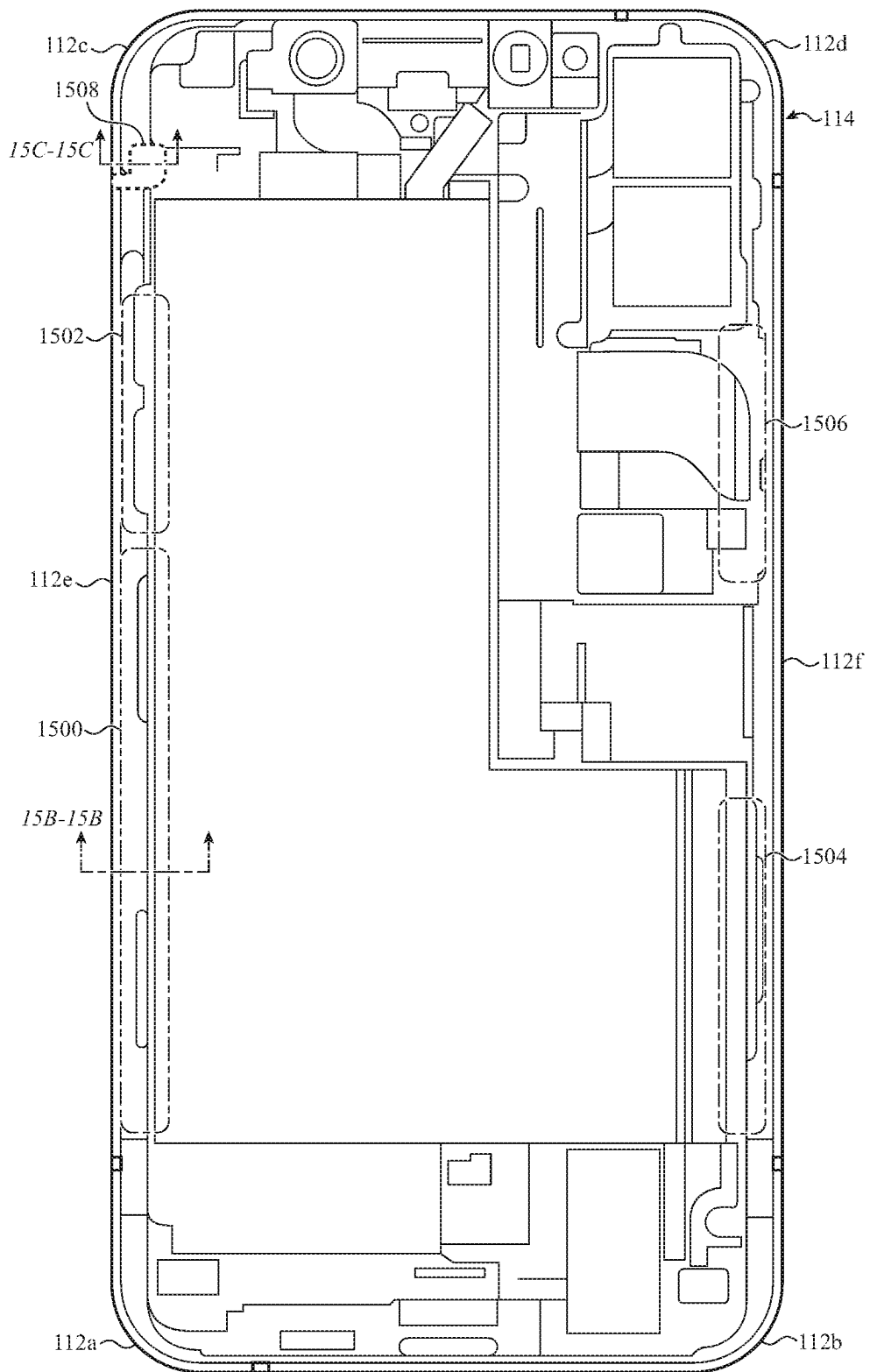


FIG. 15A

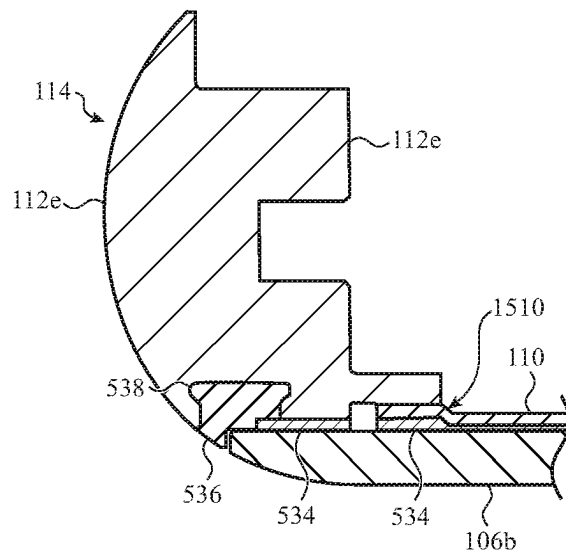


FIG. 15B

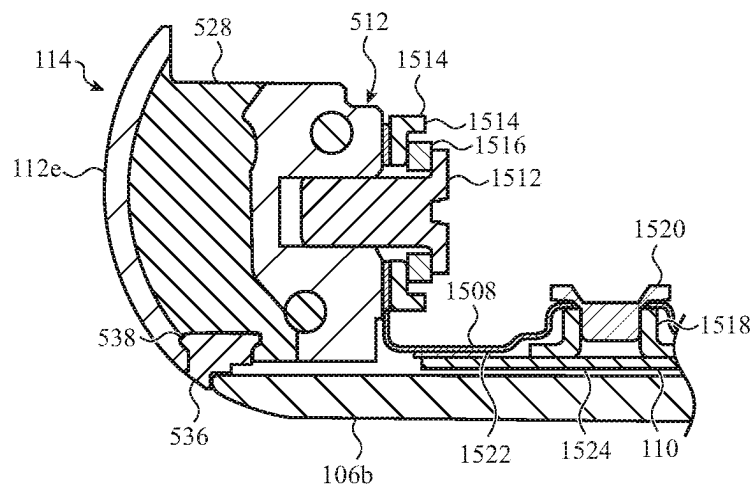


FIG. 15C

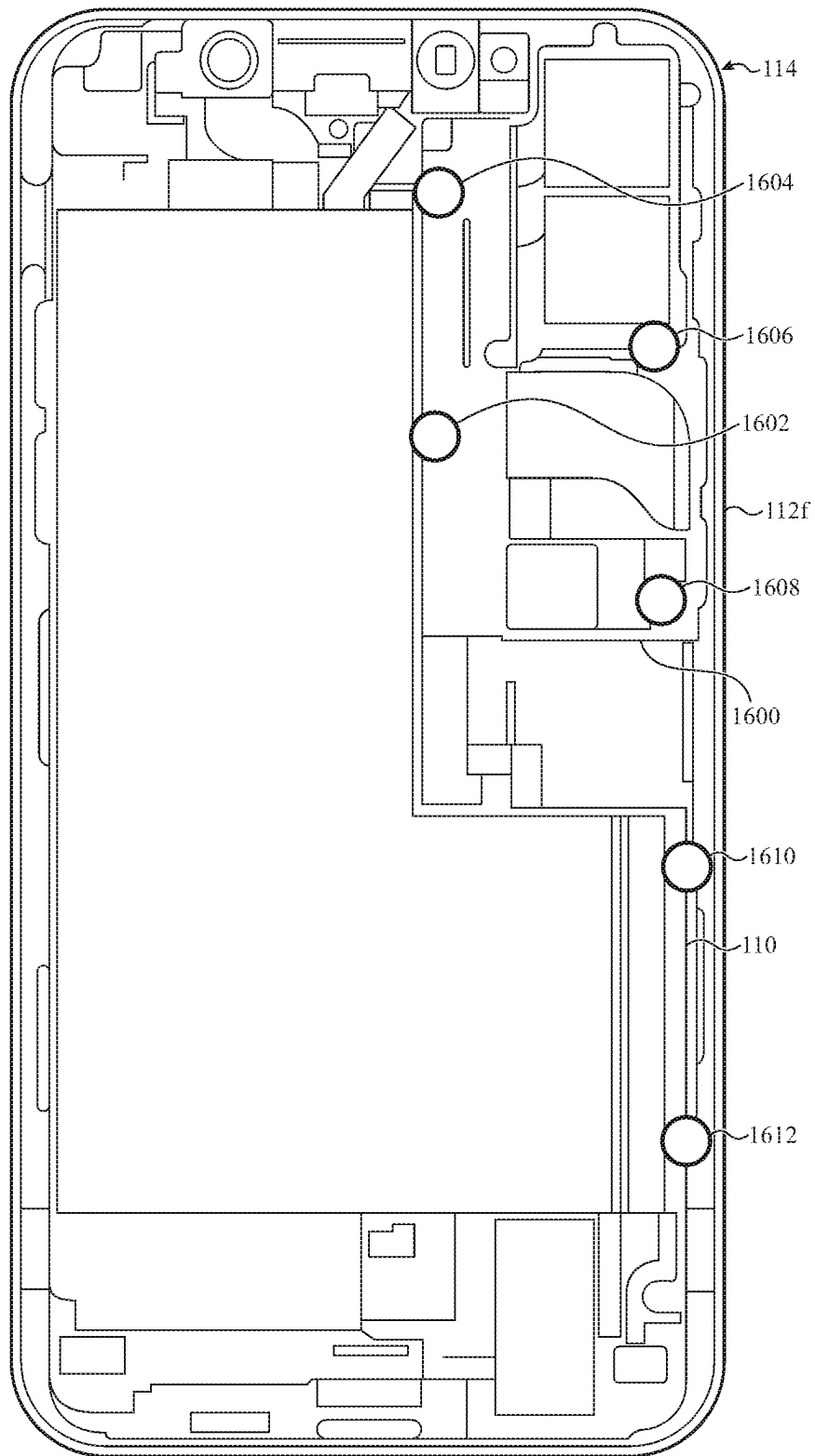


FIG. 16A

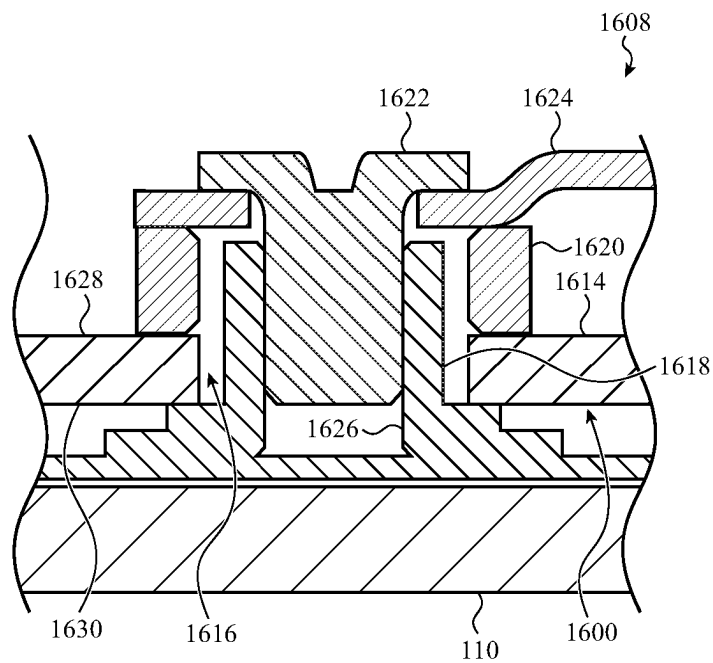


FIG. 16B

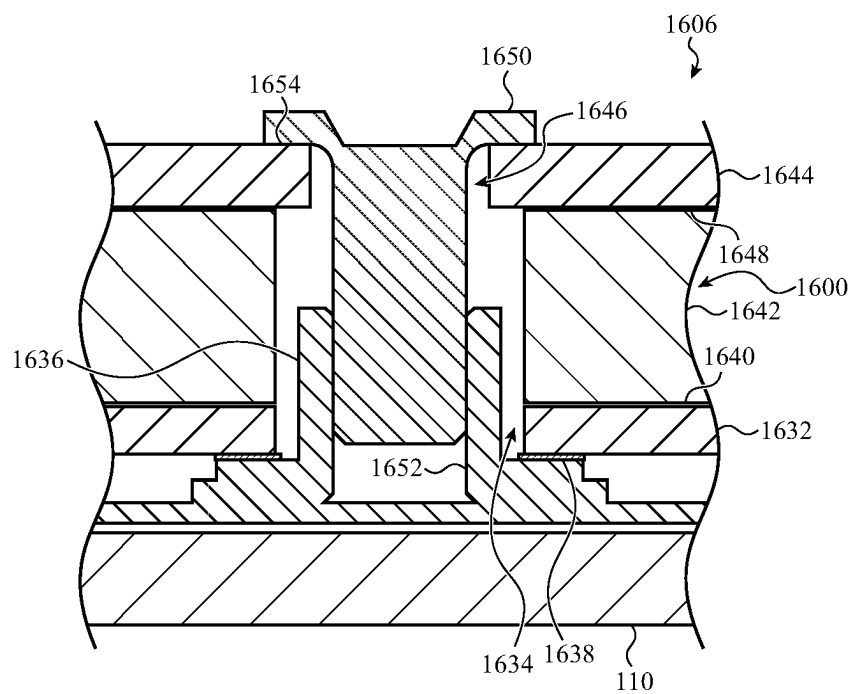


FIG. 16C

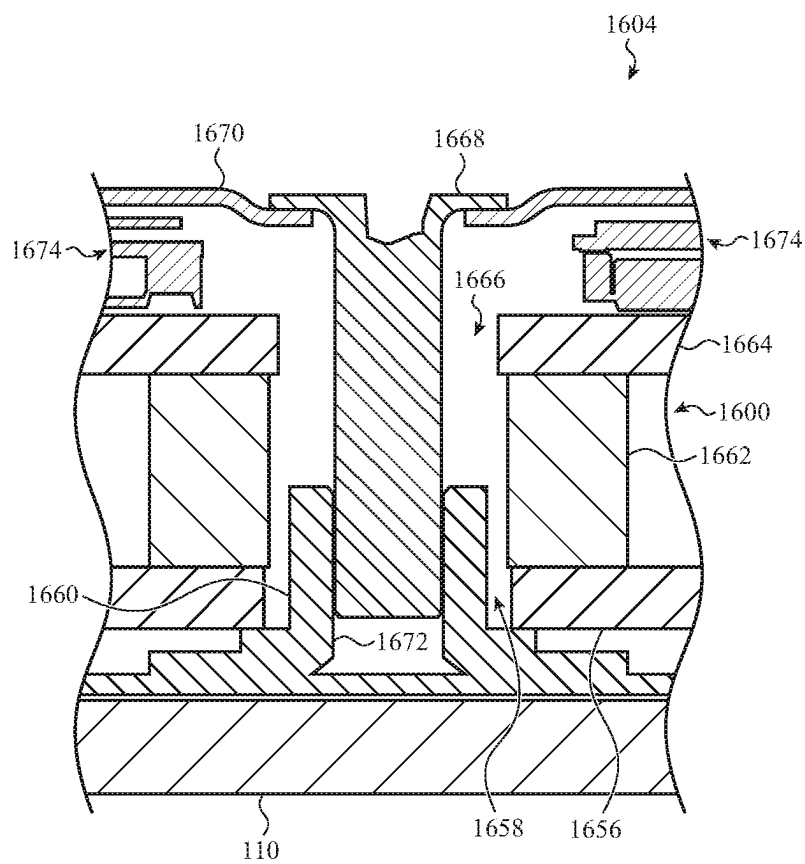


FIG. 16D

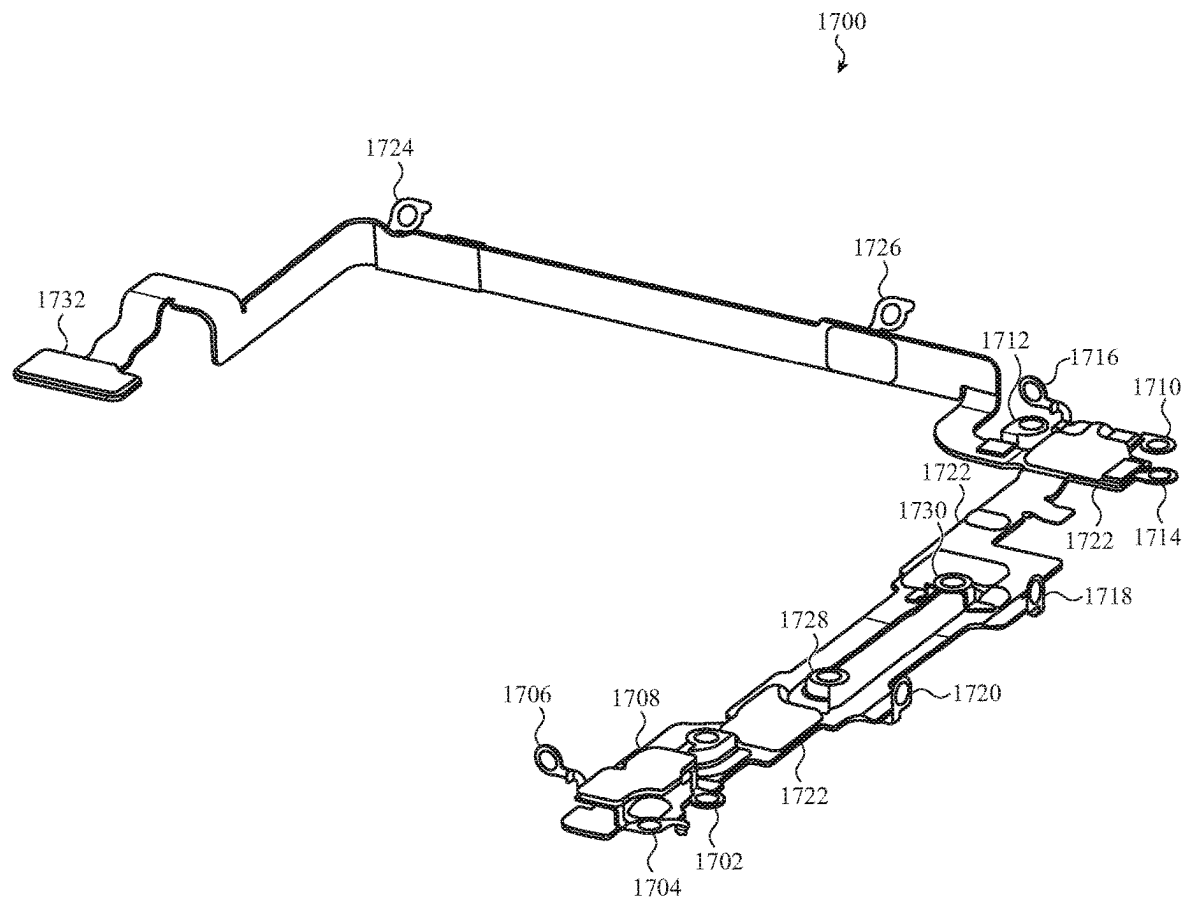


FIG. 17A

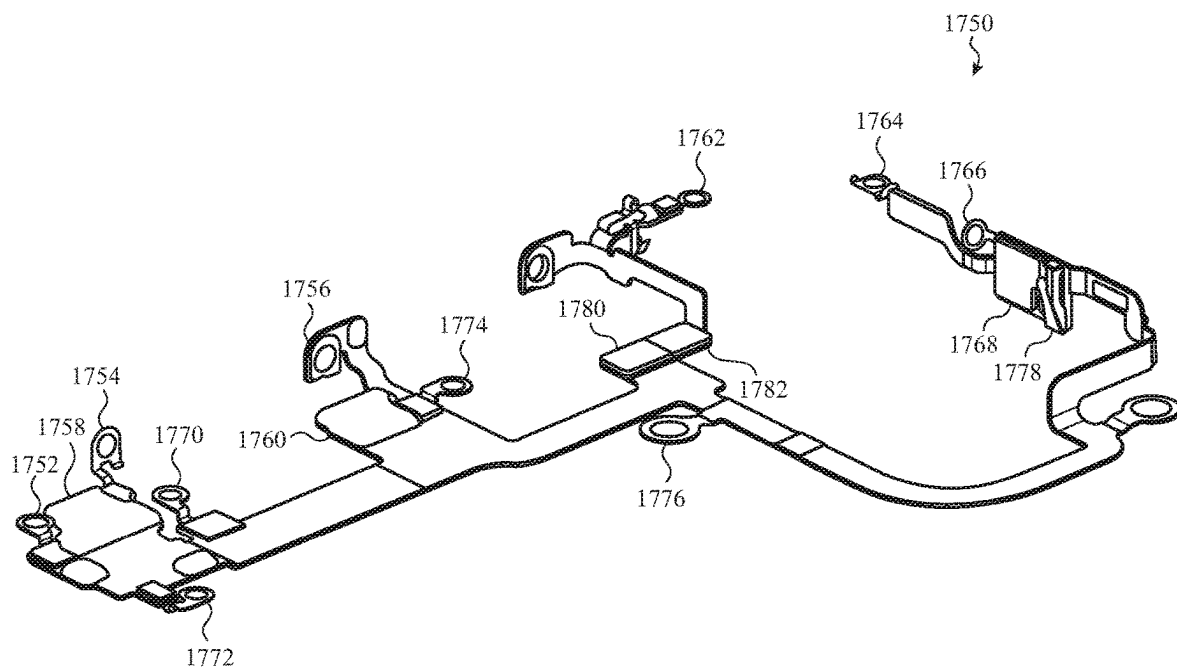


FIG. 17B

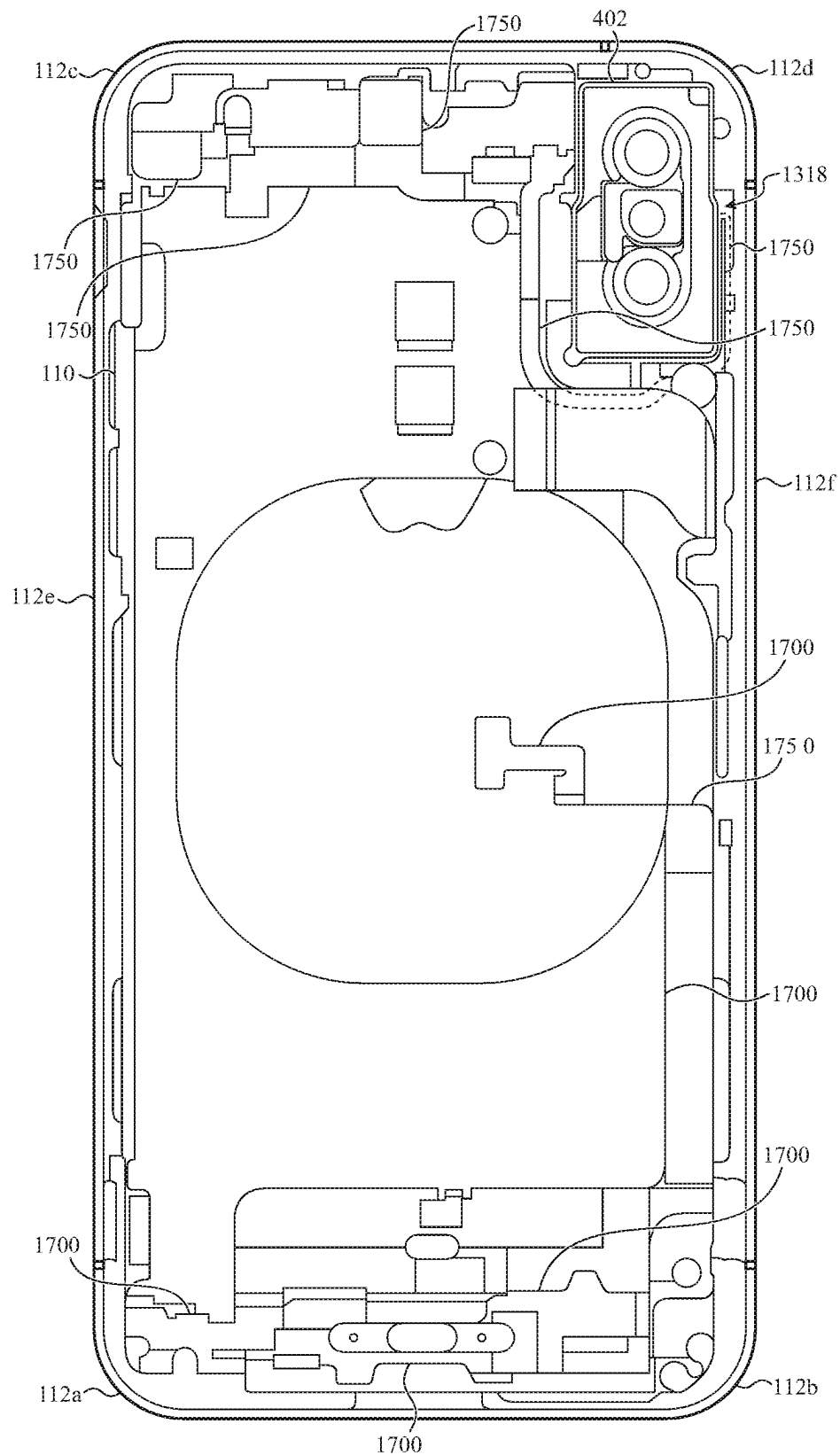


FIG. 18

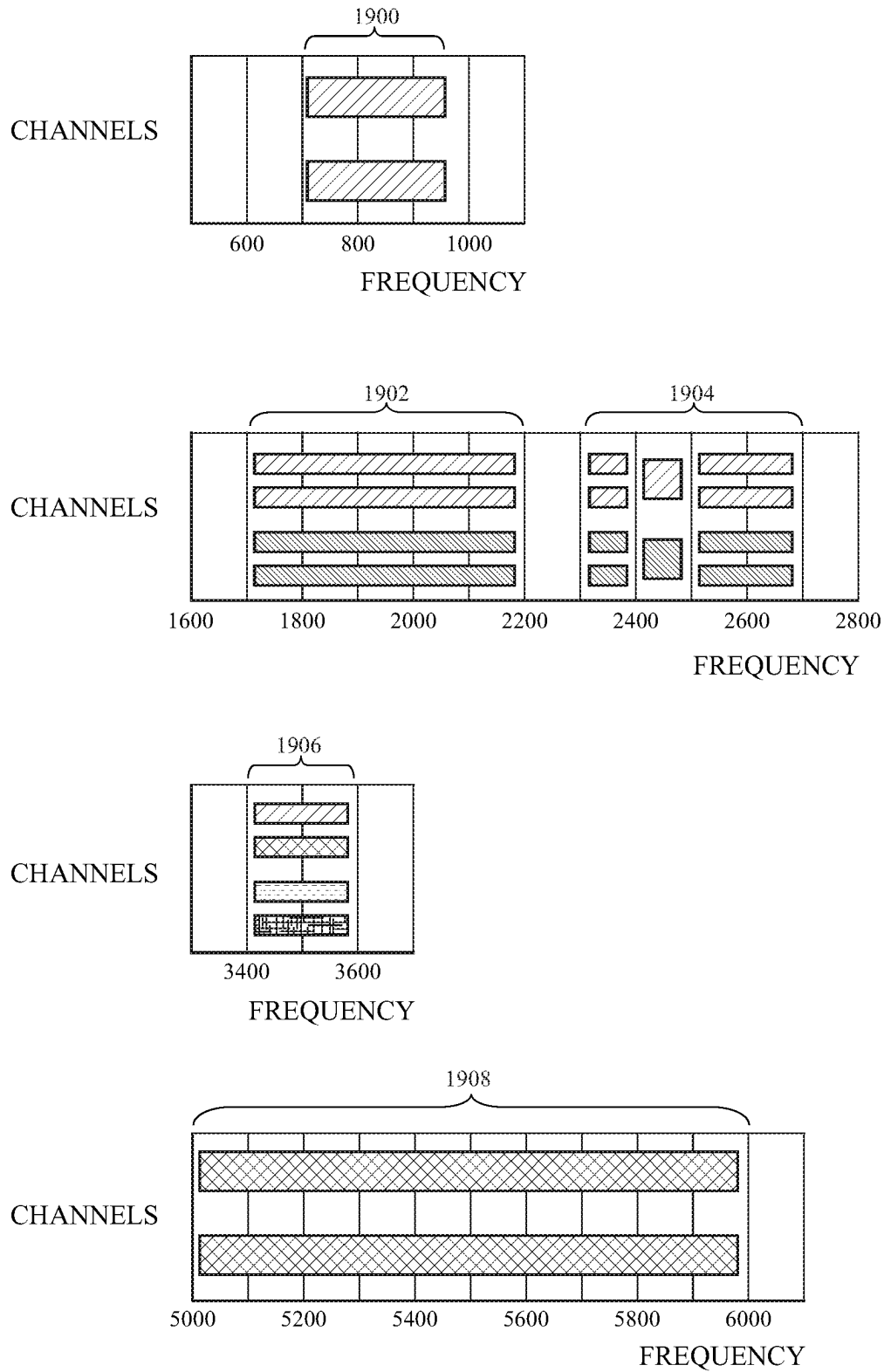


FIG. 19

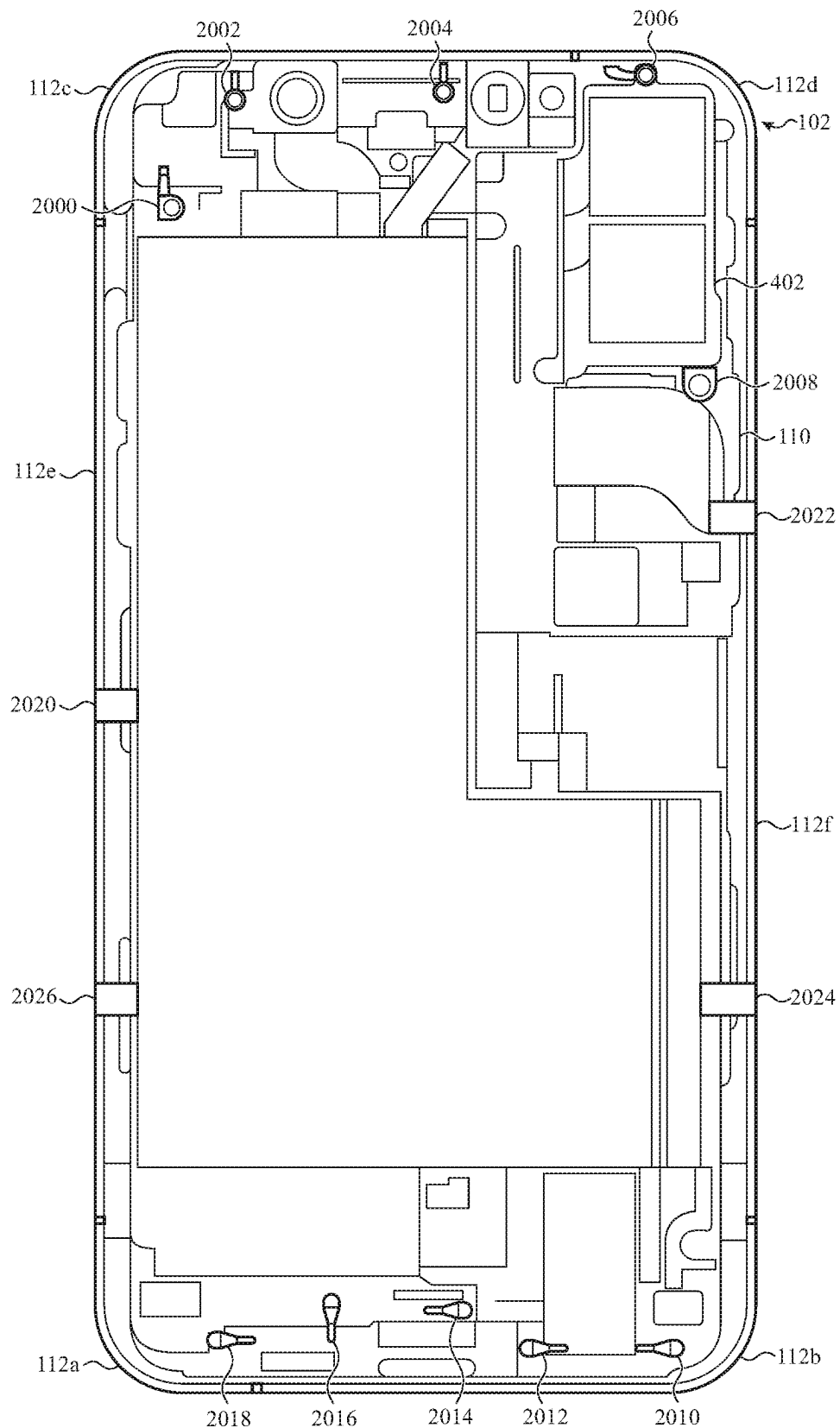


FIG. 20A

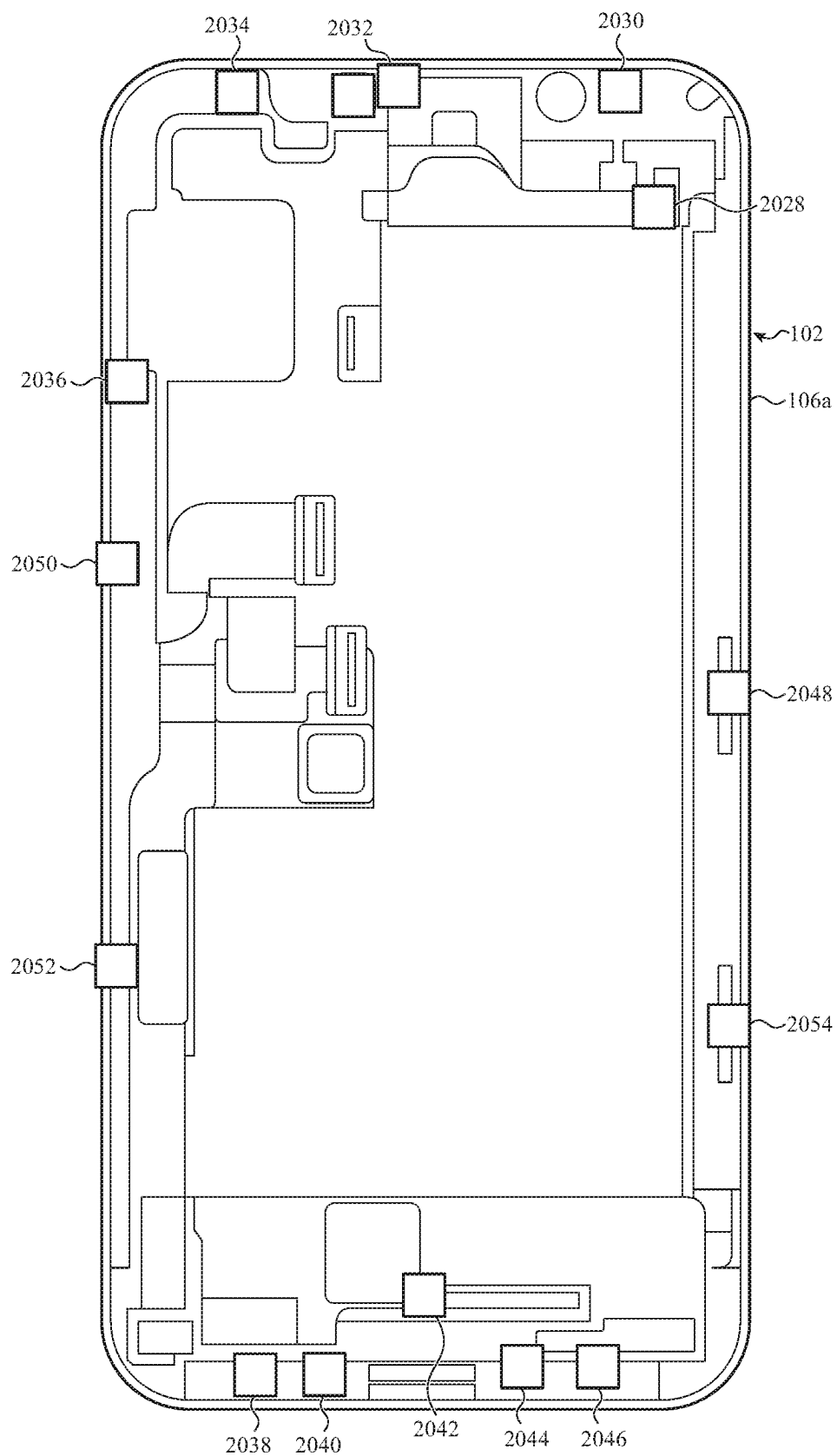


FIG. 20B

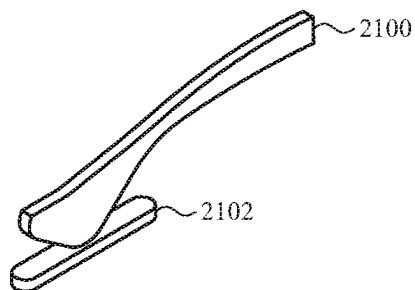


FIG. 21A

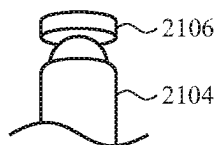


FIG. 21B

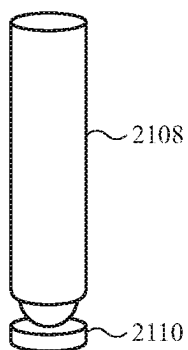


FIG. 21C

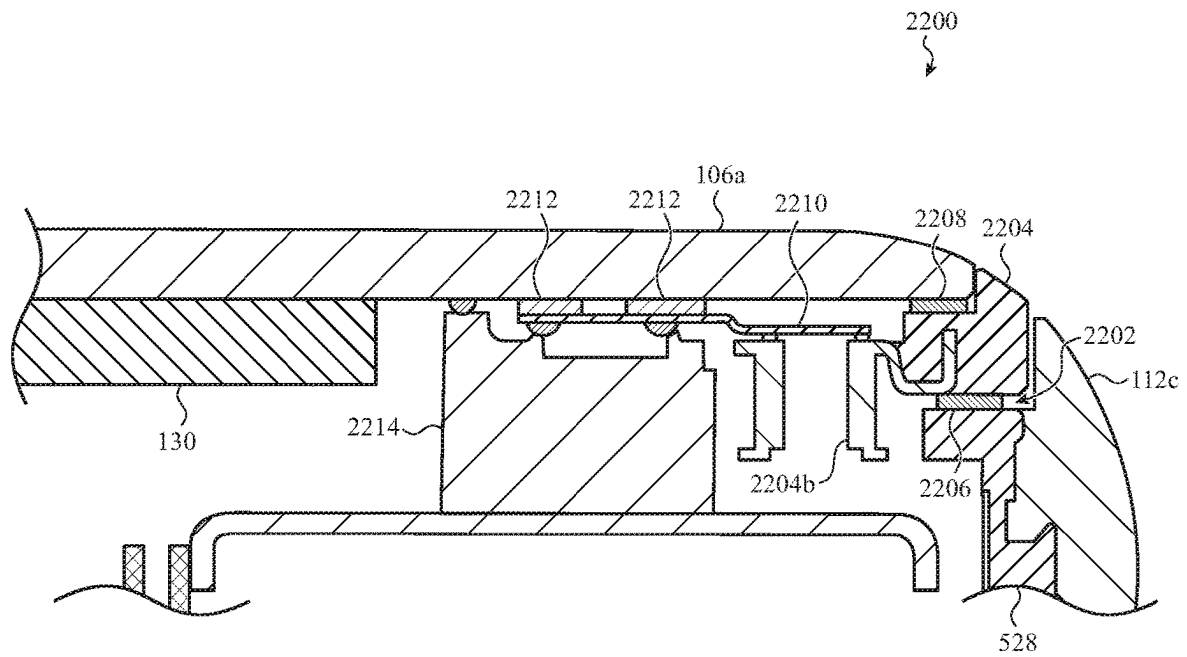
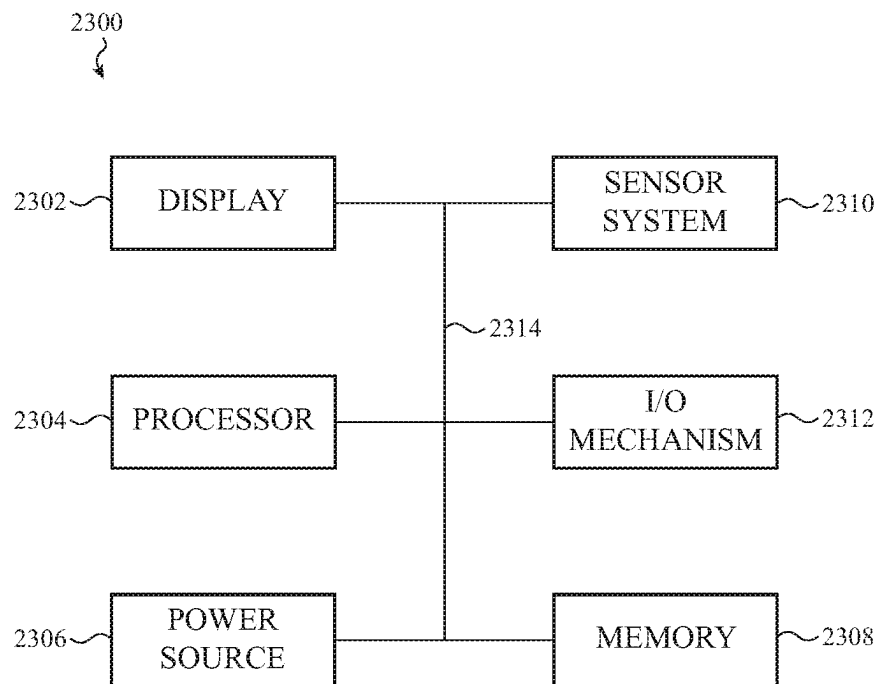


FIG. 22

**FIG. 23**

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HOUSING AND ANTENNA ARCHITECTURE FOR MOBILE DEVICE

CROSS REFERENCE TO RELATED APPLICATION(S)

This application is a continuation patent application of U.S. patent application Ser. No. 16/142,352, filed Sep. 26, 2018 and titled "Housing and Antenna Architecture for Mobile Device," which is a nonprovisional patent application of and claims the benefit of U.S. Provisional Patent Application No. 62/725,237, filed Aug. 30, 2018 and titled "Housing and Antenna Architecture for Mobile Device", the disclosures of which are hereby incorporated herein by reference in their entireties.

FIELD

The described embodiments generally relate to a housing and antenna architecture for a mobile device. More particularly, the described embodiments relate to a segmented housing in which housing segments may be positioned at each of one or more corners defining portions of an exterior surface of the device. In some embodiments, one or more of the housing segments may be operable as an antenna for the device.

BACKGROUND

Portable electronic devices have become more compact over the years. There is an increasing need to make housings that are both aesthetically pleasing and structurally robust. Some traditional housings are formed from a single material in order to simplify manufacturing and assembly. However, a single-piece housing may not provide some of the structural, functional, and/or aesthetic benefits of a multi-segment housing, as described herein. The devices, housings, and components described herein, and the corresponding methods of manufacture described herein, may be used to improve the manufacturability and function of multi-segment housings while maintaining the benefits of multi-segment housings.

SUMMARY

Some example embodiments are directed to multi-segment housings that include multiple conductive segments. The multiple conductive segments may be structurally coupled by one or more non-conductive housing components, which non-conductive housing components may define segments or splits between the conductive segments. One or more of the conductive segments may be configured to operate as an antenna, and the non-conductive housing component(s) may provide electrical insulation between a conductive segment and one or more other conductive segments or components. In some embodiments, a sidewall of a device may have a generally rectangular shape, and four different conductive segments may define four different corners about the sidewall. Each of the four different conductive segments may be configured to operate as a different antenna when the device wirelessly communicates with other devices, or different combinations of the conductive segments may be configured for wireless communication in different wireless communication modes.

In a first aspect, the present disclosure describes a device including a display and a housing. The housing may surround the display and have four corners defining portions of

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an exterior surface of the device. The housing may include a first housing segment defining at least part of a first corner of the four corners, a second housing segment defining at least part of a second corner of the four corners, a third housing segment defining at least part of a third corner of the four corners, and a non-conductive housing component that structurally couples the first housing segment to another portion of the housing. The third corner may form part of the housing diagonally opposite the second corner. The first housing segment may be configured to operate as an antenna.

In another aspect, the present disclosure describes a device including a display, a housing, and wireless communication circuitry. The housing may define a sidewall of the device surrounding a perimeter of the display. The housing may include a first conductive segment defining at least part of a first corner of the sidewall, a second conductive segment defining at least part of a second corner of the sidewall, a third conductive segment defining at least part of a third corner of the sidewall, a fourth conductive segment defining at least part of a fourth corner of the sidewall, and a non-conductive housing component that structurally couples the first conductive segment to the second conductive segment and electrically insulates the first conductive segment from the second conductive segment. The wireless communication circuitry may be coupled to at least the first conductive segment.

In still another aspect of the disclosure, a device includes a display, a housing, and wireless communication circuitry. The housing may define a sidewall of the device and at least partially define an interior volume including the display. The housing may include a first conductive antenna segment defining a first portion of the sidewall, a second conductive antenna segment defining a second portion of the sidewall, and a non-conductive housing component defining a third portion of the sidewall and electrically insulating the second conductive antenna segment from the first conductive antenna segment. The wireless communication circuitry may be disposed within the interior volume. The wireless communication circuitry may be operable in a first wireless communication with the second conductive antenna segment electrically disconnected from the first conductive antenna segment, and in a second wireless communication mode with the second conductive antenna segment electrically connected to the first conductive antenna segment.

In another aspect, the present disclosure describes a device including a display and a housing. The housing may define a sidewall of the device and an interior volume including the display. The housing may include a first housing segment defining a first portion of the sidewall and a first interlock feature extending into the interior volume. The first interlock feature may have a first interlock surface and a first hole extending into the first interlock surface. The housing may also include a second housing segment defining a second portion of the sidewall and a second interlock feature extending into the interior volume. The second interlock feature may have a second interlock surface opposite to the first interlock surface and a second hole extending into the second interlock surface. A non-conductive housing component may at least partially fill the first hole and the second hole, thereby structurally coupling the first housing segment to the second housing segment.

In yet another aspect, the present disclosure describes another device including a display and a housing. The housing may at least partially surround the display and include a first housing segment defining at least a first portion of an exterior surface of the device and a first

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interlock feature. The first interlock feature may have an interlock surface that is offset with respect to an end surface of the first housing segment, and the first interlock feature may have a first opening formed in the interlock surface. The housing may also include a second housing segment defining at least a second portion of the exterior surface of the device and a second interlock feature. The second interlock feature may have a second opening aligned with the first opening. The housing may further include a non-conductive housing component defining a third portion of the exterior surface of the device. The non-conductive housing component may extend into the first opening and the second opening.

In still another aspect of the disclosure, a device includes a display and a housing, with the housing defining a sidewall that extends around the display. The housing may include a first housing segment defining a first portion of the sidewall and a first interlock feature having a first hole, a second housing segment defining a second portion of the sidewall and a second interlock feature having a second hole, and a non-conductive housing component. The second hole may be substantially aligned with the first hole, and the non-conductive housing component may at least partially fill the first hole and the second hole, thereby structurally coupling the first housing segment to the second housing segment.

In addition to the aspects and embodiments described above, further aspects and embodiments will become apparent by reference to the drawings and by study of the following description.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure will be readily understood by the following detailed description in conjunction with the accompanying drawings, wherein like reference numerals designate like structural elements, and in which:

FIGS. 1A-1C show an example of an electronic device such as a mobile phone or tablet computer;

FIGS. 2A-2E show a number of different configurations for a multi-segment housing that forms a sidewall of a device;

FIGS. 3A-3C show example implementations of the first, second, third, and fourth housing segments described with reference to FIGS. 1A-1C & 2A, and show example positions of the housing segments with respect to a support plate;

FIG. 4 shows each of the first, second, third, fourth, fifth, and sixth housing segments described with reference to FIGS. 1A-1C & 2A in relation to the support plate described with reference to FIGS. 3A-3C, and a non-conductive housing component that structurally couples the housing segments to each other and/or the support plate;

FIGS. 5A-10C show several interior views of example implementations of the housing segments described with reference to FIGS. 2A, 3A-3C, & 4;

FIGS. 11A & 11B illustrate how exterior (sidewall) and interior gaps between housing segments disposed along a sidewall of a device may be aligned symmetrically or asymmetrically;

FIG. 12 shows an isometric view of the first housing segment described with reference to FIGS. 1A-1C, 2A, 3A-3C, 4, 7A, 7B, 9A, & 9C, with some of the non-conductive housing component(s) that abut, fill, and surround the interlock features and other interior surfaces of the first housing segment;

FIGS. 13A-13D show various details of a device forehead;

FIGS. 14A-14G show various details of a device chin;

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FIGS. 15A-15C show example areas where the fifth and sixth housing segments described with reference to FIGS. 1A-1C, 2A, 3A-3C, 4, 5A, 5B, 6A, 6C, 7A, 7C, 8A, & 8B may be structurally coupled to a support plate;

FIGS. 16A-16D illustrate various example ground connections between a support plate and a printed circuit board or logic board;

FIGS. 17A & 17B show flex circuits that may be coupled to various ones of the housing segments described with reference to FIGS. 1A-1C, 2A, 3A, 4, 5A-10C, 13A-13D, 14A-14G, & 15A-15C;

FIG. 18 shows how the flex circuits described with reference to FIGS. 17A & 17B may be placed and routed with respect to the housing segments and support plate described with reference to FIGS. 1A-1C, 2A, 3A, 4, 5A-10C, 13A-13D, 14A-14G, 15A-15C, & 16A-16D;

FIG. 19 shows a set of wireless frequency bands that may be used for wireless communication;

FIGS. 20A & 20B show corresponding ground springs and ground pads on a housing and a cover (e.g., a cover glass) mounted to the housing to enclose a device stack including a display;

FIGS. 21A-21C show various examples of low force springs and corresponding contact pads, as may be used to implement any of the ground springs and ground pads described with reference to FIGS. 20A, 20B;

FIG. 22 shows a cross-section of the device forehead shown in FIG. 13D; and

FIG. 23 shows a sample electrical block diagram of an electronic device.

The use of cross-hatching or shading in the accompanying figures is generally provided to clarify the boundaries between adjacent elements and also to facilitate legibility of the figures. Accordingly, neither the presence nor the absence of cross-hatching or shading conveys or indicates any preference or requirement for particular materials, material properties, element proportions, element dimensions, commonalities of similarly illustrated elements, or any other characteristic, attribute, or property for any element illustrated in the accompanying figures.

Additionally, it should be understood that the proportions and dimensions (either relative or absolute) of the various features and elements (and collections and groupings thereof) and the boundaries, separations, and positional relationships presented therebetween, are provided in the accompanying figures merely to facilitate an understanding of the various embodiments described herein and, accordingly, may not necessarily be presented or illustrated to scale, and are not intended to indicate any preference or requirement for an illustrated embodiment to the exclusion of embodiments described with reference thereto.

DETAILED DESCRIPTION

Reference will now be made in detail to representative embodiments illustrated in the accompanying drawings. It should be understood that the following description is not intended to limit the embodiments to one preferred embodiment. To the contrary, it is intended to cover alternatives, modifications, and equivalents as can be included within the spirit and scope of the described embodiments as defined by the appended claims.

The embodiments described herein are directed to a multi-segment housing that may include multiple conductive segments. The conductive segments may define respective portions of a sidewall or exterior surface of the device. The multiple conductive segments may be structurally

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coupled by one or more non-conductive segments or “splits”. One or more of the conductive segments may be configured to operate as an antenna (i.e., one or more conductive segments may be configured to operate as one or more antennas). One or more of the non-conductive segments may provide electrical insulation between a conductive segment and adjacent conductive segments or components.

Some example embodiments are directed to a non-conductive segment that is structurally interlocked with one or more adjacent housing segments. In particular, a non-conductive housing segment or component may be molded into a gap between a pair of housing segments, and a portion of the non-conductive housing segment that is located or positioned internal to a housing may flow into and around various features to provide a structural interlock between the pair of housing segments. As described in more detail below, the non-conductive housing segment (or split) may be molded into one or more holes, openings, recesses, or cavities in interlock features that are formed near the ends of housing segments, internal to the housing. In some implementations, a non-conductive housing segment or component may at least partially fills the holes, openings, recesses, or cavities in interlock features formed near the adjacent ends of adjacent housing segments.

As described in more detail below, one or more of the housing segments may be formed from a conductive material and may be configured to function as an antenna for an electronic device. In particular, one or more housing segments may be operably coupled to wireless communication circuitry, and may be configured as an antenna to transmit and receive wireless communication signals. In some cases, separate housing segments may define the four main corners of a device or housing. Each separate housing segment may be configured to operate as an antenna in order to facilitate single-band or multi-band wireless communication.

These and other embodiments are described with reference to FIGS. 1A-23. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes only and should not be construed as limiting.

Directional terminology, such as “top”, “bottom”, “upper”, “lower”, “front”, “back”, “over”, “under”, “above”, “below”, “left”, “right”, etc. is used with reference to the orientation of some of the components in some of the figures described below. Because components in various embodiments can be positioned in a number of different orientations, directional terminology is used for purposes of illustration only and is in no way limiting. The directional terminology is intended to be construed broadly, and therefore should not be interpreted to preclude components being oriented in different ways. The use of alternative terminology, such as “or”, is intended to indicate different combinations of the alternative elements. For example, A or B is intended to include, A, or B, or A and B.

FIGS. 1A-1C show an example of an electronic device or simply “device” 100. The device’s dimensions and form factor, including the ratio of the length of its long sides to the length of its short sides, suggest that the device 100 is a mobile phone (e.g., a smartphone). However, the device’s dimensions and form factor are arbitrarily chosen, and the device 100 could alternatively be any portable electronic device including, for example a mobile phone, tablet computer, portable computer, portable music player, health monitor device, portable terminal, or other portable or mobile device. FIG. 1A shows a front isometric view of the device 100; FIG. 1B shows a rear isometric view of the

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device 100; and FIG. 1C shows a cross-section of the device 100. The device 100 may include a housing 102 that at least partially surrounds a display 104. The housing 102 may include or support a front cover 106a or a rear cover 106b. The front cover 106a may be positioned over the display 104, and may provide a window through which the display 104 may be viewed. In some embodiments, the display 104 may be attached to (or abut) the housing 102 and/or the cover 106a.

As shown in FIGS. 1A & 1B, the housing 102 may define four corners 108 (e.g., corners 108a, 108b, 108c, and 108d) that surround the display 104 and define portions of an exterior surface of the device 100. In the current example, each of the four corners 108 of the housing 102 is positioned at a respective corner of a generally rectangular display 104. However, the relative positions of the corners 108 may vary depending on the implementation. By way of example, the corners 108 are shown to be rounded in x/y dimensions defining the front and rear surfaces of the device 100 shown in FIGS. 1A & 1B, but may alternatively be squared or have other shapes. The housing 102 may have a generally rectangular shape having a length dimension that is greater than a width dimension. In some cases the length may be greater than 100 mm and the width may be greater than 50 mm. The housing 102 may also have a thickness that ranges between 5 mm and 15 mm.

In some cases, the housing 102 may be a multi-segment housing that includes multiple conductive or metal segments that are separated by one or more non-conductive segments. In some cases the multi-segment housing, may include a support plate 110 (see, FIG. 1C) and/or additional internal structural components that are used to support internal electronic circuitry or electronic components.

The housing segments 112 of the housing 102 may form or define part or all of a sidewall 114. In particular, the housing segments 112 may define portions of a side surface (e.g., portions of an exterior surface or exterior side surface) of the device 100, which portions of the side surface may include four corners 108 of the sidewall 114. As shown in FIGS. 1A & 1B, the housing segments 112 or sidewall 114 may at least partially surround a perimeter of the display 104, and in some cases may be configured to protect the display 104 from drops of the device 100 that involve an impact to an edge or corner 108 of the sidewall 114. By way of example, the housing 102 may include six housing segments 112 that are structurally coupled to other portions of the housing 102 by a set of one or more non-conductive segments or housing components 116.

As used herein, the term “corner” may be used to refer to a portion of an exterior surface or sidewall of a device that forms a transition between adjoining sides or sidewalls. The term corner may refer to a region that includes 3-dimensional (3D) structure(s) that include portions of the sidewalls and/or portions of the front or rear covers 106a, 106b defining the front and rear surfaces, respectively. The term “corner” may also be used to refer to a portion of a sidewall 114 that extends (linearly or non-linearly) between the front and rear surfaces of a device and also joins adjacent sidewalls. In some embodiments, the corner portion of a sidewall may define a curved or arcuate contour between the front and rear surfaces. In some embodiments, the corner portion of a sidewall may define a flat side that joins the front and rear surfaces. As described herein, a generally rectangular device may be considered to have four corners that define the perimeters of the front and rear surfaces of the device with each corner connected to two adjacent corners. A generally rectangular device may also be considered to

have four corners joined by four sides, with the four corners, in combination with the four sides, defining the perimeters of the front and rear surfaces of the device.

As explained in more detail herein, one or more of the housing segments **112** may be mechanically or structurally coupled to one or more adjacent housing segments **112** by the non-conductive housing segments or components **116**, which segments or components **116** may partially or completely fill gaps between the housing segments **112**. In some cases, the non-conductive housing segments or components **116** may also couple the housing segments **112** to the support plate **110** or another internal structure. A contiguous or monolithic piece of non-conductive material (e.g., a monolithic non-conductive component) may join or form all or multiple ones of the non-conductive housing segments or components **116** (or fill all or multiple ones of the gaps between housing segments **112**), or different pieces of non-conductive material may join different sets of adjacent housing segments **112** (or fill different gaps between different pairs of adjacent housing segments **112**). At least one non-conductive housing segment or component in the set of non-conductive housing segments or components **116** may define a portion (e.g., a segment) of an exterior surface of the sidewall **114** or housing **102**. In some alternative embodiments, the housing **102** may include more or fewer housing segments separated by more or fewer gaps filled by non-conductive housing segments or components **116**. In addition to mechanically coupling housing segments **112**, the non-conductive housing segment(s) or component(s) may electrically insulate housing segments **112**.

The housing segments **112** may have various lengths or shapes, and may be positioned symmetrically or asymmetrically about the device **100** or its sidewall **114**. By way of example, and with reference to FIGS. **1A** & **1B**, the device **100** is shown to have a first housing segment **112a** defining at least part (or all) of a first corner **108a** of the sidewall **114**. A second housing segment **112b** defines at least part of a second corner **108b** of the sidewall **114**, a third housing segment **112c** defines at least part of a third corner **108c** of the sidewall **114**, and a fourth housing segment **112d** defines at least part of a fourth corner **108d** of the sidewall **114**. In some embodiments, the second and third housing segments **112b**, **112c** may traverse greater lengths along the sidewall **114** than the first and fourth housing segments **112a**, **112d**. A fifth housing segment **112e** defines at least part of a first edge of the sidewall **114**, between the first housing segment **112a** and the third housing segment **112c**, and a sixth housing segment **112f** defines at least part of a second edge of the sidewall **114**, between the second housing segment **112b** and the fourth housing segment **112d**. The third corner **108c** forms a part of the housing **102** that is diagonally opposite the second corner **108b**, and the fourth corner **108d** forms a part of the housing **102** that is diagonally opposite the first corner **108a**. The second edge forms part of the housing **102** that is opposite the first edge. The designations “first,” “second,” “third,” “fourth,” “fifth,” and “sixth” are arbitrary, and are used herein only for ease of explanation.

In this example, a different housing segment **112** forms each of the four corners **108**. However, the specific configuration of housing segments **112** may vary depending on the implementation. For example, a single housing segment may define two or more corners of a device, as described with reference to FIG. **2D**, or substantially straight or non-corner housing segments may be positioned at the top and bottom edges of a device (e.g., similarly to the fifth and sixth housing segments **112e**, **112f** positioned at the side edges of the device **100**). A housing may include more or

fewer housing segments than the housing segments **112** shown in FIGS. **1A** & **1B**, and the housing segments may be distributed in various ways about a device's sidewall, as described with reference to FIGS. **2A-2E**.

In some embodiments, one or more of the housing segments **112** may be a conductive segment formed from a metal or conductive material, and may be configured to operate as an antenna for the device **100**. Housing segments **112** that are configured to be operated as antennas may sometimes be referred to herein as conductive antenna segments. Wireless communication circuitry **118** within the device **100** may be electrically coupled to one or more of the conductive segments. For example, wireless communication circuitry **118** may be coupled to one or more (or each) of the housing segments **112** that is conductive and configured to operate as an antenna. When the housing segments **112a**, **112b**, **112c**, **112d** defining the corners **108** of the device **100** are conductive segments, the wireless communication circuitry **118** may be operable to configure the conductive segments (as antennas) for wireless communication in one or more wireless frequency bands. Configuring the conductive segments for wireless communication may enable the device **100** to communicate with other devices in one or more wireless communication modes, such as a 4x4 multiple-input multiple-output (MIMO) wireless communication mode, or other wireless communication modes that use one or more antenna, and up to four antennas, simultaneously. The wireless communication circuitry **118** may include one or more radio frequency (RF) transmitters or receivers, one or more switches, one or more modems, and so on.

In general, the housing segments **112** may be formed from a metal material including, for example, steel, stainless steel, aluminum, titanium, and/or a metal alloy. In some embodiments, the housing segments **112** may be formed from a non-metal material and may be coated or covered by a metal or metallic coating or layer. The non-conductive housing segments or components **116** may be formed from a polymer material, composite, or other non-conductive material. Example polymers include, polycarbonate, acrylonitrile butadiene styrene (ABS), polyurethane, polyether ether ketone (PEEK), polybutylene terephthalate (PBT), polyamide, or other similar materials.

In some embodiments, the non-conductive housing segments or components **116** may be formed by a polymer material having a fiber fill, and the polymer material may structurally couple the housing segments **112** in addition to forming portions of an exterior surface of the sidewall **114** (e.g., portions of the sidewall **114** that bridge or fill exterior gaps between housing segments **112**). In other embodiments, the non-conductive housing segments or components **116** may include a first portion formed from a first polymer material and a second portion formed from a second polymer material. The first polymer material may have a fiber fill and structurally couple the housing segments **112**. The second polymer material may be different from the first polymer material and form portions of an exterior surface of the sidewall **114**. Each polymer having a fiber fill may have a fiber fill including glass or other types of fibers. In some embodiments, the second polymer material may also have a fiber fill, but have a fiber fill that differs from the fiber fill of the first polymer material.

As shown in FIGS. **1A** & **1B**, the device **100** may include various other components. For example, the front of the device **100** may include one or more front-facing cameras **120**, speakers **122**, sensors **124**, microphones, or other components (e.g., audio, imaging, or sensing components)

that are configured to transmit or receive signals to/from the device **100**. In some cases, a front-facing camera **120**, alone or in combination with other sensors, may be configured to operate as a bio-authentication or facial recognition sensor. The device **100** may also include various input devices, including a mechanical or virtual button **123**, which may be located along the front surface of the device **100**. The device **100** may also include buttons or other input devices positioned along the sidewall **114** and/or rear surface of the device **100**. By way of example, the rear surface of the device **100** is shown to include a rear-facing camera **126** or other optical sensor (see, FIG. **1B**). A flash or light source may also be positioned along the rear of the device **100** (e.g., near the camera **126**).

As discussed previously, the device **100** may include a display **104** that is at least partially surrounded by the housing **102**. The display **104** may include one or more display elements including, for example, a light-emitting display (LED), organic light-emitting display (OLED), liquid crystal display (LCD), electroluminescent display (EL), or other type of display element. The display **104** may also include one or more touch and/or force sensors that are configured to detect a touch and/or a force applied to an external surface of the device **100**. The touch sensor may include a capacitive array of nodes or elements that are configured to detect a location of a touch along a surface of the cover **106**. The force sensor may include a capacitive array and/or strain sensor that is configured to detect an amount of force applied along the surface of the cover **106a**.

FIG. **1C** depicts a cross-sectional view of the device **100** of FIGS. **1A** and **1B**. As shown in FIG. **1C**, the housing **102** may include one or more non-conductive housing segments or components **116** that structurally couple the housing segments **112**. The housing **102** may also include a front cover **106a** and a rear cover **106b**, which may be structurally coupled to the non-conductive housing segments or components **116** and/or one or more of the housing segments **112**. In some cases, the rear cover **106b** may be a discrete or separate component that is attached to the non-conductive housing segments or components **116** and/or one or more of the housing segments **112**. In other cases, the rear cover **106** may be integrally formed with one or more of the housing segments **112** or non-conductive housing segments or components **116** to form a component that defines both the rear surface of the device **100** as well as one or more portions of the sidewall **116** of the device **100**.

As shown in FIG. **1C**, the sidewall **114** or housing **102** may define an interior volume **128** in which various electronic components of the device **100**, including the display **104**, may be positioned. In this example, the display **104** is at least partially positioned within the internal volume **128** and attached to an inner surface of the cover **106a**. A touch sensor, force sensor, or other sensing element may be integrated with the cover **106a** and/or the display **104** and may be configured to detect a touch and/or force applied to an outer surface of the cover **106a**. In some cases, the touch sensor, force sensor, and/or other sensing element may be positioned between the cover **106a** and the display **104**.

The touch sensor and/or force sensor may include an array of electrodes that are configured to detect a location and/or force of a touch using a capacitive, resistive, strain-based, or other sensing configuration. The touch sensor may include, for example, a set of capacitive touch sensing elements, a set of resistive touch sensing elements, or a set of ultrasonic touch sensing elements. When a user of the device touches the cover **106a**, the touch sensor (or touch sensing system) may detect one or more touches on the cover **106a** and

determine locations of the touches on the cover **106a**. The touches may include, for example, touches by a user's finger or stylus. A force sensor or force sensing system may include, for example, a set of capacitive force sensing elements, a set of resistive force sensing elements, or one or more pressure transducers. When a user of the device **100** presses on the cover **106a** (e.g., applies a force to the cover **106a**), the force sensing system may determine an amount of force applied to the cover **106a**. In some embodiments, the force sensor (or force sensing system) may be used alone or in combination with the touch sensor (or touch sensing system) to determine a location of an applied force, or an amount of force associated with each touch in a set of multiple contemporaneous touches.

As shown in FIG. **1C**, a support plate **110** may be coupled to the non-conductive housing segments or components **116** and/or one or more of the housing segments **112** and may be used to attach or mount various other components of the device **100**. For example, wireless communication circuitry, a camera(s), a bio-authentication sensor(s), a processor, and other components may be attached to the support plate **110**. As examples, the support plate **110** may be metallic or plastic (or may be formed using any of the various materials that may be used to form the housing segments **112**). In some cases, the various electronic components may be attached or integrated with one or more printed circuit boards (PCBs) or other logic boards that are attached to the support plate **110**. The processor may include a single processor or multiple processors, and may be configured to operate the touch sensing system, the force sensing system, the wireless communication circuitry, the camera(s), the bio-authentication sensor(s), or other components of the device **100**. A more detailed description of the various components of the device **100** is included below with respect to FIG. **23**.

Turning to FIGS. **2A-2E**, there are shown a number of different configurations **200** for a multi-segment housing that forms a sidewall of a device (e.g., a device such as the device **100** described with reference to FIGS. **1A-1C**). One or more conductive segments of the multi-segment housings may be configured to operate as antennas for the device.

FIG. **2A** shows a first sidewall configuration **200a** for a device (e.g., the device **100**). The sidewall **114** may include six housing segments, which housing segments may be the housing segments **112** described with reference to FIGS. **1A-1C**. Each of the housing segments **112** may be conductive or non-conductive. In some embodiments, at least one of the housing segments **112** positioned at a corner **108** of the sidewall **114** may be conductive and operated as an antenna for a device. In some embodiments, each of the housing segments **112** positioned at a corner **108** of the sidewall **114** may be conductive, and may be operated as an antenna for a device. The housing segments **112** positioned at the left and right edges of the sidewall **114** may also be conductive, and may be operated as separate antennas of a device, or as conductive antenna segments that may be electrically connected or disconnected to other conductive antenna segments of a device.

The housing segments **112** may include a first housing segment **112a** defining at least part (or all) of a first corner **108a** of the sidewall **114**, a second housing segment **112b** defining at least part (or all) of a second corner **108b** of the sidewall **114**, a third housing segment **112c** defining at least part (or all) of a third corner **108c** of the sidewall **114**, a fourth housing segment **112d** defining at least part (or all) of a fourth corner **108d** of the sidewall **114**, a fifth housing segment **112e** defining an edge disposed between the first and third housing segments **112a**, and a sixth housing

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segment **112f** defining an edge disposed between the second and fourth housing segments **112b**, **112d**. The third corner **108c** forms a part of the housing **102** that is diagonally opposite the second corner **108b**, and the fourth corner **108d** forms a part of the housing **102** that is diagonally opposite the first corner **108a**. In some embodiments, and as shown, each of the second and third housing segments **112b**, **112c** may extend along a greater portion of the sidewall **114** than each of the first and fourth housing segments **112a**, **112d**.

The first and fourth housing segments **112a**, **112d** may be substantially confined to the first and fourth corners **108a**, **108d** respectively, but in some embodiments (not shown) one or both of the first or fourth housing segments **112a**, **112d** may extend along one or more edges of the sidewall **114**. Alternatively, the first or fourth housing segment **112a**, **112d** may wrap around less than all of a corner **108a** or **108d** of the sidewall **114**.

The second and third housing segments **112b**, **112c** may wrap around the second and third corners **108b**, **108c** respectively, and may also extend along one or more edges of the sidewall **114**. For example, the second housing segment **112b** may extend along a bottom edge of the sidewall **114** (given the orientation of the sidewall **114** shown in FIG. 2A), and the third housing segment **112c** may extend along a top edge of the sidewall **114**. Alternatively, the second or third housing segment **112b**, **112c** may wrap around less than all of a corner **108b** or **108c** of the sidewall **114** and extend along one or more side edges of the sidewall **114**.

Housing segments **112** that terminate in adjacent ends along the sidewall **114** may be structurally coupled to one another by a set of one or more non-conductive housing components **116** (e.g., non-conductive housing components, **116a**, **116b**, **116c**, **116d**, **116e**, and **116f** that partially or completely fill gaps between adjacent ends of housing segments **112** about the sidewall **114**. The sidewall **114** shown in FIG. 2A has six such gaps. At least one non-conductive housing component **116** in the set of non-conductive housing components **116** may define a portion of an exterior surface of the sidewall **114** (and also an exterior surface of the housing **102** or device **100** that includes the sidewall **114**).

In some embodiments, each of the housing segments **112a**, **112b**, **112c**, **112d** positioned at a corner **108a**, **108b**, **108c**, **108d** of the sidewall **114** may be operated as a different antenna, and in some cases the housing segments **112a**, **112b**, **112c**, **112d** may be operated as different antennas simultaneously. The housing segments **112** may also be operated as antennas individually or in pairs, as may be useful for different wireless communication modes. In some examples, the first and fourth housing segments **112a**, **112d** may be used to communicate, individually or in parallel, over a same wireless frequency band or bands (e.g., the mid and high wireless frequency bands described with reference to FIG. 19), and the second and third housing segments **112b**, **112c** may be used to communicate, individually or in parallel, over a different set of one or more wireless frequency bands (e.g., the low, mid, and high wireless frequency bands described with reference to FIG. 19). The use of housing segments positioned at diagonally opposite corners of the sidewall **114** as antennas that communicate over the same wireless frequency band(s) provides a relatively maximum spatial separation between the antennas, such that the antennas are less likely to couple to one another.

Optionally, the fifth housing segment **112e** may be connected or disconnected to one of the corner housing segments (e.g., to the first housing segment **112a** or the third

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housing segment **112c**) by a circuit disposed interior to a housing including the sidewall **114**, or the sixth housing segment **112f** may be connected or disconnected to one of the corner housing segments (e.g., to the second housing segment **112b** or the fourth housing segment **112d**) by a circuit disposed interior to the housing including the sidewall **114**. Such switchable connections enable the housing segments **112** defining the sidewall **114** to be tuned to communicate over different wireless frequency bands.

FIG. 2B shows another sidewall configuration **200b** for a device (e.g., the device **100**). The sidewall **202** may include six housing segments **204**. As an example, the housing segments **204** may include a first housing segment **204a** defining at least part (or all) of a first corner **206a** of the sidewall **202**, a second housing segment **204b** defining at least part (or all) of a second corner **206b** of the sidewall **202**, a third housing segment **204c** defining at least part (or all) of a third corner **206c** of the sidewall **202**, a fourth housing segment **204d** defining at least part (or all) of a fourth corner **206d** of the sidewall **202**, a fifth housing segment **204e** defining an edge disposed between the first and third housing segments **204a**, **204c**, and a sixth housing segment **204f** defining an edge disposed between the second and fourth housing segments **204b**, **204d**. The third corner **206c** forms a part of the housing **202** that is diagonally opposite the second corner **206b**, and the fourth corner **206d** forms a part of the housing **202** that is diagonally opposite the first corner **206a**.

Housing segments **204** that terminate in adjacent ends along the sidewall **202** may be structurally coupled to one another by a set of one or more non-conductive housing components **208** (e.g., non-conductive housing components, **208a**, **208b**, **208c**, **208d**, **208e**, and **208f** that partially or completely fill gaps between adjacent ends of housing segments **204** about the sidewall **202**. The sidewall **202** shown in FIG. 2B has six such gaps. At least one non-conductive housing component **208** in the set of non-conductive housing components **208** may define a portion (e.g., a segment) of an exterior surface of the sidewall **202** (and also an exterior surface of the housing that includes the sidewall **202**). In some embodiments, the non-conductive housing components **208** may be variously configured and positioned, or formed of various materials, as described with reference to FIGS. 1A-1C.

The sidewall **202** and housing segments **204** may be formed, structurally coupled, and electrically insulated similarly to the sidewall **114** and housing segments **112** described with reference to FIG. 2A. However, the gap between the third and fourth housing segments **112c**, **112d** in FIG. 2A may be moved to the left along the top of the sidewall **202**, such that the third housing segment **204c** may be substantially confined to the third corner **206c**, and the fourth housing segment **204d** may wrap around the fourth corner **206d** and extend along the top edge of the sidewall **202** (that is, along the top edge of the sidewall **202** given the orientation of the sidewall **202** shown in FIG. 2B). In alternative embodiments (not shown), the third housing segment **204c** may also extend along one or more side edges of the sidewall **202**, or may wrap around less than all of the third corner; or the fourth housing segment **204d** may wrap around less than all of the fourth corner **206d**.

In some embodiments, each of the housing segments **204a**, **204b**, **204c**, **204d** positioned at a corner **206** of the sidewall **202** may be operated as a different antenna, and in some cases the housing segments **204a**, **204b**, **204c**, **204d** may be operated as different antennas simultaneously. The housing segments **204a**, **204b**, **204c**, **204d** may also be

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operated individually or in pairs, as may be useful for different wireless communication modes. In some examples, the first and third housing segments **204a**, **204c** may be used to communicate, individually or in parallel, over a same wireless frequency band or bands (e.g., the mid and high wireless frequency bands described with reference to FIG. 19), and the second and fourth housing segments **204b**, **204c** may be used to communicate, individually or in parallel, over a different set of one or more wireless frequency bands (e.g., the low, mid, and high wireless frequency bands described with reference to FIG. 19).

Optionally, the fifth housing segment **204e** may be connected or disconnected to one of the corner housing segments (e.g., to the first housing segment **204a** or the third housing segment **204c**) by a circuit disposed interior to a housing including the sidewall **202**, or the sixth housing segment **204f** may be connected or disconnected to one of the corner housing segments (e.g., to the second housing segment **204b** or the fourth housing segment **204d**) by a circuit disposed interior to the housing including the sidewall **202**. Such switchable connections enable the housing segments **204** defining the sidewall **202** to be tuned to communicate over different wireless frequency bands.

FIG. 2C shows another sidewall configuration **200c** for a device (e.g., the device **100**). The sidewall **210** includes six housing segments **214**. The sidewall **210** and housing segments **212** may be formed, structurally coupled, and electrically insulated similarly to the sidewall **114** and housing segments **112** described with reference to FIG. 2A. However, the manner in which the sidewall **210** is divided between the housing segments **212** differs, such that three housing segments **212a**, **212b**, **212d** are positioned near the bottom edge of the sidewall **210** (given the orientation of the sidewall **210** shown in FIG. 2C), and a single housing segment **212c** is positioned near a top edge of the sidewall **210**. Alternatively, three housing segments may be positioned near the top edge of the sidewall **210**, and a single housing segment may be positioned near the bottom edge of the sidewall **210**.

The housing segments **212** may include a first housing segment **212a** defining at least part (or all) of a first corner **214a** of the sidewall **210**, a second housing segment **212b** defining at least part (or all) of a second corner **214b** of the sidewall **210**, a third housing segment **212c** defining at least parts (or all) of third and fourth adjacent corners **214c**, **214d** of the sidewall **210** and a first edge of the sidewall **210** disposed between the third and fourth corners **214c**, **214d**, a fourth housing segment **212d** defining at least part of a second edge opposite the first edge, a fifth housing segment **212e** defining an edge disposed between the first and third housing segments **212a**, **212c**, and a sixth housing segment **212f** defining an edge disposed between the second and third housing segments **212b**, **212c**.

The first and second housing segments **212a**, **212b** may be substantially confined to the first and second corners **214a**, **214b** respectively, but in some embodiments (not shown) one or both of the first or second housing segments **212a**, **212b** may extend along one or more side edges of the sidewall **210**. Alternatively, the first or second housing segment **212a**, **212b** may wrap around less than all of a corner of the sidewall **210**.

Housing segments **212** that terminate in adjacent ends along the sidewall **210** may be structurally coupled to one another by a set of one or more non-conductive housing components **214** (e.g., non-conductive housing components, **216a**, **216b**, **216c**, **216d**, **216e**, and **216f**) that partially or completely fill gaps between adjacent ends of housing

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segments **212** about the sidewall **210**. The sidewall **210** shown in FIG. 2C has six such gaps. At least one non-conductive housing component **216** in the set of non-conductive housing components **216** may define a portion (e.g., a segment) of an exterior surface of the sidewall **210** (and also an exterior surface of the housing that includes the sidewall **210**). In some embodiments, the non-conductive housing components **216** may be variously configured and positioned, or formed of various materials, as described with reference to FIGS. 1A-1C.

The sidewall **210** and housing segments **212** may be formed, structurally coupled, and electrically insulated similarly to the sidewall **202** and housing segments **204** described with reference to FIG. 2A.

In some embodiments, each of the first, second, third, and fourth housing segments **212a**, **212b**, **212c**, **212d** may be operated as a different antenna, and in some cases the housing segments **212a**, **212b**, **212c**, **212d** may be operated as different antennas simultaneously. The housing segments **212a**, **212b**, **212c**, **212d** may also be operated individually or in pairs, as may be useful for different wireless communication modes. In some examples, the first and second housing segments **212a**, **212b** may be used to communicate, individually or in parallel, over a same wireless frequency band or bands (e.g., the mid and high wireless frequency bands described with reference to FIG. 19), and the third and fourth housing segments **212c**, **212d** may be used to communicate, individually or in parallel, over a different set of one or more wireless frequency bands (e.g., the low, mid, and high wireless frequency bands described with reference to FIG. 19).

Optionally, the fifth housing segment **212e** may be connected or disconnected to one of the corner housing segments (e.g., to the first housing segment **212a** or the third housing segment **212c**) by a circuit disposed interior to a housing including the sidewall **210**, or the sixth housing segment **212f** may be connected or disconnected to one of the corner housing segments (e.g., to the second housing segment **212b** or the third housing segment **212c**) by a circuit disposed interior to the housing including the sidewall **210**. Such switchable connections enable the housing segments **212** defining the sidewall **210** to be tuned to communicate over different wireless frequency bands.

FIG. 2D shows a sidewall configuration **200d** for a device (e.g., the device **100**). The sidewall **218** includes five housing segments **220**. The housing segments **220** may include a first housing segment **220a** defining at least part (or all) of a first corner **222a** of the sidewall **218**, a second housing segment **220b** defining at least part (or all) of a second corner **222b** of the sidewall **218**, a third housing segment **220c** defining at least parts (or all) of third and fourth adjacent corners **222c**, **222d** of the sidewall **218** and a first edge of the sidewall **218** disposed between the third and fourth corners **222c**, **222d**, a fourth housing segment **220d** defining an edge disposed between the first and third housing segments **220a**, **220c**, and a fifth housing segment **220e** defining an edge disposed between the second and third housing segments **220b**, **220c**.

Housing segments **220** that terminate in adjacent ends along the sidewall **218** may be structurally coupled to one another by a set of one or more non-conductive housing components **224** (e.g., non-conductive housing components, **224a**, **224b**, **224c**, **224d**, and **224e**) that partially or completely fill gaps between adjacent ends of housing segments **220** about the sidewall **218**. The sidewall **218** shown in FIG. 2D has five such gaps. At least one non-conductive housing component **224** in the set of non-conductive housing com-

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ponents **224** may define a portion (e.g., a segment) of an exterior surface of the sidewall **218** (and also an exterior surface of the housing that includes the sidewall **218**). In some embodiments, the non-conductive housing components **224** may be variously configured and positioned, or formed of various materials, as described with reference to FIGS. 1A-1C.

The sidewall **218** and housing segments **220** may be formed, structurally coupled, and electrically insulated similarly to the sidewall **114** and housing segments **112** described with reference to FIG. 2A. However, the third and fourth housing segments **112c**, **112d** shown in FIG. 2A are replaced with a singular housing segment **220c** having a ground connection **226** where the gap between the third and fourth housing segments **112c**, **112d** is shown in FIG. 2A. The ground connection **226** provides separation between left and right portions of the housing segment **220c** and enables the left and right portions to be operated as different antennas (e.g., similarly to the third and fourth segments **112c**, **112d** described with reference to FIG. 2A).

In some embodiments of the sidewall **218**, a portion of the second housing segment **220b** may be removed and filled with a non-conductive material **224f** to provide an apparent symmetry between the lower left and lower right portions of the sidewall **218**.

FIG. 2E shows a sidewall configuration **200e** for a device (e.g., the device **100**). The sidewall **228** includes six housing segments **230**. The housing segments **230** may include a first housing segment **230a** positioned near a first corner **232a** of the sidewall **228**, a second housing segment **230b** positioned near a second corner **232b** of the sidewall **228**, a third housing segment **230c** positioned near a third corner **232c** of the sidewall **228**, a fourth housing segment **230d** positioned near a fourth corner **232d** of the sidewall **228**, a fifth housing segment **230e** defining a first edge of the sidewall **228** and disposed between the first and third corners **232a**, **232c**, and a sixth housing segment **230f** defining a second edge of the sidewall **228** and disposed between the second and fourth corners **232b**, **232d**.

Housing segments **230** that terminate in adjacent ends along the sidewall **228** may be structurally coupled to one another by a set of one or more non-conductive housing components **234** (e.g., non-conductive housing components, **234a**, **234b**, **234c**, **234d**, **234e**, and **2340** that partially or completely fill gaps between adjacent ends of housing segments **230** about the sidewall **228**. The sidewall **228** shown in FIG. 2E has six such gaps. At least one non-conductive housing component **234** in the set of non-conductive housing components **234** may define a portion (e.g., a segment) of an exterior surface of the sidewall **228** (and also an exterior surface of the housing that includes the sidewall **228**). In some embodiments, the non-conductive housing components **234** may be variously configured and positioned, or formed of various materials, as described with reference to FIGS. 1A-1C.

The sidewall **228** and housing segments **230** may be formed, structurally coupled, and electrically insulated similarly to the sidewall **114** and housing segments **112** described with reference to FIG. 2A.

In each of the sidewall configurations **200a-e** described with reference to FIGS. 2A-2E, housing segments **112**, **204**, **212**, **220**, or **230** that are configured to be operated as primary antennas for a device may be positioned at the corners or top and bottom edges of a device's sidewall **114**, **202**, **210**, **218**, or **228**. Such a placement of antennas may be useful in that the antennas are positioned away from the sidewall edges that would normally be gripped by a user of

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a device that includes the sidewall. Housing segments **112**, **204**, **212**, **220**, or **230** that are configured to be operated in pairs, in a same wireless frequency band, may be positioned at opposite corners or opposite extents of the sidewalls **114**, **202**, **210**, **218**, or **228**.

FIGS. 3A-3C show example implementations of the first, second, third, and fourth housing segments **112a**, **112b**, **112c**, and **112d** described with reference to FIGS. 1A-1C & 2A, and show example positions of the housing segments **112** with respect to a support plate **110**. The housing segments **112** and support plate **110** may be examples of the housing segments and support plate described with reference to FIGS. 1A-1C & 2A.

Each of the housing segments **112** shown defines a rounded corner **108** of a housing sidewall **114**. In alternative embodiments, the corners **108** may be squared corners, tapered corners of an octagon, or corners have other rounded or tapered shapes.

As shown primarily with reference to FIG. 3A, the first, second, third, and fourth housing segments **112a**, **112b**, **112c**, **112d** may not overlap the support plate **110**, and may be electrically insulated from the support plate **110**. A set of one or more non-conductive housing components (not shown in FIG. 3A but shown in FIG. 4) may form a structural bridge or bridges between the support plate **110** and the housing segments **112**, and in some cases may encapsulate portions of the support plate **110**. As examples, the non-conductive housing component(s) may adhere to the support plate **110** or be adhesively bonded to the support plate **110**. In some cases, the housing segments **112** may have interlock features that extend inward from the ends of the housing segments **112**, toward the support plate **110** or an interior volume defined at least in part by the housing segments **112**, as shown in later figures (e.g., FIGS. 5A-10C). The non-conductive housing component(s) may extend into, through, or around such interlock features, such that the non-conductive housing component(s) may better hold, grab or retain the housing segments **112**. Having a separation between the housing segments **112** and the support plate **110** can allow the housing segments **112** to resonate more freely when operated as antennas. The support plate **110** may be separated from some housing segments **112** (or some portions of housing segments **112**) more than from other housing segments **112** (or other portions of housing segments **112**). In alternative embodiments, the support plate **110** may extend under one or more of the housing segments **112** but be electrically insulated from the housing segments **112**, or the support plate **110** may be grounded to one or more of the housing segments **112** at selected points (e.g., at a ground connection, such as the ground connection described **226** with reference to FIG. 2D).

In some embodiments, the housing segments **112e**, **112f** disposed along the long sides of the device **100** described with reference to FIGS. 1A-1C may be conductive, and may be welded or otherwise structurally and electrically coupled to the left and right sides **306a**, **306b** (e.g., the long sides) of the support plate **110**. In other embodiments, the housing segments **112e**, **112f** may be conductive, and may be integrally formed as extensions of the support plate **110** (e.g., in a pan configuration). In still other embodiments, the housing segments **112e**, **112f** may be non-conductive, and may be formed as extensions of the non-conductive housing component that structurally couples the first, second, third, and fourth housing segments **112a**, **112b**, **112c**, **112d**. In the latter embodiments, the support plate **110** may also be non-

conductive, and may be part of a monolithic component that structurally couples the housing segments 112.

As also shown in FIG. 3A, the support plate 110 may define portions (e.g., antenna portions) or entirety of one or more slot antenna features 302 (e.g., slot antenna features 302a, 302b, 302c, and 302d). By way of example, a slot antenna feature 302 is shown near each of the four main corners 304 (e.g., corners 304a, 304b, 304c, and 304d) of the support plate 110. A fifth housing segment may be coupled to the left side 306a of the support plate 110 (i.e., the left edge as shown in FIG. 3A) and define further portions (e.g., further antenna portions) of the slot antenna feature 302a or 302c. Similarly, a sixth housing segment may be coupled to the right side 306b of the support plate 110 and define further portions (e.g., further antenna portions) of the right-side slot antenna feature 302b or 302d. In some embodiments, the fifth housing segment may be electrically connected to, or disconnected from, the first or third housing segment 112a or 112c, thereby adding the left-side slot antenna feature 302a or 302c to the first or third housing segment 112a or 112c and enabling an antenna including the first or third housing segment 112a or 112c to resonate in a different wireless frequency band. In some embodiments, the sixth housing segment may be electrically connected to, or disconnected from, the second or fourth housing segment 112b or 112d, thereby adding the right-side slot antenna feature 302b or 302d to the second or fourth housing segment 112b or 112d and enabling an antenna including the second or fourth housing segment 112b or 112d to resonate in a different wireless frequency band, such as the B42 wireless frequency band. In some cases the fifth and sixth housing segments may be welded (e.g., spot welded or laser welded) to the support plate 110 along the left and right sides 306a, 306b of the support plate 110.

In some embodiments, the support plate 110 or housing segments 112 may additionally or alternatively define all or portions of other antenna tuning features,

FIG. 3A further shows potential locations of antennas 324, 326 that may be housed within the interior volume defined by the housing segments 112. In some embodiments, the interior antennas 324, 326 may be positioned at or near the corners defined by the first and third housing segments 112a, 112c. In other embodiments, the interior antennas 324, 326 may be positioned elsewhere. In some embodiments, the interior antennas 324, 326 may be used in combination with antennas incorporating the second and fourth housing segments 112b, 112d to operate in the B42 wireless frequency band. The interior antennas 324, 326 may be positioned near the left side of the sidewall 114 to provide good separation (and decoupling) from the antennas incorporating the second and fourth housing segments 112b, 112d.

In some embodiments, different portions 324a, 324b of the interior antenna 324 may be operated as different antennas to facilitate wireless communication in one or more wireless communication modes, in one or more wireless frequency bands.

FIGS. 3B & 3C show example locations of feed and ground connectors to the housing segments 112a and 112b, which feed and ground connectors enable the housing segments 112a, 112b to be operated as antennas. The locations of feed and ground connectors described with reference to FIGS. 3B & 3C can be replicated for the third and fourth housing segments 112c, 112d, or feed and ground connectors for the third and fourth housing segments 112c, 112d may be located in alternate locations.

As shown in FIG. 3B, feed and ground connectors 308, 310 for the first housing segment 112a may be located

interior to the sidewall 114, on opposite sides of the first corner 108a. In the embodiment shown, the feed connector 308 may be located more toward the apex of the first corner 108a than the leftmost end 312a of the first housing segment 112a. The ground connector 310 may be located more toward the rightmost end 312b of the first housing segment 112a. Alternatively, the locations of the feed connector 308 and ground connector 310 may be swapped, with the feed connector 308 being located more toward the rightmost end 312b of the first housing segment 112a. The arrangement shown in FIG. 3B may be advantageous in that the ground connector 310, along with the gap between adjacent ends 312b, 314a of the first and second housing segments 112a, 112b, helps define a demarcation point between the antennas provided by the first and second housing segments 112a, 112b. The arrangement shown also enables a switchable coupling of the fifth housing segment and slot antenna feature 302a to the first housing segment 112a, to extend the length of the resonate portion of an antenna including the first housing segment 112a.

As also shown in FIG. 3B, feed and ground connectors 316, 318 for the second housing segment 112b may be located interior to the sidewall 114 near the second corner 108b. In the embodiment shown, the ground connector 318 may be located closer to the rightmost end 314b of the second housing segment 112b than the feed connector 316. Alternatively, the feed and ground connectors 316, 318 may be swapped, with the feed connector 316 being located more toward the rightmost end 314b of the second housing segment 112b. The arrangement shown in FIG. 3B enables a switchable coupling of the sixth housing segment and slot antenna feature 302b to the second housing segment 112b, to extend the length of the resonate portion of an antenna including the second housing segment 112b.

As shown, the resonate portion of the first housing segment 112a may resonate within frequencies of the mid and high bands described with reference to FIG. 19. The second housing segment 112b may have two resonate portions, with the leftmost resonate portion resonating within frequencies of the low and high wireless frequency bands described with reference to FIG. 19, and the rightmost resonant portion resonating within frequencies of the mid wireless frequency band described with reference to FIG. 19. The lengths of the arrows extending from the housing segments 112a, 112b represent the relative voltages along the housing segments 112a, 112b. Longer length arrows indicate increased voltage and areas of better antenna efficiency in various wireless frequency bands. As shown, the portions of the first and second housing segments 112a, 112b with highest efficiency are at various ends of the housing segments 112a, 112b. To achieve maximum possible efficiencies, it is therefore desirable to electrically insulate these ends (e.g., ends 312a, 312b, 314a, and 314b) from surrounding conductors, and to decouple (e.g., lower the capacitance of) these ends 312a, 312b, 314a, 314b with respect to surrounding conductors.

The antenna configuration described with reference to FIG. 3B provides good separation between the portions of the first and second housing segments 112a, 112b that resonate within the mid wireless frequency band. FIG. 3C shows alternative locations of feed and ground connectors 320, 322 for the second housing segment 112b. The alternative feed and ground connectors 320, 322 are located approximately in the middle of the lower edge of the sidewall 114 (and in some cases, somewhat closer to the first corner 108a, as shown). The feed connector 320 may be located closer to the second corner 108b, and the ground connector 322 may be located closer to the first corner 108a.

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This alternative configuration for feed and ground connectors **320**, **322** of the second housing segment **112b** may provide good or better low and high wireless frequency band efficiency, but may increase the likelihood of coupling between the portions of the housing segments **112a**, **112b** that resonate in the mid wireless frequency band, absent good electrical insulation between the adjacent ends **312b**, **314a** of the first and second housing segments **112a**, **112b**. In some embodiments, wireless communication circuitry may switchably connect the second housing segment **112b** to the feed and ground connectors **316**, **318** described with reference to FIG. 3B, or to the feed and ground connectors **320**, **322** described with reference to FIG. 3C. One or the other set of connectors may be used as necessary in response to particular trigger conditions to improve antenna efficiency or other parameters.

FIG. 4 shows each of the first, second, third, fourth, fifth, and sixth housing segments **112a**, **112b**, **112c**, **112d**, **112e**, and **112f** described with reference to FIGS. 1A-1C, 2A, & 3A-3C in relation to the support plate **110** described with reference to FIGS. 1C & 3A-3C. FIG. 4 also shows an example non-conductive housing component including a portion **400** that structurally couples the housing segments **112** to each other and/or the support plate **110**. By way of example, the non-conductive housing component may include a first portion **400** (e.g., a fiber-filled polymer material) that structurally couples the housing segments **112** to each other and the support plate **110**, and second portions **116** (e.g., a polymer material without fiber fill), that fills the outer portions of gaps between the housing segments **112** and forms portions of a smooth exterior surface of the sidewall **114**. In some embodiments, the first portion **400** may at least partially encapsulate portions of the support plate **110**. The second portions **116** of the non-conductive housing component may be color-matched (or not color-matched) to the exterior surfaces of the housing segments **112**. Alternatively, the non-conductive housing component may include a single portion that both structurally couples the housing segments **112** to each other and the support plate **110**, and forms portions of the exterior surface of the sidewall **114**.

FIG. 4 also shows a camera brace **402** that is structurally coupled to the upper right corner of the support plate **110**. When the support plate **110** and camera brace **402** are metallic, the camera brace **402** may be welded to the support plate **110** for strength and to provide an electrical coupling between the support plate **110** and camera brace **402**. The electrical coupling may enable the support plate **110** and camera brace **402** to be coupled to a common ground, which may improve the performance of the housing segments **112** when the housing segments **112** are operated as antennas. The camera brace **402** may provide a housing for one or more camera modules, such as one or more rear-facing camera modules (i.e., cameras with a field of view extending from the rear or non-display side of a device).

Turning now to FIGS. 5A-10C, there are shown several interior views of example implementations of the housing segments **112** described with reference to FIGS. 1A-1C, 2A, 3A-3C, & 4. The various views show example details of interlock features defined by the housing segments **112**. In particular, the various views show example details of the interlock features that extend from adjacent ends of respective adjacent housing segments **112** around the sidewall **114** (e.g., details of the interlock features that extend from adjacent ends of respective adjacent conductive housing segments, which adjacent conductive housing segments are separated by a non-conductive housing component). As

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shown in the figures, the interlock features may extend into an interior volume of a device. The "A" views in FIGS. 5A-10C provide isometric views of the interlock features that extend from adjacent ends of adjacent housing segments **112**. The "B" and "C" views illustrate respective cross-sections of the two different interlock features shown in the corresponding "A" view, with various holes in the interlock features filled by a non-conductive housing component that structurally couples the two adjacent housing segments **112** shown in the "A" view. The various housing segments **112**, interlock features, sub-features thereof, and techniques for forming the housing segments **112**, interlock features, and sub-features thereof, as described in FIGS. 5A-10C, may be applied to join various of the housing segments described with reference to FIGS. 1A-4, as will be understood by a person of ordinary skill in the art after reading this disclosure.

FIGS. 5A-5C show examples of interlock features **500**, **502** that extend into an interior volume **544** defined at least partly by the sidewall **114**. The interlock features **500**, **502** may extend inward, into the interior volume **544**, from adjacent ends of the fifth and third housing segments **112e**, **112c** described with reference to FIGS. 1A-1C, 2A, 3A-3C, & 4. A portion of the fifth housing segment **112e** is shown on the left, and a portion of the third housing segment **112c** is shown on the right. As previously described, the fifth and third housing segments **112e**, **112c** may be separated by a gap along the sidewall **114** that is filled by a non-conductive housing component. In some embodiments, each of the gaps defined between adjacent ends of adjacent housing segments **112** about the sidewall **114** may have the same dimensions. In other embodiments, the gaps may have different dimensions. To provide good structural rigidity, the gaps may have a relatively small width about the sidewall **114**. However, when the housing segments **112** are configured to be operated as antennas, and where possible, it may be desirable to increase the widths of the gaps interior to an exterior surface of the sidewall **114**. The increased separation may decrease the capacitance between adjacent housing segments **112** and reduce the likelihood that the adjacent housing segments **112** couple to one another and interfere with the resonance of, or decrease the efficiency of, the housing segments **112** (e.g., when the housing segments **112** are operated as antennas).

A first interlock feature **500** may include a first protrusion **540** that extends inward from an end of the fifth housing segment **112e**, into the interior volume **544**. A second interlock feature **502** may include a second protrusion **542** that extends inward from an end of the third housing segment **112b**, into the interior volume **544**. The first interlock feature **500** feature may be located near a button that protrudes through a cavity **504** in the fifth housing segment **112e** and is operable from the exterior of a device including the fifth and third housing segments **112e**, **112c**. Because of the button, the first interlock feature **500** may be somewhat thinner than the second interlock feature **502**.

As shown, the interlock features **500**, **502** and their protrusions **540**, **542** may be integrally defined by (e.g., molded with or machined into) the fifth and third housing segments **112e**, **112c** respectively. Alternatively, the interlock features **500**, **502** and their protrusions **540**, **542** may be structurally coupled to the fifth and third housing segments **112e**, **112c** in other ways, such as by welds or fasteners. Each of the interlock features **500**, **502** and protrusions **540**, **542** may be set slightly back from a gap along the sidewall (i.e., an exterior or sidewall gap **506**) to form an interior gap **508** between the housing segments **112e**, **112c** that has a greater width than the exterior gap **506**. For example, a first end

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surface 546 of the fifth housing segment 112e may be positioned opposite a second end surface 548 of the third housing segment 112c to define the exterior gap 506. The first interlock feature 500 may have a first interlock surface 550 positioned opposite a second interlock surface 552 of the second interlock feature 502 to define the interior gap 508. The first interlock surface 550 may be defined by the first protrusion 540, and the second interlock surface 552 may be defined by the second protrusion 542. The exterior gap 506 may have a first gap width, and the interior gap 508 may have a second gap width. In some embodiments, and as shown in FIG. 5A, the second gap width may be greater than the first gap width. By way of example, the entirety of the second interlock feature 502 may be set back from the exterior gap 506 by an offset 510b, whereas an outermost portion of the first interlock feature 500 may be set back from the exterior gap 506 by an offset 510a, and an innermost portion of the first interlock feature 500 may widen into the interior gap 508 (e.g., part of the first interlock feature 500 may overlap the offset 510a). This widening can increase the structural rigidity of the first interlock feature 500 given its overall narrow width, and/or can provide material for forming a boss protrusion 512 (e.g., a screw boss).

A number of through holes or blind holes (also referred to as openings or recesses at times herein) may be formed in each of the interlock features 500, 502 to enable a non-conductive housing component to extend into, through, or around the interlock features 500, 502 (thereby increasing the strength of the structural coupling between the fifth housing segment 112e, the non-conductive housing component, and the third housing segment 112c). With reference to FIGS. 5A & 5B, a first hole 514 (e.g., a blind hole) may extend into the first interlock surface 550 of the first interlock feature 500. In some embodiments, the first hole 514 may be end-milled or drilled into the first interlock surface 550. In some embodiments, the first hole 514 may be a blind hole so that a metal wall extends from the exterior surface of the sidewall 114 to the innermost point of the first interlock feature 500 (i.e., to the innermost point of the first interlock feature 500 with respect to the interior volume 544). Making the first hole 514 a blind hole can maintain separation between the first hole 514 and the button cavity 504, which in some cases may be open to the exterior of a device. This can enable the button cavity 504 to be sealed separately from the first hole 514 and tends to decrease the likelihood that moisture or contaminants will enter into the gap between the fifth and third housing segments 112e, 112c via the cavity 504.

In some embodiments, the first hole 514 may have a portion with a shape or contour that corresponds to a shape or contour of the exterior surface of the sidewall 114, thereby defining a wall or wall portion between the first hole 514 and the exterior surface of the sidewall 114 that has a substantially uniform thickness (see FIG. 5C). This can enable more non-conductive material to extend into the first interlock feature 500 while maintaining the structural rigidity of the sidewall 114. In some embodiments, all portions of the wall surrounding the first hole 514 may have a substantially uniform thickness. When the exterior surface of the sidewall 114 has a curved contour, the first hole 514 may in some cases be kidney-shaped.

The first hole 514 may be end-milled so that it has a smooth wall. This can be important when a device is dropped, in that a drop of a device can cause sharp features to function as knives or chisels that tend to crack non-conductive housing components such as the non-conductive

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housing component that extends into the first interlock feature 500. Interlock features that are thinner (such as the first interlock feature 500) may be more susceptible to such cracking. In alternative embodiments, the first hole 514 may be drilled, or may not be kidney-shaped, or may be a through hole.

With further reference to FIG. 5A, a second hole 554 may extend into an upper surface 530 of the first interlock feature 500 or first protrusion 540. In some embodiments, the second hole 554 may be end-milled or drilled into the upper surface 530. The second hole 554 may be transverse to the first hole 514 and intersect the first hole 514 (i.e., the second hole 554 may be a transverse hole with respect to the first hole 514). The non-conductive housing component that structurally couples the fifth and third housing segments 112e, 112c may be insert molded into the first and second holes 514, 554, and in some cases may enter one of the holes and exit the other of the holes, thereby extending into the first interlock feature 500 in at least two orthogonal directions.

A boss protrusion 512 (e.g., a screw boss) for mounting a button assembly to the fifth housing segment 112e may be integrated with (e.g., molded or machined into) the first interlock feature 500 or first protrusion 540. However, the boss protrusion 512 may require extending the extent of the first interlock feature 500 into the gap 508 between the fifth and third housing segments 112e, 112c (e.g., the boss protrusion 512 may be offset from the first interlock surface 550 toward the first end surface 546). A threaded hole 516 may be tapped into or defined by the boss protrusion 512 as a blind hole, thereby increasing the structural rigidity of the first interlock feature 500 and boss protrusion 512 and enabling a separation of moisture sealing issues pertaining to the threaded hole 516 and other holes formed in the first interlock feature 500. Alternatively, the threaded hole 516 may be a through hole.

In addition to the first and second holes 514, 554, additional holes 518, 520, 558, 560 may be formed in the first interlock feature 500 or first protrusion 540, as shown in FIGS. 5A & 5B. The additional holes 518, 520, 558, 560 may provide additional surface area for the non-conductive housing component to hold or grab onto, retain, or conform to, thereby improving the strength of the structural coupling between the fifth and third housing segments 112e, 112c. In some embodiments, the additional holes 518, 520, 558, 560 may be drilled. In some embodiments, some of the additional holes 518, 520, 558, 560 may be blind holes and/or intersect. For example, the holes 518 and 558 may intersect, and the holes 520 and 560 may intersect. The intersecting holes may provide paths through which the material(s) of the non-conductive housing component may be molded. Forming the holes 518 and 520 as blind holes again enables a separation of moisture sealing issues, and may increase the structural rigidity of the first interlock feature 500. In alternative embodiments, one or more of the additional holes 518, 520, 558, 560 may be a through hole.

With reference to FIGS. 5A & 5C, a first hole 522 (e.g., a round through hole) may extend into the second interlock surface 552 of the second interlock feature 502. In some embodiments, the first hole 522 may be drilled or otherwise cut into the second interlock feature 502. The first hole 522 may be opposite to or substantially aligned with the first hole 514 in the first interlock feature 500 or protrusion 540. As defined herein substantially aligned holes or components are disposed along a common axis and may be fully aligned or partly aligned. In some embodiments, partly aligned holes or components may have cross-sections that overlap by at least

25%, or 50%, or even 75%. A second hole **524** (e.g., a round hole) may extend into an upper surface **532** of the second interlock feature **502** or second protrusion **542**, and may be drilled into or otherwise cut into the upper surface **532**. The second hole **524** may be transverse to the first hole **522** (e.g., the second hole **524** may be a first transverse hole that intersects the first hole **522** (e.g., perpendicularly)). A third hole **526** (e.g., a round hole) may extend into a lower surface **564** of the second interlock feature **502** or second protrusion **542** (see FIG. 5C), and may be drilled or otherwise cut into the lower surface **564**. The third hole **526** may also be transverse to the first hole **522** (e.g., the third hole **526** may be a second transverse hole that intersects the first hole **522** (e.g., perpendicularly)). The second and third holes **524**, **526** may have the same diameter or different diameters, and in some cases may be formed as a single through hole. In contrast to the first interlock feature **500**, which extends into the interior gap **508**, the second interlock feature **502** may not extend past the second interlock surface **552**.

As shown in FIGS. 5B & 5C, a non-conductive housing component **528** may at least partially fill the first, second, and other holes formed in the first and second interlock features **500**, **502**, thereby structurally coupling the fifth housing segment **112e** and the third housing segment **112c**. In some cases, the non-conductive component **528** may at least partially fill each of the first, second, and additional holes **514**, **554**, **518**, **520**, **558**, **560** in the first interlock feature **500**, and the first, second, and third holes **522**, **524**, **526** in the second interlock feature **502**. The non-conductive housing component **528** may not extend over the upper surface **530** of the first interlock feature **500**, but may extend over the upper surface **532** of the second interlock feature **502**. Surrounding as many surfaces of the interlock features **500**, **502** as possible with the material(s) of the non-conductive housing component **528** may tend to increase the strength of the structural coupling between the interlock features **500**, **502** and the non-conductive housing component **528**. In some embodiments, a shelf or shelves may be cut into the upper surfaces of the fifth and third housing segments **112e**, **112c** or first and second interlock features **500**, **502**. For example, a shelf **556** may be cut into the upper surface **532** of the second interlock feature. The shelves, including the shelf **556** may serve various purposes. For example, the shelves may reduce the capacitive coupling between a housing component **112** and another component, or the shelves may increase the surface area over which the non-conductive component **528** may adhere to a housing segment **112**. In some embodiments, holes may be formed in an upper surface or shelf of a housing segment **112** to enable the non-conductive component **528** to extend into and through portions of a housing segment **112** away from an interlock feature. For example, holes **562**, **566** may be cut into the shelf **556**.

In some embodiments, a front cover (e.g., the front cover **106a** described with reference to FIGS. 1A-1C) may be bonded to upper surfaces of the first and second interlock features **500**, **502** or housing segments **112e**, **112c**, or upper surfaces of the non-conductive housing component **528** (e.g., as shown in FIG. 5C, where the non-conductive housing component **528** extends over the upper surface **532** of the second interlock feature **502**). In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. 1A-1C) may be bonded to lower surfaces of the first and second interlock features **500**, **502** or housing segments **112e**, **112c** by an adhesive **534** (see FIGS. 5B & 5C). A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112e**, **112c** and

extending along the sidewall **114**. The seal **536** and adhesive **534** may help prevent moisture from entering a device between the housing segments **112e**, **112c** and rear cover **106b**.

In some embodiments, the non-conductive housing component **528** may be formed by a polymer material having a fiber fill, and the polymer material may at least partially fill various holes in the first and second interlock features in addition to forming an exterior surface of the sidewall **114** (e.g., a portion of the sidewall **114** that bridges or fills the exterior gap **506**). In other embodiments, the non-conductive housing component **528** may include a first portion formed from a first polymer material and a second portion formed from a second polymer material. The first polymer material may have a fiber fill and at least partially fill various holes in the first and second interlock features. The second polymer material may be different from the first polymer material and form an exterior surface of the sidewall **114** (e.g., a portion of the sidewall **114** that bridges or fills the exterior gap **506**). Each polymer having a fiber fill may have a fiber fill including glass or other types of fibers. In some embodiments, the second polymer material may also have a fiber fill, but have a fiber fill that differs from the fiber fill of the first polymer material.

The structures of the first and second interlock features **500**, **502**, and more generally all of the interlock features described herein, may be configured to reduce strain on the housing segments **112** and non-conductive housing component in bending modes induced by a device drop.

FIGS. 6A-6C show examples of the interlock features **600**, **602** that extend into an interior volume **634** defined at least partly by the sidewall **114**. The interlock features **600**, **602** may extend inward, into the interior volume **634**, from adjacent ends of the fourth and sixth housing segments **112d**, **112f** described with reference to FIGS. 1A-1C, 2A, 3A-3C, & 4. A portion of the fourth housing segment **112d** is shown on the left, and a portion of the sixth housing segment **112f** is shown on the right. As previously described, the fourth and sixth housing segments **112d**, **112f** may be separated by a gap along the sidewall **114** that is filled by a non-conductive housing component. The non-conductive housing component may be part of the same non-conductive housing component that structurally couples the housing segments **112** described with reference to FIGS. 5A-5C, or a different non-conductive housing component.

A first interlock feature **600** may include a first protrusion **636** that extends inward from an end of the fourth housing segment **112d**, into the interior volume **634**. A second interlock feature **602** may include a second protrusion **638** that extends inward from an end of the sixth housing segment **112f**, into the interior volume **634**. Each of the first and second interlock features **600**, **602** may be located between the camera brace **402** described with reference to FIG. 4 and the sidewall **114** formed by the fourth and sixth housing segments **112d**, **112f**. Because of the camera brace **402** (or other components), the first and second interlock features **600**, **602** may be somewhat shallower than other interlock features (i.e., they may extend inward from the sidewall **114** to a lesser extent).

As shown, the interlock features **600**, **602** and their protrusions **636**, **638** may be integrally defined by (e.g., molded with or machined into) the fourth and sixth housing segments **112d**, **112f** respectively. Alternatively, the interlock features **600**, **602** and their protrusions **636**, **638** may be structurally coupled to the fourth and sixth housing segments **112d**, **112f** in other ways, such as by welds or fasteners. Each of the interlock features **600**, **602** and

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protrusions **636**, **638** may be set slightly back from a gap along the sidewall **114** (i.e., an exterior or sidewall gap **606**) to form an interior gap **608** between the housing segments **112d**, **112f** that has a greater width than the exterior gap **606**. For example, a first end surface **640** of the fourth housing segment **112d** may be positioned opposite a second end surface **642** of the sixth housing segment **112f** to define the exterior gap **606**. The first interlock feature **600** may have a first interlock surface **644** positioned opposite a second interlock surface **646** of the second interlock feature **602** to define the interior gap **608**. The first interlock surface **644** may be defined by the first protrusion **636**, and the second interlock surface **646** may be defined by the second protrusion **638**. The exterior gap **606** may have a first gap width, and the interior gap **608** may have a second gap width. In some embodiments, and as shown in FIG. 6A, the second gap width may be greater than the first gap width. By way of example, the first interlock feature **600** may be set back from the exterior gap **606** by a first offset, and the second interlock feature **602** may be set back from the exterior gap **606** by a second offset.

A number of holes may be formed in each of the interlock features **600**, **602** to enable a non-conductive housing component to extend into, through, or around the interlock features **600**, **602** (thereby increasing the strength of the structural coupling between the fourth housing segment **112d**, the non-conductive housing component, and the sixth housing segment **112f**). With reference to FIGS. 6A & 6B, a first hole **610** (e.g., a blind hole) may extend into the first interlock surface **644** of the first interlock feature **600**. In some embodiments, the first hole **610** may be formed by a set of multiple partially overlapping drill holes. Drilling multiple partially overlapping holes may save cost over forming the first hole **610** using an end mill. In some embodiments, the partially overlapping holes may be drilled close enough to reduce the ridges where holes overlap and thereby reduce the tendency of the ridges to operate as knives or chisels when a device is dropped. In alternative embodiments, the first hole **610** may be formed by an end mill or other means. The first hole **610** may have an out-of-round shape that enables it to have a greater diameter perpendicular to the front (display) and rear surfaces of a device while having a smaller diameter extending transverse to the sidewall **114**. The importance of a smooth wall, as discussed with reference to the first hole **514** shown in FIGS. 5A & 5B, may be less important given the increased width of the first interlock feature **600** shown in FIGS. 6A & 6B.

Additional holes **612**, **614** may also be formed in the first interlock feature **600** or protrusion **636**. The additional holes **612**, **614** may be transverse to the first hole **610**, and in some cases, first, second, and third intersecting holes **610**, **612**, and **614** may be oriented along respective x, y, and z axes. This may enable a non-conductive housing component to extend into and through the first interlock feature **600** along three axes, which can strengthen the structural coupling between the non-conductive housing component and the first interlock feature **600**. The second hole **612** may extend into an interior surface **648** of the first interlock feature **600** or protrusion **636**, and may be a blind hole given that it extends toward the sidewall **114**. The third hole **614** may extend into an upper surface **650** of the first interlock feature **600** or protrusion **636**, and may be a through hole extending perpendicular to the front and rear surfaces of a device. As shown, the first interlock surface **644**, interior surface **648**, and upper surface **650** may be orthogonal surfaces.

A boss protrusion **616** (e.g., a screw boss), which in some cases may serve as an antenna feed connector to which a flex

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circuit may be coupled, may be machined into or structurally coupled to the fourth antenna segment **112d** adjacent the first interlock feature **600**. The boss protrusion **616** may be tapped with or otherwise define a threaded hole **618** that extends perpendicular to the front and rear surfaces of a device. In some embodiments, the boss protrusion **616**, or boss protrusion **616** in combination with the threaded hole **618**, may be formed using a hole cutter (e.g., a computer numerical control (CNC) hole cutter). The hole cutter may form the boss protrusion **616**, as well as a trough, lip, or ledge **620** surrounding and recessed from an upper surface of the boss protrusion **616**. The ledge **620** may enable non-conductive material(s) such as those of the non-conductive housing component that is used to structurally couple the first and second interlock features **600**, **602**, and/or a polyurethane applied as a seal between plastic and metal portions of the housing that includes the first and second interlock features **600**, **602** and non-conductive housing component, to be applied around the boss protrusion **616** without extending over an upper surface of the boss protrusion **616**. Stated differently, the ledge **620** enables non-conductive material(s) to be disposed around the boss protrusion **616** without compromising good electrical contact between a flex circuit or other element and the boss protrusion **616**. In some cases, a non-conductive material may be applied around the boss protrusion **616**, and then an upper portion of the boss protrusion **616** may be removed, or the upper surface of the boss protrusion **616** may be planed, to ensure that the upper surface of the boss protrusion **616** is free of any non-conductive material.

With reference to FIGS. 6A & 6C, a first hole **622** may extend into the second interlock surface **646** of the second interlock feature **602**. The first hole **622** may be drilled or otherwise cut into the second interlock feature **602**. The first hole **622** may be a conical hole (e.g., a hole formed using a conical drill bit). A conical drill bit enables the size of the hole **622** to have a larger diameter toward the second interlock surface **646**, but a smaller diameter where the hole extends behind a boss protrusion **624**. Similarly to how the first hole **610** in the first interlock feature **600** is formed, the first hole **622** in the second interlock feature **602** may be formed by drilling multiple partially overlapping holes with the conical drill bit. Alternatively, the first hole **622** may be formed by drilling a single conical hole, or multiple holes using a uniform diameter drill bit, or multiple holes using different drill bits (e.g., a conical drill bit in combination with a uniform diameter drill bit, or a set of drill bits having different uniform diameters). In some embodiments, the partially overlapping holes may be drilled close enough to reduce the ridges where holes overlap, thereby reducing the tendency of the ridges to operate as knives or chisels when a device is dropped. The first hole **622** may be a through hole, as shown, or a blind hole. A second hole **626** (e.g., a round hole) may extend into an inner surface **652** of the second interlock feature **602** or protrusion **638**. The second hole **626** may be drilled or otherwise cut into the second interlock feature **602**, and may be transverse to the first hole **622** (e.g., the second hole **626** may be a transverse hole that intersects the first hole **622** (e.g., perpendicularly)).

A boss protrusion **624**, which in some cases may serve as a flex circuit connection point, may be machined in or structurally coupled to the sixth antenna segment **112f**, and may be integrated with the second interlock feature **602** or protrusion **638**. The boss protrusion **624** may have a surface that is angled (e.g., at a 30 degree angle, $\pm 10\%$) with respect to an imaginary line extending perpendicular to the front and rear surfaces of a device. In alternative embodiments, the

surface may be angled between 25 and 35 degrees, between 20 and 40 degrees, or between 0 and 90 degrees. The boss protrusion 624 may be angled so that a screw may be threaded into a threaded hole 628 within the boss protrusion 624 after the sixth housing segment 112f has been attached to a support plate 220 adjacent a camera brace 402, or so that the threaded hole 628 may be tapped after the sixth housing segment 112f is structurally coupled to the support plate 110 adjacent the camera brace 402. Examples of the support plate 110 and camera brace 402 are described with reference to FIG. 4. The boss protrusion 624 may also be angled to reduce the inward extension of the second interlock feature 602 or protrusion 638 from the sidewall 114, while also maintaining an upper surface 630 (of the second interlock feature 602) to which a device cover or other element may be mated or sealed. The angle of the angled surface of the boss protrusion 624 may be selected to balance inward extension of the second interlock feature 602 or protrusion 638 (or inward extension of the boss protrusion 624) and the area of the sealing surface 630. The boss protrusion 624 may be formed similarly to the boss protrusion 616 formed in the first interlock feature 600, but in some cases may have or define a threaded hole 628 with a shallower thread depth, so that the threaded hole 628 does not intersect the first hole 622 formed in the second interlock feature 602.

The second interlock feature 602 may form part of (or abut) an antenna tuning feature (e.g., a slot antenna feature 632 formed between the sixth housing segment 112f and a support plate to which the fourth and sixth housing segments 112d, 112f are structurally coupled). In some embodiments, the antenna tuning feature may be defined by a variable thickness of the sixth housing segment 112f along the sidewall 114. For example, the slot antenna feature 632 may extend from the second interlock feature 602 along the sixth housing segment 112f, and may be defined by a thinned portion of the sixth housing segment 112f. The wider portions of the sixth housing segment 112f, adjacent each end of the thinned portion, may taper into the thinned portion (e.g., with an arc or other profile), as shown, or there may be an abrupt transition (e.g., a step) from each wider portion to the thinned portion.

As shown in FIGS. 6B & 6C, a non-conductive housing component 528 may at least partially fill or extend into the first and second interlock features 600, 602. Surrounding as many surfaces of the interlock features 600, 602 as possible with the non-conductive material(s) of a non-conductive housing component 528 may tend to increase the strength of the structural coupling between the interlock features 600, 602 and the non-conductive housing component 528. In some embodiments, the non-conductive housing component 528 may extend into holes 610, 612, 614, 622, and 626.

In some embodiments, a shelf or shelves may be cut into the upper surfaces of the fourth and sixth housing segments 112d, 112f or first and second interlock features 600, 602. For example, a shelf 654 may be cut into the upper surface 650 of the first interlock feature 600. In some embodiments, holes may be formed in an upper surface or shelf of a housing segment 112 to enable the non-conductive component 528 to extend into and through portions of a housing segment 112 away from an interlock feature. For example, hole 656 may be cut into the shelf 654.

In some embodiments, a front cover (e.g., the front cover 106a described with reference to FIGS. 1A-1C) may be bonded to upper surfaces of the first and second interlock features 600, 602 or housing segments 112d, 112f, or upper surfaces of the non-conductive housing component 528 (e.g., as shown in FIGS. 6B & 6C, where the non-conductive

housing component 528 extends over the upper surfaces of the first and second interlock features 600, 602). In some embodiments, a rear cover (e.g., the rear cover 106b described with reference to FIGS. 1A-1C) may be bonded to lower surfaces of the first and second interlock features 600, 602 by an adhesive 534. A seal 536 may be inserted into a groove 538 formed in a lower surface of each housing segment 112d, 112f and extending parallel to the sidewall 114. The seal 536 and adhesive 534 may help prevent moisture from entering a device between the housing segments 112d, 112f and rear cover 106b.

FIGS. 7A-7C show examples of the interlock features 700, 702 that extend into an interior volume 732 defined at least partly by the sidewall 114. The interlock features 700, 702 may extend inward, into the interior volume 732, from adjacent ends of the first and fifth housing segments 112a, 112e described with reference to FIGS. 1A-1C, 2A, 3A-3C, & 4. A portion of the first housing segment 112a is shown on the left, and a portion of the fifth housing segment 112e is shown on the right. As previously described, the first and fifth housing 112a, 112e segments may be separated by a gap along the sidewall 114 that is filled by a non-conductive housing component. The non-conductive housing component may be part of the same non-conductive housing component that structurally couples the housing segments 112 described with reference to FIGS. 5A-5C and/or 6A-6C, or a different non-conductive housing component.

A first interlock feature 700 may include a first protrusion 734 that extends inward from an end of the first housing segment 112a, into the interior volume 732. A second interlock feature 702 may include a second protrusion 736 that extends inward from an end of the fifth housing segment 112e, into the interior volume 732. As shown, the interlock features 700, 702 and their protrusions 734, 736 may be integrally defined by (e.g., molded with or machined into) the first and fifth housing segments 112a, 112e respectively. Alternatively, the interlock features 700, 702 and their protrusions 734, 736 may be structurally coupled to the first and fifth housing segments 112a, 112e in other ways, such as by welds or fasteners. Each of the interlock features 700, 702 and protrusions 734, 736 may be set slightly back from a gap along the sidewall 114 (i.e., an exterior or sidewall gap 706) to form an interior gap 708 between the housing segments 112a, 112e that has a greater width than the exterior gap 706. For example, a first end surface 738 of the first housing segment 112a may be positioned opposite a second end surface 740 of the fifth housing segment 112e to define the exterior gap 706. The first interlock feature 700 may have a first interlock surface 742 positioned opposite a second interlock surface 744 of the second interlock feature 702 to define the interior gap 708. The first interlock surface 742 may be defined by the first protrusion 734, and the second interlock surface 744 may be defined by the second protrusion 736. The exterior gap 706 may have a first gap width, and the interior gap 708 may have a second gap width. In some embodiments, and as shown in FIG. 7A, the second gap width may be greater than the first gap width.

By way of example, the entirety of the first interlock feature 700 may be set back from the exterior gap 706 by an offset 710a, whereas an outermost portion of the second interlock feature 702 may be set back from the exterior gap 706 by an offset 710b, and an innermost portion of the second interlock feature 702 may widen into the interior gap 708 (e.g., part of the second interlock feature 702 may overlap the offset 710b).

A number of holes may be formed in each of the interlock features 700, 702 to enable a non-conductive housing com-

ponent to extend into, through, or around the interlock features **700**, **702** (thereby increasing the strength of the structural coupling between the first housing segment **112a**, the non-conductive housing component, and the fifth housing segment **112e**). With reference to FIGS. **7A** & **7B**, a first hole **712** (e.g., a round through hole) may extend into the first interlock surface **742** of the first interlock feature **700**. In some embodiments, the first hole **712** may be formed by drilling a single hole into the first interlock feature **700**. Additional holes **714**, **716** may also be formed in the first interlock feature **700**, as shown in FIG. **7B**. The additional holes **714**, **716** may provide additional surface area for the non-conductive housing component to hold or grab onto, retain, or conform to, thereby improving the strength of the structural coupling between the first and fifth housing segments **112a**, **112e**. As an example, a second hole **714** (e.g., a round hole) may extend into an upper surface **746** of the first interlock feature **700** or protrusion **734**. The second hole **714** may be drilled or otherwise cut into the first interlock feature **700**, and may be transverse to (e.g., perpendicularly intersect) the first hole **712**. A third hole **716** (e.g., a round hole) may extend into a lower surface **748** of the first interlock feature **700** or protrusion **734**. The third hole **716** may be drilled or otherwise cut into the first interlock feature **700**, and may be transverse to (e.g., perpendicularly intersect) the first hole **712**. The second and third holes **714**, **716** may have the same diameter or different diameters, and in some cases may be formed as a single through hole. The first interlock surface **742** of the first interlock feature **700** may be flat.

With reference to FIGS. **7A** & **7C**, a first hole **718** may extend into the second interlock surface **744** of the second interlock feature **702**. In some embodiments, the first hole **718** may be drilled or otherwise cut into the second interlock feature **702**. The first hole **718** may be formed as a through hole using a uniform diameter drill bit. A second hole **720** (e.g., a round hole) may extend into an upper surface **726** of the second interlock feature **702** or protrusion **734**. In some embodiments, the second hole **720** may be drilled or otherwise cut into the second interlock feature **702**, and may be transverse to (e.g., perpendicularly intersect) the first hole **718**. A third hole **722** (e.g., a round hole) may extend into a lower surface **752** of the second interlock feature **702** or protrusion **736**. In some embodiments, the third hole **722** may be drilled or otherwise cut into the second interlock feature **702**, and may be transverse to (e.g., perpendicularly intersect) the first hole **718**. The second and third holes **720**, **722** may have the same diameter or different diameters, and in some cases may be formed as a single through hole.

A boss protrusion **724** (e.g., a screw boss), which in some cases may serve as a flex circuit connection point, may be machined into or structurally coupled to the fifth antenna segment **112e**, and may be integrated with the second interlock feature **702**. The boss protrusion **724** may have a surface that is angled (e.g., at a 10 degree angle, $\pm 10\%$) with respect to an imaginary line extending perpendicular to the front and rear surfaces of a device. In alternative embodiments, the surface may be angled between 5 and 15 degrees, or between 0 (aligned with the imaginary line) and 20 degrees, or between 0 and 90 degrees. The boss protrusion **724** may be angled to reduce the inward extension of the second interlock feature **702** from the sidewall **114**, while also maintaining an upper surface **726** (of the second interlock feature **702**) to which a device cover or other element may be mated or sealed. The boss protrusion **724** may be formed similarly to the boss protrusions **616**, **624** formed in the first and second interlock features **600**, **602** described

with reference to FIGS. **6A-6C**. Alternatively, the boss protrusion **724** may be formed by a protruding piece of metal that is CNC'd off after non-conductive material(s) have been deposited into and around the first and second interlock features **700**, **702**. A threaded hole **728** may be tapped into the boss protrusion **724** after the protruding piece of metal is CNC'd off. CNC'ing off the piece of protruding metal after the non-conductive material(s) have been deposited may prevent the boss protrusion **724** from becoming buried in the non-conductive material(s). Similarly to the boss protrusion **624**, the boss protrusion **724** may have a threaded hole **728** with a shallower thread depth, so that the threaded hole **728** does not intersect the first hole **718** formed in the second interlock feature **702**.

The second interlock feature **702** may form part of (or abut) an antenna tuning feature (e.g., a slot antenna feature **730** formed between the fifth housing segment **112e** and a support plate to which the first and fifth housing segments **112a**, **112e** are structurally coupled). In some embodiments, the antenna tuning feature may be defined by a variable thickness of the fifth housing segment **112e** along the sidewall **114**. For example, the slot antenna feature **730** may extend from the second interlock feature **702** along the fifth housing segment **112e**, and may be defined by a thinned portion of the fifth housing segment **112e**. The wider portions of the fifth housing segment **112e**, adjacent each end of the thinned portion, may taper into the thinned portion (e.g., with an arc or other profile), as shown, or there may be an abrupt transition (e.g., a step) from each wider portion to the thinned portion.

As shown in FIGS. **7B** & **7C**, a non-conductive housing component **528** may at least partially fill or extend into the first and second interlock features **700**, **702**. Surrounding as many surfaces of the interlock features **700**, **702** as possible with the non-conductive material(s) of a non-conductive housing component **528** may tend to increase the strength of the structural coupling between the interlock features **700**, **702** and the non-conductive housing component **528**. In some embodiments, the non-conductive housing component **528** may extend into holes **712**, **714**, **716**, **718**, **720**, and **722**.

In some embodiments, a shelf or shelves may be cut into the upper surfaces of the first and fifth housing segments **112a**, **112e** or first and second interlock features **700**, **702**. For example, a shelf **754** may be cut into the upper surface **746** of the first interlock feature **700**. In some embodiments, holes may be formed in an upper surface or shelf of a housing segment **112** to enable the non-conductive component **528** to extend into and through portions of a housing segment **112** away from an interlock feature. For example, holes **756**, **758**, and **760** may be cut into the shelf **754**.

In some embodiments, a front cover (e.g., the front cover **106a** described with reference to FIGS. **1A-1C**) may be bonded to upper surfaces of the first and second interlock features **700**, **702** or housing segments **112a**, **112e**, or upper surfaces of the non-conductive housing component **528** (e.g., as shown in FIGS. **7B** & **7C**, where the non-conductive housing component **528** extends over the upper surfaces of the first and second interlock features **700**, **702**). In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. **1A-1C**) may be bonded to lower surfaces of the first and second interlock features **700**, **702** by an adhesive **534**. A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112a**, **112e** and extending parallel to the sidewall **114**. The seal **536** and adhesive **534** may help prevent moisture from entering a device between the housing segments **112a**, **112e** and rear cover **106b**.

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FIGS. 8A-8C show examples of the interlock features **800**, **802** that extend into an interior volume **834** defined at least partly by the sidewall **114**. The interlock features **800**, **802** may extend inward, into the interior volume **834**, from adjacent ends of the sixth and second housing segments **112f**, **112b** described with reference to FIGS. 1A-1C, 2A, 3A-3C, & 4. A portion of the sixth housing segment **112f** is shown on the left, and a portion of the second housing segment **112b** is shown on the right. As previously described, the sixth and second housing segments **112f**, **112b** may be separated by a gap along the sidewall **114** that is filled by a non-conductive housing component. The non-conductive housing component may be part of the same non-conductive housing component that structurally couples the housing segments **112** described with reference to FIGS. 5A-7C, or a different non-conductive housing component.

A first interlock feature **800** may include a first protrusion **836** that extends inward from an end of the sixth housing segment **112f**, into the interior volume **834**. A second interlock feature **802** may include a second protrusion **838** that extends inward from an end of the second housing segment **112b**. As shown, the interlock features **800**, **802** and their protrusions may be integrally defined by (e.g., molded with or machined into) the sixth and second housing segments **112f**, **112b** respectively. Alternatively, the interlock features **800**, **802** and their protrusions **836**, **838** may be structurally coupled to the sixth and second housing segments **112f**, **112b** in other ways, such as by welds or fasteners. Each of the interlock features **800**, **802** and protrusions **836**, **838** may be set slightly back from a gap along the sidewall **114** (i.e., an exterior or sidewall gap **806**) to form an interior gap **808** between the housing segments **112f**, **112b** that has a greater width than the exterior gap **806**. For example, a first end surface **840** of the sixth housing segment **112f** may be positioned opposite a second end surface **842** of the second housing segment **112b** to define the exterior gap **806**. The first interlock feature **800** may have a first interlock surface **844** positioned opposite a second interlock surface **846** of the second interlock feature **802** to define the interior gap **808**. The first interlock surface **844** may be defined by the first protrusion **836**, and the second interlock surface **846** may be defined by the second protrusion **838**. The exterior gap **806** may have a first gap width, and the interior gap **808** may have a second gap width. In some embodiments, and as shown in FIG. 8A, the second gap width may be greater than the first gap width.

By way of example, the entirety of the second interlock feature **802** may be set back from the exterior gap **806** by an offset **810b**, whereas an outermost portion of the first interlock feature **800** may be set back from the exterior gap **806** by an offset **810a**, and an innermost portion of the first interlock feature **800** may widen into the interior gap **808** (e.g., part of the first interlock feature **800** may overlap the offset **810a**).

A number of holes may be formed in each of the interlock features **800**, **802** to enable a non-conductive housing component to extend into, through, or around the interlock features **800**, **802** (thereby increasing the strength of the structural coupling between the fifth housing segment **112e**, the non-conductive housing component, and the second housing segment **112b**). For example, each of the interlock features **800**, **802** may include a first hole **812** or **814** (e.g., a through hole) that extends into the first interlock surface **844** or the second interlock surface **846**. The first hole **812** or **814** may be formed by drilling a single hole into each interlock feature **800**, **802** or protrusion **836**, **838**. Additional holes may also be formed in each interlock feature **800**, **802**

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or protrusion **836**, **838**, as shown in FIGS. 7B & 7C. The additional holes may provide additional surface area for non-conductive housing component(s) to hold or grab onto, retain, or conform to, thereby improving the strength of the structural coupling between the sixth and second housing segments **112f**, **112b**. As an example, a second hole **816** or **818** (e.g., a round hole) may extend into an upper surface **848** or **850** of each interlock feature **800**, **802** or protrusion **836**, **838**. The second holes **816**, **818** may be drilled or otherwise cut into each interlock feature **800**, **802**, and may be transverse to (e.g., perpendicularly intersect) the first hole **812** or **814**. A third hole **820** or **822** (e.g., a round hole) may extend into a lower surface **852** or **854** of each interlock feature **800**, **802** or protrusion **836**, **838**. The third holes **820**, **822** may be drilled or otherwise cut into each interlock feature **800**, **802**, and may be transverse to (e.g., perpendicularly intersect) the first hole **812** or **814**. The second and third holes **816-822** may have the same diameter or different diameters, and in some cases may be formed as respective single through holes. The formation of all three holes **812-822** by means of drilling (e.g., instead of by end milling) may save cost and reduce cycle time without sacrificing the integrity of the structural coupling between the sixth and second housing segments **112f**, **112b**.

A boss protrusion **824** (e.g., a screw boss), which in some cases may serve as a flex circuit connection point (e.g., an antenna tuning connector point), may be machined in or structurally coupled to the sixth antenna segment **112f**, and may be integrated with the first interlock feature **802**. The boss protrusion **824** may have a surface **826** that is angled (e.g., at a 10 degree angle, $\pm 10\%$) with respect to an imaginary line extending perpendicular to a front and rear surfaces of a device. In alternative embodiments, the surface may be angled between 5 and 15 degrees, between 0 (aligned with the imaginary line) and 20 degrees, or between 0 and 90 degrees. The boss protrusion **824** may be angled to reduce the inward extension of the first interlock feature **800** from the sidewall **114**, while also maintaining an upper surface **828** (of the first interlock feature **800**) to which a device cover or other element may be mated or sealed. The boss protrusion **824** may be formed similarly to the boss protrusion **724** described with reference to FIGS. 7A & 7C. Similarly to the boss protrusion **724**, the boss protrusion **824** may have a threaded hole **830** with a shallower thread depth, so that the threaded hole **830** does not intersect the first hole **812** formed in the first interlock feature **800**.

The first interlock feature **800** may form part of (or abut) an antenna tuning feature (e.g., a slot antenna feature **832** formed between the sixth housing segment **112f** and a support plate to which the sixth and second housing segments **112f**, **112b** are structurally coupled). In some embodiments, the antenna tuning feature may be defined by a variable thickness of the sixth housing segment **112f** along the sidewall **114**. For example, the slot antenna feature **832** may extend from the first interlock feature **800** along the sixth housing segment **112f**, and may be defined by a thinned portion of the sixth housing segment **112f**. The wider portions of the sixth housing segment **112f**, adjacent each end of the thinned portion, may taper into the thinned portion (e.g., with an arc or other profile), as shown, or there may be an abrupt transition (e.g., a step) from each wider portion to the thinned portion.

As shown in FIGS. 8B & 8C, a non-conductive housing component **528** may at least partially fill or extend into the first and second interlock features **800**, **802**. Surrounding as many surfaces of the interlock features **800**, **802** as possible with the non-conductive material(s) of a non-conductive

housing component **528** may tend to increase the strength of the structural coupling between the interlock features **800**, **802** and the non-conductive housing component **528**. In some embodiments, the non-conductive housing component **528** may extend into holes **812**, **814**, **816**, **818**, **820**, and **822**.

In some embodiments, a shelf or shelves may be cut into the upper surfaces of the sixth and second housing segments **112f**, **112b** or first and second interlock features **800**, **802**. For example, a shelf **856** may be cut into the upper surface **850** of the second interlock feature **802**. In some embodiments, holes may be formed in an upper surface or shelf of a housing segment **112** to enable the non-conductive component **528** to extend into and through portions of a housing segment **112** away from an interlock feature. For example, holes **858**, **860**, and **862** may be cut into the shelf **856**.

In some embodiments, a front cover (e.g., the front cover **106a** described with reference to FIGS. **1A-1C**) may be bonded to upper surfaces of the first and second interlock features **800**, **802** or housing segments **112f**, **112b**, or upper surfaces of the non-conductive housing component **528** (e.g., as shown in FIGS. **8B** & **8C**, where the non-conductive housing component **528** extends over the upper surfaces of the first and second interlock features **800**, **802**). In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. **1A-1C**) may be bonded to lower surfaces of the first and second interlock features **800**, **802** by an adhesive **534**. A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112f**, **112b** and extending parallel to the sidewall **114**. The seal **536** and adhesive **534** may help prevent moisture from entering a device between the housing segments **112f**, **112b** and rear cover **106b**.

FIGS. **9A-9C** show examples of the interlock features **900**, **902** that extend into an interior volume **928** defined at least partly by the sidewall **114**. The interlock features **900**, **902** may extend inward, into the interior volume **928**, from adjacent ends of the second and first housing segments **112b**, **112a** described with reference to FIGS. **1A-1C**, **2A**, **3A-3C**, & **4**. A portion of the second housing segment **112b** is shown on the left, and a portion of the first housing segment **112a** is shown on the right. As previously described, the second and first housing segments **112b**, **112a** may be separated by a gap along the sidewall **114** that is filled by a non-conductive housing component. The non-conductive housing component may be part of the same non-conductive housing component that structurally couples the housing segments **112** described with reference to FIGS. **5A-8C**, or a different non-conductive housing component.

A first interlock feature **900** may include a first protrusion **930** that extends inward from an end of the second housing segment **112b**, into the interior volume **928**. A second interlock feature **902** may include a second protrusion **932** that extends inward from an end of the first housing segment **112a**, into the interior volume **928**. The first interlock feature **900** may be located near a port **904** formed through the second housing segment **112b** (e.g., a pressure port, a speaker port, etc.). Because of the port **904**, the first interlock feature **900** may be somewhat thinner than the second interlock feature **902**. To compensate for the thinness of the first interlock feature **900** and improve its performance in a crushing mode, the first interlock feature **900** may protrude farther inward from the sidewall **114** than other interlock features (e.g., farther inward than the second interlock feature **902**). In some embodiments, the second interlock feature **902** may also extend farther inward from the sidewall **114**, similarly to the first interlock feature **900**.

As shown, the interlock features **900**, **902** may be integrally defined by (e.g., molded with or machined into) the second and first housing segments **112b**, **112a** respectively. Alternatively, the interlock features **900**, **902** may be structurally coupled to the second and first housing segments **112b**, **112a** in other ways, such as by welds or fasteners. Each of the interlock features **900**, **902** and protrusions **930**, **932** may be set slightly back from a gap along the interior of the sidewall (i.e., an exterior or sidewall gap **906**) to form an interior gap **908** between the housing segments **112b**, **112a** that has a greater width than the exterior gap **906**. For example, a first end surface **934** of the second housing segment **112b** may be positioned opposite a second end surface **936** of the first housing segment **112a** to define the exterior gap **906**. The first interlock feature **900** may have a first interlock surface **938** positioned opposite a second interlock surface **940** of the second interlock feature **902** to define the interior gap **908**. The first interlock surface **938** may be defined by the first protrusion **930**, and the second interlock surface **940** may be defined by the second protrusion **932**. The exterior gap **906** may have a first gap width, and the interior gap **908** may have a second gap width. In some embodiments, and as shown in FIG. **9A**, the second gap width may be greater than the first gap width.

A number of holes may be formed in each of the interlock features **900**, **902** to enable a non-conductive housing component to extend into, through, or around the interlock features **900**, **902** (thereby increasing the strength of the structural coupling between the second housing segment **112b**, the non-conductive housing component, and the first housing segment **112a**). With reference to FIGS. **9A** & **9B**, a first hole **912** (e.g., a through hole) may extend into the first interlock surface **938** of the first interlock feature **900**. In some embodiments, the first hole **912** may be formed by a set of multiple partially overlapping drill holes. Drilling multiple partially overlapping holes may save cost over forming the first hole **912** using an end mill. In some embodiments, the partially overlapping holes may be drilled close enough to reduce the ridges where holes overlap and thereby reduce the tendency of the ridges to operate as knives or chisels when a device is dropped. In alternative embodiments, the first hole **912** may be formed by an end mill or other means.

In addition to the first hole **912**, additional holes **914**, **916**, **918**, **920** may extend into additional surfaces of the first interlock feature **900** or protrusion **930**, as shown in FIGS. **9A** & **9B**. The additional holes **914-920** may provide additional surface area for the non-conductive housing component(s) to hold or grab onto, retain, or conform to, thereby improving the strength of the structural coupling between the second and first housing segments **112b**, **112a**. In some embodiments, the additional holes **914-920** may be drilled. The additional holes **914-920** may include second and third holes **914**, **916** drilled into an upper surface **942** of the first interlock feature **900**, transverse to (e.g., perpendicularly intersecting) the first hole **912**; a fourth hole **918** drilled into an inner surface **944** of the first interlock feature **900**, transverse to (e.g., perpendicularly intersecting) the first hole **912**; and a fifth hole **920** drilled into a lower surface **946** of the first interlock feature **900**, transverse to the sidewall **114** and perpendicularly intersecting the first hole **912**. A non-conductive housing component may extend into, and through each of the holes **912-920**, and around various portions of the first interlock feature **900**.

A non-conductive housing component may extend through the first interlock **900** in five directions of an x/y/z coordinate space (e.g., in all directions but through the

sidewall **114**). This can increase the structural rigidity of the first interlock feature **900**, which can be useful given its thinner width.

With reference to FIGS. **9A** & **9C**, a first hole **922** (e.g., a round through hole) may extend into the second interlock surface **940** of the second interlock feature **902** or protrusion **932**. In some embodiments, the first hole **922** may be drilled or otherwise cut into the second interlock feature **902**. A second hole **924** (e.g., a round hole) may extend into an upper surface **948** of the second interlock feature **902** or protrusion **932**. The second hole **924** may also be drilled, and may be transverse to (e.g., perpendicularly intersect) the first hole **922**. A third hole **926** (e.g., a round hole) may extend into a lower surface **950** of the second interlock feature **902** or protrusion **932**. The third hole **926** may be drilled or otherwise cut into the second interlock feature **902** and be transverse to (e.g., perpendicularly intersect) the first hole **922**. The second and third holes **924**, **926** may have the same diameter or different diameters, and in some cases may be formed as a single through hole.

As shown in FIGS. **9B** & **9C**, a non-conductive housing component **528** may at least partially fill or extend into the first and second interlock features **900**, **902**. Surrounding as many surfaces of the interlock features **900**, **902** as possible with the non-conductive material(s) of a non-conductive housing component **528** may tend to increase the strength of the structural coupling between the interlock features **900**, **902** and the non-conductive housing component **528**. In some embodiments, the non-conductive housing component **528** may extend into holes **912**, **914**, **916**, **918**, **920**, **922**, **924**, and **926**.

In some embodiments, a shelf or shelves may be cut into the upper surfaces of the second and first housing segments **112b**, **112a** or first and second interlock features **900**, **902**. For example, a shelf **952** may be cut into the upper surface **948** of the second interlock feature **902**. In some embodiments, holes may be formed in an upper surface or shelf of a housing segment **112** to enable the non-conductive component **528** to extend into and through portions of a housing segment **112** away from an interlock feature. For example, holes **954** and **956** may be cut into the shelf **952**.

In some embodiments, a front cover (e.g., the front cover **106a** described with reference to FIGS. **1A-1C**) may be bonded to upper surfaces of the first and second interlock features **900**, **902** or housing segments **112b**, **112a**, or upper surfaces of the non-conductive housing component **528** (e.g., as shown in FIGS. **9B** & **9C**, where the non-conductive housing component **528** extends over the upper surfaces of the first and second interlock features **900**, **902**). In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. **1A-1C**) may be bonded to lower surfaces of the first and second interlock features **900**, **902** by an adhesive **534**. A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112b**, **112a** and extending parallel to the sidewall **114**. The seal **536** and adhesive **534** may help prevent moisture from entering a device between the housing segments **112b**, **112a** and rear cover **106b**.

FIGS. **10A-10C** show examples of the interlock features **1000**, **1002** that extend into an interior volume **1028** defined at least partly by the sidewall **114**. The interlock features **1000**, **1002** may extend inward, into the interior volume **1028**, from adjacent ends of the third and fourth housing segments **112c**, **112d** described with reference to FIGS. **1A-1C**, **2A**, **3A-3C**, & **4**. A portion of the third housing segment **112c** is shown on the left, and a portion of the fourth housing segment **112d** is shown on the right. As previously

described, the third and fourth housing segments **112c**, **112d** may be separated by a gap along the sidewall **114** that is filled by a non-conductive housing component. The non-conductive housing component may be part of the same non-conductive housing component that structurally couples the housing segments **112** described with reference to FIGS. **5A-9C**, or a different non-conductive housing component. Similarly, the interior volumes described with reference to FIGS. **5A**, **6A**, **7A**, **8A**, **9A**, & **10A** may be the same or different interior volumes.

A first interlock feature **1000** may include a first protrusion **1030** that extends inward from an end of the third housing segment **112c**, into the interior volume **1028**. A second interlock feature **1002** may include a second protrusion **1032** that extends inward from an end of the fourth housing segment **112d**, into the interior volume **1028**. As shown, the interlock features **1000**, **1002** and their protrusions **1030**, **1032** may be integrally defined by (e.g., molded with or machined into) the third and fourth housing segments **112c**, **112d** respectively. Alternatively, the interlock features **1000**, **1002** may be structurally coupled to the third and fourth housing segments **112c**, **112d** in other ways, such as by welds or fasteners.

Each of the interlock features **1000**, **1002** may be set slightly back from a gap along the interior of the sidewall **114** (i.e., an exterior or sidewall gap **1006**) to form an interior gap **1008** between housing segments **112c**, **112d** that has a greater width than the exterior gap **1006**. For example, a first end surface **1034** of the third housing segment **112c** may be positioned opposite a second end surface **1036** of the fourth housing segment **112d** to define the exterior gap **1006**. The first interlock feature **1000** may have a first interlock surface **1038** positioned opposite a second interlock surface **1040** of the second interlock feature **1002** to define the interior gap **1008**. The first interlock surface **1038** may be defined by the first protrusion **1030**, and the second interlock surface **1040** may be defined by the second protrusion **1032**. The exterior gap **1006** may have a first gap width, and the interior gap **1008** may have a second gap width. In some embodiments, and as shown in FIG. **10A**, the second gap width may be greater than the first gap width.

A number of holes may be formed in each of the interlock features **1000**, **1002** to enable a non-conductive housing component to extend into, through, or around the interlock features **1000**, **1002** (thereby increasing the strength of the structural coupling between the third housing segment **112c**, the non-conductive housing component, and the fourth housing segment **112d**). With reference to FIGS. **10A** & **10B**, a first hole **1010** (e.g., a through hole) may extend into the first interlock surface **1038** of the first interlock feature **1000** or protrusion **1030**. The first hole **1010** may be formed by a set of multiple partially overlapping drill holes. In some embodiments, the partially overlapping holes may be drilled close enough to reduce the ridges where holes overlap and thereby reduce the tendency of the ridges to operate as knives or chisels when a device is dropped. In alternative embodiments, the first hole **1010** may be formed by an end mill or other means.

In addition to the first hole **1010**, additional holes **1012**, **1014** may be formed in the first interlock feature **1000** or protrusion **1030**. The additional holes **1012**, **1014** may provide additional surface area for the non-conductive housing component(s) to hold or grab onto, retain, or conform to, thereby improving the strength of the structural coupling between the third and fourth housing segments **112c**, **112d**. In some embodiments, the additional holes **1012**, **1014** may be drilled. The additional holes **1012**, **1014** may include a

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second hole **1012** that extends into an upper surface **1018** of the first interlock feature **1000** or protrusion **1030**, transverse to (e.g., perpendicularly intersecting) the first hole **1010**, and a third hole **1014** that extends into an inner surface **1042** of the first interlock feature **1000** or protrusion **1030**, transverse to the sidewall **114** and first hole **1010** (e.g., perpendicularly intersecting the first hole **1010**). A non-conductive housing component may extend into and through each of the holes **1010-1014**, and around various portions of the first interlock feature **1000**. A shelf **1016** may be cut into the upper surface **1018** of the first interlock feature **1000**. In some cases, the shelf **1016** may be curved. In some cases, the shelf **1016** may intersect the second hole **1012**. As shown, the shelf **1016** may intersect the second hole **1012** perpendicularly. The shelf **1016** may increase the separation or reduce the capacitance between the third housing segment **112c** and a conductive component(s) that are routed near the third housing segment **112c**. In some embodiments, the shelf **116** may be covered by a non-conductive housing component **528**, as shown in FIG. **10B**.

With reference to FIGS. **10A** & **10C**, a first hole **1020** (e.g., a round through hole) may extend into the second interlock surface **1040** of the second interlock feature **1002** or protrusion **1032**. In some embodiments, the first hole **1020** may be drilled or otherwise cut into the second interlock feature **1002**. A second hole **1022** (e.g., a round hole) may extend into an upper surface **1044** of the second interlock feature **1002** or protrusion **1032**, and may also be drilled or otherwise cut into the second interlock feature **1002**. The second hole **1022** may be transverse to (e.g., perpendicularly intersect) the first hole **1020**. A third hole **1024** (e.g., a round hole) may extend into a lower surface **1046** of the second interlock feature **1002** or protrusion **1032**. In some embodiments, the third hole **1024** may be drilled or otherwise cut into the second interlock feature **1002**, transverse to (e.g., perpendicularly intersecting) the first hole **1020**. The second and third holes **1022**, **1024** may have the same diameter or different diameters, and in some cases may be formed as a single through hole.

As shown in FIG. **10A**, a boss protrusion **1026** (e.g., a screw boss) may be integrated with the second interlock feature **1002** or protrusion **1032**. In some embodiments, the boss protrusion **1026** may be formed as described with reference to FIG. **6A**.

As shown in FIGS. **10B** & **10C**, a non-conductive housing component **528** may at least partially fill or extend into the first and second interlock features **1000**, **1002**. Surrounding as many surfaces of the interlock features **1000**, **1002** as possible with the non-conductive material(s) of a non-conductive housing component **528** may tend to increase the strength of the structural coupling between the interlock features **1000**, **1002** and the non-conductive housing component **528**. In some embodiments, the non-conductive housing component **528** may extend into holes **1010**, **1012**, **1014**, **1020**, **1022**, and **1024**.

In some embodiments, a shelf or shelves may be cut into a housing segment **112** (e.g., into the fourth housing segment **112d**). For example, a shelf **1048** may be cut into the fourth housing segment **112d** above the upper surface **1044** of the second interlock feature **802**. In some embodiments, holes may be formed in the shelf **1048** to enable the non-conductive component **528** to extend into and through the shelf **1048**. For example, holes **1050** and **152** may be cut into the shelf **1048**.

In some embodiments, a front cover (e.g., the front cover **106a** described with reference to FIGS. **1A-1C**) may be bonded to upper surfaces of the first and second interlock

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features **1000**, **1002** or housing segments **112c**, **112d**, or upper surfaces of the non-conductive housing component **528** (e.g., as shown in FIGS. **10B** & **10C**, where the non-conductive housing component **528** extends over the upper surfaces of the first and second interlock features **1000**, **1002**). In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. **1A-1C**) may be bonded to lower surfaces of the first and second interlock features **1000**, **1002** by an adhesive **534**. A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112c**, **112d** and extending parallel to the sidewall **114**. The seal **536** and adhesive **534** may help prevent moisture from entering a device between the housing segments **112c**, **112d** and rear cover **106b**.

In some embodiments, one or more surfaces of the interlock features described with reference to FIGS. **5A-10C** may be etched, machined, or treated to make the surface textured or porous. For example, in some embodiments, some surfaces of the interlock features may be etched to form pores that are 2-3 microns deep and 2-3 microns wide. Such pores provide additional blind holes for non-conductive housing components to flow into, thereby increasing the structural bond between interlock features and non-conductive housing components. In areas where a wall of an interlock feature is thin, the surface of the wall may not be etched, or etching may be controlled to ensure that pores do not break through the wall (e.g., to avoid sealing issues for different holes or cavities).

FIGS. **11A** & **11B** illustrate how exterior (sidewall) and interior gaps between the housing segments **112** described with reference to FIGS. **2A**, **3A-3C**, **4**, **5A**, **6A**, **7A**, **8A**, **9A**, & **10A** may be aligned symmetrically or asymmetrically. FIG. **11A** shows corresponding exterior and interior gaps **1100**, **1102** that are aligned symmetrically. FIG. **11B** shows corresponding exterior and interior gaps **1124**, **1126** that are aligned asymmetrically.

By way of example, FIG. **11A** shows a generic representation of two adjacent housing segments **1104**, **1106** that may form part of a sidewall **1108** of a housing. A first interlock feature **1110** may have a protrusion **1130** that extends inward from an end of the first housing segment **1104**, into an interior volume **1132** defined at least partly by the sidewall **1108**. A second interlock feature **1112** may have a protrusion **1134** that extends inward from an end of the second housing segment **1106**, into the interior volume **1132**. An exterior gap **1100** may be defined between the first and second housing segments **1104**, **1106**. The interlock features **1110**, **1112** may be set back from the exterior gap **1100** to form an interior gap **1102** that has a greater width than the exterior gap **1100**. For example, a first end surface **1136** of the first housing segment **1104** may be positioned opposite a second end surface **1138** of the second housing segment **1106** to define the exterior gap **1100**. The first interlock feature **1110** may have a first interlock surface **1140** positioned opposite a second interlock surface **1142** of the second interlock feature **1112** to define the interior gap **1102**. The first interlock surface **1140** may be defined by the first protrusion **1130**, and the second interlock surface **1142** may be defined by the second protrusion **1134**. The exterior gap **1100** may have a first gap width, and the interior gap **1102** may have a second gap width. In some embodiments, and as shown in FIG. **11A**, the second gap width may be greater than the first gap width.

The first interlock surface **1140** may be offset from the first end surface **1136** by a first offset **1114a**, and the second interlock surface **1142** may be offset from the second end surface **1138** by a second offset **1114b**, which offsets **1114a**, **1114b** may be the same such that the exterior and interior

gaps **1100**, **1102** are aligned symmetrically. A non-conductive housing component may overlap interior surfaces of the housing segments **1104**, **1106** along the sidewall **1108** (i.e., extend along the offsets **1114a**, **1114b**) and fill the exterior and interior gaps **1100**, **1102**.

In some embodiments, the non-conductive housing component may be formed by a polymer material having a fiber fill, and the polymer material may at least partially fill various holes in the first and second interlock features **1110**, **1112** in addition to forming an exterior surface of the sidewall **1108** (e.g., a portion of the sidewall **1108** that bridges or fills the exterior gap **1100**). In other embodiments, the non-conductive housing component may include a first portion formed from a first polymer material and a second portion formed from a second polymer material. The first polymer material may have a fiber fill and at least partially fill various holes in the first and second interlock features. The second polymer material may be different from the first polymer material and form an exterior surface of the sidewall **1108** (e.g., a portion of the sidewall **1108** that bridges or fills the exterior gap **1100**). Each polymer having a fiber fill may have a fiber fill including glass or other types of fibers. In some embodiments, the second polymer material may also have a fiber fill, but have a fiber fill that differs from the fiber fill of the first polymer material.

Also by way of example, FIG. **11B** shows a generic representation of two adjacent housing segments **1116**, **1118** that may form part of a sidewall **1108** of a housing. A first interlock feature **1120** may have a protrusion **1144** that extends inward from an end of the first housing segment **1116**, into an interior volume **1146** defined at least partly by the sidewall **1108**. A second interlock feature **1122** may have a protrusion **1148** that extends inward from an end of the second housing segment **1118**, into the interior volume **1146**. An exterior gap **1124** may be defined between the first and second housing segments **1116**, **1118**. The interlock features **1120**, **1122** may be set back from the exterior gap **1124** to form an interior gap **1126** that has a greater width than the exterior gap **1124**. For example, a first end surface **1150** of the first housing segment **1116** may be positioned opposite a second end surface **1152** of the second housing segment **1118** to define the exterior gap **1124**. The first interlock feature **1120** may have a first interlock surface **1154** positioned opposite a second interlock surface **1156** of the second interlock feature **1122** to define the interior gap **1126**. The first interlock surface **1154** may be defined by the first protrusion **1144**, and the second interlock surface **1156** may be defined by the second protrusion **1148**. The exterior gap **1124** may have a first gap width, and the interior gap **1126** may have a second gap width. In some embodiments, and as shown in FIG. **11B**, the second gap width may be greater than the first gap width.

The first interlock surface **1154** may be offset from the first end surface **1150** by a first offset **1128a**, and the second interlock surface **1156** may be offset from the second end surface **1152** by a second offset **1128b**, which offsets **1128a**, **1128b** may differ such that the exterior and interior gaps **1124**, **1126** are aligned asymmetrically. A non-conductive housing component may overlap interior surfaces of the housing segments **1116**, **1118** along the sidewall **1108** (i.e., extend along the offsets **1128a**, **1128b**) and fill the exterior and interior gaps **1124**, **1126**.

Asymmetrically aligned exterior and interior gaps **1124**, **1126** may enable adjacent interlock features **1120**, **1122** to be shifted in position along a housing sidewall **1108** while still providing sufficient separation between conductive housing segments **1116**, **1118** that may be operated as antennas.

Sufficient separation may be needed to mitigate the likelihood that the housing segments **1116**, **1118** couple to one another (thereby enabling the conductive housing segments to resonate independently) and/or to reduce the capacitance between the conductive housing segments.

In some embodiments, the thickness of a housing segment at a boundary of an exterior gap may be defined as a function of the width of the exterior gap, or conversely, the width of the exterior gap may be defined as a function of the thickness of the housing segment at the boundary of the exterior gap. For example, the thickness of the housing segment **1104** or **1116** at the boundary of the exterior gap **1100** (see FIGS. **11A** & **11B**) may be defined as a function of the width of the exterior gap **1100** or **1124**.

In some embodiments, the thickness of a housing segment at a boundary of an interior gap may be defined as a function of the width of the interior gap, or conversely, the width of the exterior gap may be defined as a function of the thickness of the housing segment at the boundary of the interior gap. For example, the width of the interior gap **1102** or **1126** may be defined as a function of the thickness of the housing segment **1104** or **1116** at the boundary of the interior gap **1102** or **1126** (see FIGS. **11A** & **11B**).

In some embodiments, the width of the exterior or interior gaps described with reference to FIGS. **11A** & **11B** may be otherwise defined to reduce capacitance or coupling between the ends of adjacent conductive housing segments while also maintaining good structural rigidity of the sidewall **1108**.

Turning to FIG. **12**, there is shown an isometric view of the first housing segment **112a** described with reference to FIGS. **1A-1C**, **2A**, **3A-3C**, **4**, **7A-7B**, **9A**, & **9C** and portions **116a**, **116c**, **1204** of the non-conductive housing component **528** that abut, fill, and surround the interlock features **902**, **700** and other interior surfaces of the first housing segment **112a**. As shown, a first portion **1204** of the non-conductive housing component **528** may form a gusset **1200** that extends from under the lower surfaces of the first housing segment **112a** (including lower surfaces of its interlock features **902**, **700**) to over, under, or onto a support plate (e.g., over, under, onto, or encapsulating the support plate **110** described with reference to FIG. **4**). The first portion **1204** of the non-conductive housing component **528** may also extend at least partially into the interlock features **902**, **700**, into other features of the first housing segment **112a**, and along an interior surface of the sidewall **114**. The first portion **1204** of the non-conductive housing component **528** may also extend into the interlock features **900**, **702** extending from adjacent ends of other housing segments (e.g., the second housing segment and the fifth housing segment **112b**, **112e** described with reference to other figures). A strengthening rib or buttress **1202a** may be formed between the gusset **1200** and portions of the non-conductive housing component **528** that extend along the interior surface of the sidewall **114**. The gusset **1200** and/or buttress **1202a** may improve the rigidity of the first housing segment **112a** and its structural couplings to adjacent housing segments, and in some cases may span that portion of a corner that lies between an interlock feature **900** at an end of the second housing segment to an interlock feature **702** at an end of the fifth housing segment. Other buttresses, such as buttress **1202b**, may also be formed by the first portion **1204** of the non-conductive housing component **528**. In some embodiments, an adhesive may bond the first portion **1204** of the non-conductive housing component (e.g., the gusset **1200**) to the support plate **110**. The gusset **1200** and buttresses **1202a**, **1202b** may be replicated in other corners of the sidewall **114**, and may provide extra structural support for a

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housing segment mounted at the corner of a device (and in particular, a housing segment that wraps just around a corner of a device).

As also shown in FIG. 12, second portions 116e, 116c of the non-conductive housing component 528 may fill portions of the gaps between housing segments 112 and form exterior surface portions of the sidewall 114. In some embodiments, the first portion 1204 of the non-conductive housing component 528 may provide more structural rigidity, and the second portions 116e, 116c of the non-conductive housing component 528 may have a more uniform consistency than the first portion 1204 and provide a smoother exterior surface along the sidewall 114.

In some embodiments, the entirety of the non-conductive housing component 528 may be formed by a polymer material having a fiber fill, and the polymer material may at least partially fill various holes in the interlock features 900, 902, 700, 702, in addition to forming exterior surface portions of the sidewall 114. In other embodiments, the non-conductive housing component 528 may include the first portion 1204, which may be formed from a first polymer material, and the second portions 116e, 116c, which may be formed from a second polymer material. The first polymer material may have a fiber fill and at least partially fill various holes in the first and second interlock features. The second polymer material may be different from the first polymer material and form an exterior surface of the sidewall 114. Each polymer having a fiber fill may have a fiber fill including glass or other types of fibers. In some embodiments, the second polymer material may also have a fiber fill, but have a fiber fill that differs from the fiber fill of the first polymer material.

Boss protrusions 1204, 1206 (e.g., screw bosses) may be formed (e.g., machined into) portions of the first housing segment 112a that extend inward from a lower surface of the first housing segment 112a (i.e., into a surface of the first housing segment 112a that is oriented toward the rear of a device). In some embodiments, the boss protrusions 1204, 1206 may be formed using the previously-described hole cutter. The boss protrusions 1204, 1206 may be tapped to receive screws that attach a flex circuit to each of the boss protrusions 1204, 1206, thereby attaching the flex circuit to the first housing segment 112a. The boss protrusion 1204 to the left in FIG. 12 may provide a ground connector, and the boss protrusion 1206 to the right may provide an antenna feed connector (alternatively referred to as just a “feed connector”). The boss protrusions 1204, 1206 may be formed on separate inward extensions of the first housing segment 112a, to increase the length of the conductive path (or increase the length of the resonant portion of the first housing segment 112a) therebetween.

FIGS. 13A-13D show various details of a device forehead (e.g., the portion of a device extending under, over, or above a top edge of a device display), and FIGS. 14A-14G show various details of a device chin (e.g., the portion of a device extending under, over, or below a bottom edge of a device display).

FIG. 13A shows a plan view of the third and fourth housing segments 112c, 112d described with reference to FIGS. 1A-1C, 2A, 3A, 4, 5A, 5C, 6A, 6B, & 10A-10C, and portions of the fifth and sixth housing segments 112e, 112f. FIG. 13A also shows a non-conductive housing component 528 that structurally couples all of these components, including portions 116a, 116b, 116f of the non-conductive housing component 528 that differ from another portion 1354 (e.g., a more rigid portion) of the non-conductive

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housing component 528 and form non-conductive segments or portions of the sidewall 114.

As shown, the support plate 110 may have a deep recess 1300 extending inward from an edge 1302 of the support plate 110 that is closer to the sidewall 114. The recess 1300 may accommodate the positioning of components attached to a flex circuit, and may enable the flex circuit to be positioned very close to the sidewall 114.

When the housing segments 112 are conductive and used as antennas, slots in the support plate 110 (e.g., slot 1304) may couple to the housing segments 112 and undesirably alter antenna operation or decrease antenna efficiency. In some embodiments, a parasitic slot in the support plate 110 (e.g., slot 1304) may be electrically closed or shorted (e.g., by welding a conductive component 1306 such as a strap between opposite edges of the slot 1304, at or near an open end 1308 of the slot 1304, or at other points between opposite edges of the slot 1304). Parasitic slots are slots that detract from antenna performance, and differ from slot antenna features that can be used to tune antenna performance.

FIG. 13A shows the camera brace 402 described with reference to FIG. 4. In some embodiments, the camera brace 402 may be grounded, and a ground connector 1310 of the fourth housing segment 112d may be coupled to ground via the grounded camera brace 402. Electrically coupling the ground connector 1310 of the fourth housing segment 112d to ground via the camera brace 402 may be easier than coupling the ground connector 1310 to a ground on a flex circuit given the small clearances between the camera brace 402 and adjacent housing segments 112c, 112d, 112f. However, because a device may be dropped on the corner including the fourth housing segment 112d, a welded or other rigid connection between the camera brace 402 and ground connector 1310 may have a tendency to break when the device is dropped. To mitigate the likelihood of the ground between the camera brace 402 and the ground connector 1310 being disrupted, the ground connector 1310 may be electrically coupled to the camera brace 402 using a compliant conductive component 1312, such as a compliant conductive tab or strap that extends from one of the camera brace 402 or ground connector 1310 and is welded to the other of the ground connector 1310 or camera brace 402. Alternatively, the compliant conductive component 1312 (e.g., a wire, strap, or thin metal plate) may be welded or otherwise electrically coupled to each of the camera brace 402 and ground connector 1310.

As shown in FIG. 13B, a ground pad 1314, to which a ground spring may be mated, may be formed on the camera brace 402. In some embodiments, the ground pad 1314 may be formed on a side 1316 of the camera brace 402 facing the sixth housing segment 112f (e.g., in a channel 1318 between the side 1316 of the camera brace 402 and the sixth housing segment 1120). To improve the electrical continuity of the ground connection and reduce ground noise, the ground pad 1314 may include a gold plate (or gold-plated plate) that is welded to the stainless steel or other conductive material that forms the camera brace 402. The ground spring that is mated to the ground pad 1314 may be formed on a flex circuit, which flex circuit is disposed in the channel 1318 with the ground spring facing and contacting the ground pad 1314 (the ground spring is not shown in FIG. 13B, but is shown in FIG. 17B). Gold-on-gold contact (e.g., a gold or gold-plated ground pad 1314 and ground spring) may be useful here and in other locations where spring contacts, and especially low-force spring contacts, are electrically coupled to a conductive pad. When an RF signal passes through a

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spring junction, the spring junction can introduce harmonics that interfere with wireless communication (e.g., particular wireless frequency bands). Gold-on-gold contact can reduce the likelihood or amplitude of such harmonics.

FIG. 13C shows internal structures and connections of the device forehead, as viewed while looking from the front of the device toward the rear of the device. The connections include various antenna connections made between the third, fourth, and fifth housing segments 112c, 112d, 112e and one or more antenna flex circuits. Examples of the antenna flex circuits are described in more detail with reference to FIGS. 17A & 17B. The antenna flex circuits may extend grounds to the ground connections of the housing segments 112c, 112d, 112e, carry signals to and from antenna feed connectors, and/or carry antenna tuning components (e.g., components that may be used to tune the resonance, frequency, or bandwidth of a housing segment that is operated as an antenna).

As shown in FIG. 13C, the third housing segment 112c may include a ground connector 1320, a feed connector 1322, and a tuning connector 1324. The ground connector 1320 may be electrically coupled to the support plate 110 (and thereby to ground) by a flex circuit that connects to both the ground connector 1320 and a ground connector 1326 on the support plate 110. In some embodiments, the ground connector 1320 may be positioned at the corner defined by the third housing segment 112c. The feed connector 1322 may be positioned in from the corner, along a top edge 1328 of a device. The tuning connector 1324 may be positioned near the center of the top edge 1328. Both the feed connector 1322 and the tuning connector 1324 may be electrically coupled to a flex circuit. In some embodiments, each of the ground connector 1320, the feed connector 1322, the tuning connector 1324, and the support plate ground connector 1326 may be electrically coupled to a common flex circuit, such as the flex circuit described with reference to FIG. 17B. The same flex circuit may also provide electrical connectors for grounding the fifth housing segment 112e (e.g., at a ground connector 1330), and tuning components that connect to the tuner ground connector 1332.

As also shown in FIG. 13C, the fourth housing segment 112d may also include a ground connector 1334, a feed connector 1336, and a tuning connector 1338. The ground connector 1334 may be electrically coupled to the camera brace 402 (e.g., using the compliant conductive component 1312 described with reference to FIG. 13A). A flex circuit, such as the flex circuit described with reference to FIG. 17B, may carry a ground spring that is mated to a ground pad on the camera brace 402 (e.g., the ground pad 1314 described with reference to FIG. 13B). The ground potentials of the flex circuit and the fourth housing segment 112d may therefore be electrically coupled through the camera brace 402. Other ground connections for the camera brace 402 may be made, for example, through the support plate 110 or camera module bias springs 1340, 1342 disposed at points along the perimeter of the camera brace 402. The ground and feed connectors 1334, 1336 for the fourth housing segment 112d may be disposed near opposite ends of the fourth housing segment 112d, and in some embodiments, the ground connector 1334 may be positioned more toward the top edge 1328 of the device, and the feed connector 1336 may be positioned more toward a side edge 1344 of the device. The tuning connector 1338 for the fourth housing segment 112d may be on the sixth housing segment 112f and may take the form of the boss protrusion described with reference to FIG. 6A. The tuning connector 1338 may be coupled to the flex circuit described with reference to FIG.

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17B, which flex circuit may carry tuning components such as a circuit including a switch that may be operated to connect or disconnect the slot antenna feature 302d (see, FIG. 3A) defined between the sixth housing segment 112f and the support plate 110 to the fourth housing segment 112d.

The third housing segment 112c may in some cases have a Fargo feed connector 1346. The Fargo feed connector 1346 may be located along the top edge 1328 of the device, near the end of the third housing segment 112c that is adjacent the fourth housing segment 112d. The Fargo feed connector 1346 may alternatively be located in other places along the third housing segment 112c.

FIG. 13C shows example positions of a speaker 1348, camera 1350, and bio-authentication sensor 1352 (e.g., an infrared camera) that may be mounted in the device forehead. Ground connections 1354, 1356 for these components, or other components located in the device forehead, may be provided near the top edge 1328 of the device as shown.

FIG. 13D shows additional internal structures and connections of the device forehead, as viewed while looking from the rear of a device toward the front cover 106a of the device. The structures include the camera 1350 and bio-authentication sensor 1352 described with reference to FIG. 13C. A cross-section of the device forehead is shown in FIG. 22.

Turning now to the device chin, FIG. 14A shows a plan view of the first and second housing segments 112a, 112b described with reference to FIGS. 1A-1C, 2A, 3A, 4, 7A, 7B, 8A, 8C, & 9A-9C, and portions of the fifth and sixth housing segments 112e, 112f. FIG. 14A also shows a non-conductive housing component 528 that structurally couples all of these components, including portions 116c, 116d, 116e of the non-conductive housing component 528 that differ from another portion 1474 of the non-conductive housing component and form non-conductive segments or portions of the sidewall 114.

As shown, the support plate 110 may have one or more deep recesses 1400, 1402 extending inward from an edge 1404 of the support plate 110 that is closer to the sidewall 114. The recesses 1400, 1402 may accommodate the positioning of components attached to a flex circuit, and may enable the flex circuit to be positioned very close to the sidewall 114.

A first slot antenna feature 302a may be formed between the fifth housing segment 112e and the support plate 110, and a second slot antenna feature 302b may be formed between the sixth housing segment 112f and the support plate.

FIG. 14B shows internal structures and connections of the device chin, as viewed while looking from the front of a device toward the rear of the device. The connections include various antenna connections made between the first, second, fifth, and sixth housing segments 112a, 112b, 112e, 112f and one or more antenna flex circuits. Examples of the antenna flex circuits are described in more detail with reference to FIG. 17A. The antenna flex circuits may extend grounds to the ground connectors 1406, 1408 of the housing segments 112a, 112b, carry signals to and from the antenna feed connectors 1410, 1412, and carry antenna tuning components that are coupled to the tuning connectors 1414, 1416.

As shown in FIG. 14B, the first housing segment 112a may include a ground connector 1406, a feed connector 1410, and a tuning connector 1414. The ground connector 1406 may be electrically coupled to the support plate 110 (and thereby to ground) by a flex circuit that connects to both

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the ground connector **1406** and a ground connector **1418** on the support plate **110**. In some embodiments, the ground connector **1406** may be positioned closer to the left edge **1420** of the device (away from the second housing segment **112b**), and the feed connector **1410** may be positioned closer to the lower edge **1422** of the device. The tuning connector **1414** for the first housing segment **112a** may be on the fifth housing segment **112e** and take the form of the boss protrusion **724** described with reference to FIG. 7A. The tuning connector **1414** may be coupled to the flex circuit described with reference to FIG. 17A, which flex circuit may carry tuning components such as a circuit including a switch that may be operated to connect or disconnect the slot antenna feature **302a** (see, FIG. 14A) defined between the fifth housing segment **112e** and the support plate **110** to the first housing segment **112a**. Points along the slot antenna feature **302a** may be grounded to the support plate at connectors **1414** and **1424**.

As also shown in FIG. 14B, the second housing segment **112b** may also include a ground connector **1408**, a feed connector **1412**, and a tuning connector **1416**. The ground connector **1408** may be electrically coupled to the support plate **110** (and thereby to ground) by a flex circuit that connects to the ground connector **1408** and one or more ground connectors **1426**, **1428**, **1430**, **1432** on the support plate **110** or sixth housing segment **112f**. In some embodiments, the ground connector **1408** may be positioned closer to the right edge **1434** of the device (away from the first housing segment **112a**), and the feed connector **1412** may be positioned closer to the lower edge **1422** of the device. The tuning connector **1416** may be positioned near the center of the lower edge **1422**. Both the feed connector **1412** and the tuning connector **1416** may be electrically coupled to a flex circuit. In some embodiments, each of the ground connector **1408**, the feed connector **1412**, the tuning connector **1416**, and the support plate ground connectors **1426**, **1428**, **1430**, **1432** may be electrically coupled to a common flex circuit, such as the flex circuit described with reference to FIG. 17A. The same flex circuit may also provide tuning components that connect to the tuning connector **1416**.

The second housing segment **112b** may in some cases include an alternative ground, feed, and/or tuning connector (e.g., the feed/tuning connector **1436**). These alternative connectors may be coupled to the same flex circuit as the other ground, feed, and tuning connectors **1408**, **1412**, **1416**, but may be positioned closer to the first housing segment **112a** than the other ground, feed, and tuning connectors **1408**, **1412**, **1416**. an additional tuning connector **1438** on the sixth housing segment **112f**, which may take the form of the boss protrusion **824** described with reference to FIG. 8A. The tuning connector **1438** may be coupled to the flex circuit described with reference to FIG. 17A, which flex circuit may carry tuning components such as a circuit including a switch that may be operated to connect or disconnect the slot antenna feature **302b** (see, FIG. 14A) defined between the sixth housing segment **112f** and the support plate **110** to the second housing segment **112b**.

FIG. 14C shows an exterior isometric view of the second housing segment **112b**. A number of ports **1440** (through holes) may be formed in the second housing segment **112b**. The ports may include ports such as an ambient pressure sensing port **1440a** (e.g., a barometric pressure sensing port) of the device, a second one or more ports **1440b** that function as speaker or microphone ports, and a power port **1440c** for receiving a power cord. Other ports may also be provided, or some of the ports **1440** shown may be repurposed. For example, the second housing segment **112b** may include an

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audio jack, video port, or audio/visual (A/V) port. In some embodiments, one or more of the ports **1440** shown may not be provided or have a different shape. By way of example, all of the ports **1440** but the power port **1440c** are shown to have a round shape along the exterior surface of the second housing segment **112b**. The power port **1440c** is shown to be oblong. FIG. 14C shows four speaker ports **1440b** disposed to the left of the power port **1440c** and six speaker ports **1440b** disposed to the right of the power port **1440c**. Alternatively, the same number of speaker ports **1440b** may be provided on either side of the power port **1440c**, or a different number of speaker ports **1440b** may be provided.

FIG. 14D shows an interior isometric view of the second housing segment **112b**. As shown, walls **1442**, **1444**, **1446** may surround individual ones or sets of some of the ports **1440** formed in the second housing segment **112b**. The walls **1442**, **1444**, **1446** may extend inward from the second housing segment **112b** and provide a set of surfaces that may function as sealing surfaces. For example, gaskets or other components may be mated to a first sealing surface **1448** formed around the speaker ports **1440b** located to the left of the power port **1440c**, and to a second sealing surface **1450** formed around the speaker ports **1440b** and ambient pressure sensing port **1440a** located to the right of the power port **1440c**. In some embodiments, portions **1452**, **1454** of the second housing segment **112b** may be removed to alter (e.g., reduce) the capacitance between the second housing segment **112b** and other conductive structures in the chin area. For example, portions **1452**, **1454** of the second housing segment **112b** (e.g., portions **1452**, **1454** on either side of the power port **1440c**) may be removed to reduce the capacitance between the second housing segment **112b** and a grounded element attached to a cover fitted to the housing segments **112** that form the sidewall **114**. When portions **1452**, **1454** of the upper surface of the second housing segment **112b** are removed, other portions **1456**, **1458** may remain to provide a support surface for the cover and retain the structural and sealing surface integrity of the second housing segment **112b**.

FIG. 14E shows the second housing segment **112b** from the same angle as FIG. 14D, but additionally provides an example of how a conductive housing component (e.g., conductive housing components **528** and **116e**) may be disposed around the sealing surfaces **1448**, **1450** described with reference to FIG. 14D. In some embodiments, the same non-conductive housing component **528** used to structurally couple adjacent interlock features of adjacent housing segments **112** may also be disposed around the walls **1442**, **1444**, **1446** that provide the sealing surfaces **1448**, **1450**.

FIG. 14F shows a cross-section of the second housing segment **112b**, taken through one of the ports (e.g., **1440a** or **1440b**) disposed to the right of the power port **1440c** in FIG. 14E. FIG. 14G shows a cross-section of the second housing segment **112b** taken through one of the ports **1440b** disposed to the left of the power port **1440c** in FIG. 14E. As shown in FIG. 14F, a portion of the ambient pressure sensing port **1440a** (and similarly, all of the ports **1440** to the right of the power port **1440c** in FIG. 14E) may have an inner surface **1476** or bore that extends toward a rear cover **106b** of a device as the inner surface **1476** extends into the device. The upper wall **1460** of the port **1440a** (i.e., that portion of the wall that faces the front cover of the device) may also extend toward the rear cover **106b** as the upper wall **1460** extends into the device. The angled port **1440a**, and in particular the angled upper wall **1460** of the port **1440a**, may help to further reduce the capacitance between the second housing segment **112b** and a conductive element attached to a front

cover supported by the second housing segment **112a**. The ambient pressure sensing port **1440a** (and other ports **1440**) may be angled downward (i.e., toward the rear cover **106b**) at an angle that balances capacitance reduction with access to the port **1440a** (e.g., assuming that one or more components need to be inserted into the port **1440a** from interior to the device). In some embodiments, the sealing surface **1450** around the inner inlet to the port **1440a** may extend perpendicular to an inner surface of the rear cover **106b**.

In some embodiments, the entirety of a port **1440** may be angled toward the rear cover **106b**. In other embodiments, and as shown in FIGS. **14F** & **14G**, only a portion of a port **1440** may be angled toward the rear cover **106b**. In some cases, a port **1440** may be formed as a result of multiple drilling or end-milling operations. For example, the port shown in FIG. **14F** may be formed by drilling a first hole **1462** perpendicular to the sidewall **114**, from outside the sidewall **114**. A second hole **1464** may be drilled perpendicular to the sidewall **114** from inside the sidewall **114**. The first and second holes **1462**, **1464** may be offset, and may or may not intersect. A third hole **1466** may be drilled from inside the sidewall **114**, at an angle between perpendicular and parallel to the first and second holes **1462**, **1464**. In some embodiments, the third hole **1466** may be drilled at an angle of about 30 degrees ($\pm 10\%$) with reference to the first and second holes **1462**, **1464**. The third hole **1466** may intersect both the first hole **1462** and the second hole **1464**, but may not extend through to the exterior surface of the second housing segment **112b**. In alternative embodiments, the third hole **1466** may be drilled at an angle of between 15 degrees and 45 degrees, or at another angle.

As shown in FIG. **14G**, a portion of a speaker port **1440b** (and similarly, all of the ports **1440** to the left of the power port **1440c** in FIG. **14E**) may have an inner surface **1478** or bore that extends toward a rear cover **106b** of a device as the inner surface **1478** extends into the device. The upper wall **1480** of the port **1440b** may also extend toward the rear cover **106b** as the upper wall **1480** extends into the device. In general, the speaker port **1440b** may be angled similarly to the ambient pressure sensing port **1440a** described with reference to FIG. **14F**. However, in contrast to the sealing surface **1450** around the inner inlet of the ambient pressure sensing port **1440a**, the sealing surface **1448** around the speaker port **1440b** may be sloped with respect to the rear cover **106b** of the device. The configuration shown in FIG. **14G** may also reduce the capacitance between the second housing segment **112b** and other conductive structures in the chin area. The slope of the sealing surface **1448** may be sloped to different degrees, or not sloped, to balance capacitance reduction and maintenance of structural rigidity (e.g., the port cross-section shown in FIG. **14F** may have greater structural rigidity than the port cross-section shown in FIG. **14G**, but the port cross-section shown in FIG. **14G** may provide further capacitance reduction over the port cross-section shown in FIG. **14F**).

In some embodiments, the port shown in FIG. **14G** may be formed by drilling a first hole **1468** perpendicular to the sidewall **114**, from outside the sidewall **114**. A second hole **1470** may be drilled perpendicular to the sidewall **114** from inside the sidewall **114**. The first and second holes **1468**, **1470** may be offset, and may or may not intersect. A third hole **1472** may be drilled from inside the sidewall **114**, at an angle between perpendicular and parallel to the first and second holes **1468**, **1470**. In some embodiments, the third hole **1472** may be drilled at an angle of about 30 degrees ($\pm 10\%$) with reference to the first and second holes **1468**, **1470**. The third hole **1472** may intersect both the first hole

1468 and the second hole **1470**, but may not extend through to the exterior surface of the second housing segment **112b**. In alternative embodiments, the third hole **1472** may be drilled at an angle of between 15 degrees and 45 degrees, or at another angle.

In each of FIGS. **14F** & **14G**, a non-conductive housing component **528** extends along an inner, upper surface of the sidewall **114** and forms a ledge **1482** and border **1484** that may support and surround a display (e.g., the display **104** described with reference to FIGS. **1A-1C**). The portions of the non-conductive housing component **528** that extend along the inner, upper surface of the sidewall **114** (e.g., border **1484**) can help absorb crushing loads on the sidewall **114** and/or transfer crushing loads upward along the sidewall—away from a display.

Referring now to FIGS. **15A-15C**, there are shown example areas **1500**, **1502**, **1504**, **1506** where the fifth and sixth housing segments **112e**, **112f** described with reference to FIGS. **1A-1C**, **2A**, **3A-3C**, **4**, **5A**, **5B**, **6A**, **6C**, **7A**, **7C**, **8A**, & **8B** may be structurally coupled to the support plate **110** described with reference to FIGS. **4**, **13A**, & **14A**. As shown in FIG. **15A**, the fifth housing segment **112e** may be structurally and electrically coupled to the support plate **110** by a pair of welds at areas **1500**, **1502** along the left edge of the support plate **110**, and the sixth housing segment **112f** may be structurally and electrically coupled to the support plate **110** by a pair of welds at areas **1504**, **1506** along the right edge of the support plate **110**. In some embodiments, the welds may be laser welds. In some embodiments, the welds may be longitudinal welds, as shown. In some embodiments, the longitudinal welds may be replaced by, or supplemented with, spot welds.

The fifth housing segment **112e** may have an aperture in which a button assembly may be mounted, as discussed with reference to FIG. **6A**. At the end of the fifth housing segment **112e** adjacent the third housing segment **112c**, there may be little room for a weld and the fifth housing segment **112e** may be structurally and electrically coupled to the support plate **110** by a clip **1508**. Example cross-sections of the welds and clip **1508** shown in FIG. **15A** are shown in FIGS. **15B** & **15C** respectively.

FIG. **15B** shows an example cross-section of one of the welds **1510** described with reference to FIG. **15A**. The weld **1510** may be a weld that directly welds the fifth or sixth housing segment **112e** or **112f** to the support plate **110**. In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. **1A-1C**) may be bonded to a lower surface of the fifth or sixth housing segment **112e** or **112f** by an adhesive **534**. A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112e**, **112f** and extend parallel to the sidewall **114**. The seal **536** and adhesive **534** may help prevent moisture from entering a device between the housing segments **112e**, **112f** and rear cover **106b**.

FIG. **15C** shows an example cross-section of the clip **1508** described with reference to FIG. **15A**. By way of example, the clip **1508** may be aligned with and surround a threaded hole in the boss protrusion **512** described with reference to FIG. **5A**, and may be electrically coupled to the fifth housing segment **112e** by a screw **1512** that is inserted through a hole in the clip **1508** and threaded into the boss protrusion **512**. The screw **1512** may also retain a bracket **1514** for a button assembly and a washer **1516**. The clip **1508** may extend toward the rear cover **106b** of the device and bend one or more times to extend over a portion of the support plate **110** and over and around a boss protrusion **1518** formed in or mechanically coupled to the support plate **110**. The clip **1508**

may be retained to the boss protrusion **1518** by a screw **1520** that is inserted through a hole in the clip **1508** and threaded into the boss protrusion **1518**. A portion of the clip **1508** that extends over the support plate **110** may be welded to the support plate **110** (e.g., by weld **1522**) to provide extra rigidity for the structural coupling of the fifth housing segment **112e** to the support plate **110**.

In some embodiments, a rear cover (e.g., the rear cover **106b** described with reference to FIGS. 1A-1C) may be bonded to a lower surface of the support plate **110** by an adhesive **1524**. A seal **536** may be inserted into a groove **538** formed in a lower surface of each housing segment **112e**, **112f** and extend parallel to the sidewall **114**. The seal **536** and adhesive **1524** may help prevent moisture from entering a device between the housing segments **112e**, **112f** and rear cover **106b**.

FIGS. 16A-16D illustrate various example ground connections between the support plate **110** described with reference to FIGS. 3A-3C, 4, 13A, 14A, & 15A and a printed circuit board or logic board **1600**. As shown in FIG. 16A, the ground connections may include a first ground connection **1602**, which may be configured as a single ground connection between the support plate **110** and the logic board **1600**; a second ground connection **1604**, which may be configured as a double ground connection between the support plate **110** and the logic board **1600**; a third ground connection **1606**, which may be configured as a single Stockholm ground connection between the support plate **110** and the logic board **1600**; and a fourth ground connection **1608**, which may be configured as a single ground connection between the support plate **110** and the logic board **1600**, and may be coupled to the sixth housing segment **112f**. In some embodiments, the antenna flex circuit described with reference to FIG. 17B may be coupled to each of the second ground connection **1604** and the third ground connection **1606**. The antenna flex circuit described with reference to FIG. 17A may be coupled to first and second ground connections **1610**, **1612** along the lower part of the sixth housing segment **112f**.

FIG. 16B shows an example of the single ground connection **1608** between the support plate **110** and the logic board **1600**. As shown, the logic board **1600** may include a single layer **1614** having a hole **1616** that receives a boss protrusion **1618**. The boss protrusion **1618** may be mechanically and electrically coupled to the support plate **110**. The logic board **1600** may abut the periphery of the boss protrusion **1618**. A spacer **1620** may be placed around the boss protrusion **1618** and rest on the logic board **1600**. A screw **1622** may be inserted through a ground connection eye **1624** on the flex circuit shown in FIG. 17B and screwed into a hole **1626** in the boss protrusion **1618**, slightly compressing the ground connection eye **1624** against the spacer **1620**, compressing the spacer **1620** against the logic board **1600**, and compressing the logic board **1600** against the periphery of the boss protrusion **1618**. The single ground connection **1608** may ground conductive traces on the major surfaces **1628**, **1630** of the logic board **1600** to the support plate **110**.

FIG. 16C shows an example of the single Stockholm ground connection **1606** between the support plate **110** and the logic board **1600**. As shown, the logic board **1600** may include a first layer **1632** or logic board having a hole **1634** that receives a boss protrusion **1636**. The boss protrusion **1636** may be mechanically and electrically coupled to the support plate **110**. The first layer **1632** or logic board may abut the periphery of the boss protrusion **1636** and have a dielectric **1638** on its lower surface, and a layer **1640** (e.g., a conductive layer or a dielectric layer) on its upper surface.

The dielectric **1638** may electrically insulate the first layer **1632** or logic board from the support plate **110**. A logic board interposer **1642** (e.g., a conductive or dielectric interposer) may be placed around the boss protrusion **1636** and rest on the layer **1640** on the upper surface of the first layer **1632** or logic board. The logic board **1600** may further include a second layer **1644** or logic board having a hole **1646** aligned with the boss protrusion **1636**. The second layer **1644** or logic board may have a layer **1648** (e.g., a conductive or dielectric layer) on its lower surface. The layer **1648** may rest on the logic board interposer **1642**. A screw **1650** may be inserted through aligned holes in the second layer **1644** or logic board, the logic board interposer **1642**, and the first layer **1632** or logic board, and may be screwed into a hole **1652** in the boss protrusion **1636**, slightly compressing the first layer **1632** or logic board against the logic board interposer **1642**, compressing the logic board interposer **1642** against the first layer **1632** or logic board, and compressing the first layer **1632** or logic board against the periphery of the boss protrusion **1636**. The Stockholm ground connector **1606** may ground conductive traces **1654** (e.g., copper traces) on the upper surface of the second layer **1644** or logic board to the support plate **110**, but electrically insulate the first layer **1632** or logic board from the support plate **110** (or only electrically connect the first layer **1632** or logic board to ground through the logic board interposer **1642** and second layer **1644** or logic board).

FIG. 16D shows an example of the double ground connection **1604** between the support plate **110** and the logic board **1600**. As shown, the logic board **1600** may include a first layer **1656** or logic board having a hole **1658** that receives a boss protrusion **1660**. The boss protrusion **1660** may be mechanically and electrically coupled to the support plate **110**. The first layer **1656** or logic board may abut the periphery of the boss protrusion **1660**. A logic board interposer **1662** may be placed around the boss protrusion **1660** and rest on the first layer **1656** or logic board. The logic board **1600** may further include a second layer **1664** or logic board that rests on the logic board interposer **1662** and has a hole **1666** aligned with the boss protrusion **1660**. A screw **1668** may be inserted through a ground connection eye **1670** on the flex circuit shown in FIG. 17B and through aligned holes in the second layer **1664** or logic board, the logic board interposer **1662**, and the first layer **1656** or logic board, and may be screwed into a hole **1672** in the boss protrusion **1660**, slightly compressing the ground connection eye **1670** against the second layer **1664** or logic board (with other conductive components **1674** sometimes forming part of a conductive path between the ground connection eye **1670** and the second layer **1664** or logic board), compressing the second layer **1664** or logic board against the logic board interposer **1662**, compressing the logic board interposer **1662** against the first layer **1656** or logic board, and compressing the first layer **1656** or logic board against the periphery of the boss protrusion **1660**. The double ground connection **1604** may ground each of the first layer **1656** or logic board and the second layer **1664** or logic board to the support plate **110**.

FIGS. 17A & 17B show examples of flex circuits **1700**, **1750** that may be coupled to various ones of the housing segments **112** described with reference to FIGS. 1A-1C, 2A, 3A, 4, 5A-10C, 13A-13D, 14A-14G, & 15A-15C. The flex circuits **1700**, **1750** may connect to the ground, feed, and tuning connectors of the housing segments **112**, and may propagate signals to/from the ground, feed, and tuning connectors to operate the housing segments **112** as antennas. The flex circuit **1700** shown in FIG. 17A may include a

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ground connector **1702**, feed connector **1704**, and tuning connector **1706** that may be respectively electrically coupled to the ground connector **1406**, feed connector **1410**, and tuning connector **1414** of the first housing segment **112a**, as described with reference to FIG. **14B**. The flex circuit **1700** may also include tuning components **1708** that may be electrically coupled to the tuning connector **1706**, feed connector **1704**, and/or ground connector **1702**.

The flex circuit **1700** may also include ground connectors **1710**, **1712**, feed connectors **1714**, and tuning connectors **1716**, **1718**, **1720** that may be respectively electrically coupled to the ground connectors **1408**, **1430**, feed connectors **1412**, and tuning connectors **1438**, **1416**, **1436** of the second housing segment **112b**, as described with reference to FIG. **14B**. The flex circuit **1700** may also include tuning components **1722** that may be electrically coupled to the tuning connectors **1716**, **1718**, **1720**, feed connectors **1714**, and/or ground connectors **1710**, **1712**.

The flex circuit **1700** may further include one or more ground connectors **1724**, **1726** that may be electrically coupled to the sixth housing segment **112f** described with reference to other figures, or one or more ground connectors **1728**, **1730** that may be electrically coupled to the support plate **110** described with reference to other figures, and/or a board-to-board (B2B) connector **1732** that may electrically couple the flex circuit **1700**, its connectors, and its tuning components to the logic board **1600** described with reference to FIGS. **16A-16D**. As shown, some portions of the flex circuit **1700** may be bent and routed orthogonal to other portions of the flex circuit **1700** within a device.

The flex circuit **1750** shown in FIG. **17B** may include a ground connector **1752**, feed connector **1754**, and tuning connector **1756** that may be respectively electrically coupled to the ground connector **1320**, feed connector **1322**, and tuning connector **1324** of the third housing segment **112c**, as described with reference to FIG. **13C**. The flex circuit **1750** may also include tuning components **1758**, **1760** that may be electrically coupled to the tuning connector **1756**, feed connector **1754**, and/or ground connector **1752**.

The flex circuit **1750** may also include a ground connector **1762**, feed connector **1764**, and tuning connector **1766** that may be respectively electrically coupled to the ground connector **1334**, feed connector **1336**, and tuning connector **1338** of the fourth housing segment **112d**, as described with reference to FIG. **13C**. The flex circuit **1750** may also include tuning components **1768** that may be electrically coupled to the tuning connector **1766**, feed connector **1764**, and/or ground connector **1762**.

The flex circuit **1750** may further include one or more ground connectors **1770**, **1772**, **1774**, **1776** that may be electrically coupled to the support plate **110** described with reference to other figures, a ground spring **1778** that may be electrically coupled to the ground pad **1314** described with reference to FIG. **13B**, and/or one or more B2B connectors **1780**, **1782** that may electrically couple the flex circuit **1750**, its connectors, and its tuning components to the logic board **1600** described with reference to FIGS. **16A-16D**. The flex circuit **1750** may further include a Fargo feed connector **1784** and a Fargo logic board connector **1786**. The Fargo feed connector **1784** may be electrically connected to the Fargo feed connector **1346** described with reference to FIG. **13C**, and the Fargo logic board connector **1786** may be electrically connected to the Stockholm ground connector **1606** described with reference to FIG. **16A**. The Fargo connectors **1784**, **1786** may be used to establish a near-field communication (NFC) inductive loop usable for NFC communications.

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As shown in FIG. **17B**, some portions of the flex circuit **1750** may be bent and routed orthogonal to other portions of the flex circuit **1750** within a device.

FIG. **18** shows how the flex circuits **1700**, **1750** described with reference to FIGS. **17A** & **17B** may be placed and routed with respect to the housing segments **112** and support plate **110** described with reference to FIGS. **1A-1C**, **2A**, **3A**, **4**, **5A-10C**, **13A-13D**, **14A-14G**, **15A-15C**, & **16A-16D**. As shown, a portion of the flex circuit **1750** described with reference to FIG. **17B** may be oriented perpendicular to the support plate **110** and routed through the channel **1318** between the camera brace **402** and sixth housing segment **112f**. In some embodiments, portions of the flex circuits **1700**, **1750** may be adhesively bonded to the support plate **110**.

Turning now to FIG. **19**, there is shown a set of wireless frequency bands **1900**, **1902**, **1904**, **1906**, **1908** that may be used for wireless communication. The wireless frequency bands include a low wireless frequency band **1900** extending from about 600-950 MHz, a mid wireless frequency band **1902** extending from about 1700-2200 MHz, a high wireless frequency band **1904** extending from about 2300-2800 MHz, a B42 wireless frequency band **1906** extending from about 3400-3600 MHz (currently defined for use in the European Union (EU) and Japan), and a 5G, Wi-Fi, or B46 band **1908** extending from about 5000-6000 MHz.

In some embodiments of the devices and housings described herein, multiple housing segments disposed along a sidewall of a device housing may be operated individually or simultaneously as antennas, in the wireless frequency bands described with reference to FIG. **19** or in other wireless frequency bands. For example, in some cases, wireless communication circuitry may be operable to communicate in different wireless communication modes, using the same or different combinations of antennas (including, in some cases, housing segments that are configured to operate as antennas).

In some embodiments, wireless communication circuitry may be configured to operate in a first wireless communication mode (e.g., a 2x2 MIMO wireless communication mode). In the first wireless communication mode, the wireless communication circuitry may be configured to use the second and third housing segments **112b**, **112c** described with reference to FIG. **2A** or **3A** as different antennas for wireless communication. In some embodiments, wireless communication in the first wireless communication mode may occur in the low wireless frequency band **1900**. Alternatively, the wireless communication circuitry may be operated in a second wireless communication mode (e.g., a 4x4 MIMO wireless communication mode). In the second wireless communication mode, the wireless communication circuitry may be configured to use the first, second, third, and fourth housing segments **112a**, **112b**, **112c**, **112d** described with reference to FIG. **2A** or **3A** as different antennas for wireless communication. In some embodiments, wireless communication in the second wireless communication mode may occur in the mid wireless frequency band **1902** or the high wireless frequency band **1904**. Alternatively, the wireless communication circuitry may be operated in a third wireless communication mode (e.g., another 4x4 MIMO wireless communication mode). In the third wireless communication mode, the wireless communication circuitry may be configured to use the second housing segment **112b** in combination with the sixth housing segment **112f**, a first interior antenna **324** (described with reference to FIG. **3A**), a second interior antenna **326** (described with reference to FIG. **3A**), and the fourth housing segment **112d** in combi-

nation with the sixth housing segment **112f** as different antennas for wireless communication. In some embodiments, wireless communication in the third wireless communication mode may occur in the B42 wireless frequency band **1906**. Alternatively, the wireless communication circuitry may be operated in a fourth wireless communication mode (e.g., another 2x2 MIMO wireless communication mode). In the fourth wireless communication mode, the wireless communication circuitry may be configured to use the first and second interior antennas **324a**, **324b** described with reference to FIG. 3A for wireless communication in the 5G, Wi-Fi, or B46 wireless frequency band **1908**.

In general, it can be useful to configure antennas with the greatest physical separation to operate simultaneously in a wireless communication mode that requires two antennas. Thus, two-antenna wireless communication modes may be supported by antennas positioned on diagonally opposite corners of a device, when possible.

The housing segments described with reference to FIGS. 2B-2E may also be used in different combinations (or individually) to communicate in one or more of the wireless frequency bands described with reference to FIG. 19.

FIGS. 20A & 20B show corresponding ground springs **2000**, **2002**, **2004**, **2006**, **2008**, **2010**, **2012**, **2014**, **2016**, **2018** and ground pads **2028**, **2030**, **2032**, **2034**, **2036**, **2038**, **2040**, **2042**, **2044**, **2046** on the housing **102** (e.g., on the housing segments **112** or support plate **110**), and a cover (e.g., the front cover **106a** described with reference to FIGS. 1A-1C) mounted to the housing **102**, to enclose a device stack including a display. The display may be viewable through the cover **106a**. FIG. 20A shows example locations of ground springs **2000-2018** that are electrically coupled to the housing **102**, and FIG. 20B shows example locations of ground pads **2028-2046** that are coupled to the cover **106a**.

The ground springs **2000-2018** and ground pads **2028-2046** may be positioned about the periphery of the support plate **110** or cover **106a**, and may be generally located in the device forehead and device chin. A ground spring **2000-2018** that is mechanically and electrically coupled to the support plate **110**, a housing segment **112**, or the camera brace **402** may electrically contact a respective ground pad **2028-2046** that is mechanically coupled to the cover, and vice versa. Distribution of the ground springs **2000-2018** and ground pads **2028-2046** throughout the device forehead and chin can provide a good ground reference for the housing segments **112** when the housing segments are configured to operate as antennas. The ground springs and ground pads may in some cases be configured to provide low inductance connections between a ground plane on the cover **106a** and a ground plane provided by the support plate **110**. In some embodiments, providing relatively equal spacing between ground springs in the device forehead and device chin can provide a more uniform ground. A uniform ground can improve the resonance of the housing segments **112**.

The ground springs **2000-2018** may be mechanically and electrically coupled to the support plate **110**, housing segments **112**, or camera brace **402** by welds, solder, conductive adhesive, or other types of fasteners. In some embodiments, one or more of the ground springs **2000-2018** may be a preformed, stamped, or bent metal spring. In some embodiments, the ground springs **2000-2018** and ground pads **2028-2046** may be gold or gold plated.

Corresponding mechanical snaps **2020**, **2022**, **2024**, **2026**, **2048**, **2050**, **2052**, **2054** may be coupled to the housing **102** (e.g., to the fifth and sixth housing segments **112e**, **112f** or portions of the support plate **110** adjacent the fifth and sixth housing segments **112e**, **1120** and to the cover **106a**, and

may engage one another to mechanically couple the cover **106a** to the housing segments **112**.

FIGS. 21A-21C show various examples of low force springs and corresponding contact pads, as may be used to implement any of the ground springs and ground pads described with reference to FIGS. 20A, 20B, or other figures. The examples of low force springs include a wiping contact **2100** (FIG. 22A), a point contact **2104** (FIG. 22B), and a magsafe pin **2108** (FIG. 22C). Each of the low force springs **2100**, **2104**, **2108** may contact a respective contact pad **2102**, **2106**, or **2110**. In some embodiments, each of the low force springs **2100**, **2104**, **2108** and their corresponding contact pads **2102**, **2106**, **2110** may be made of gold or be gold plated.

FIG. 22 shows a cross-section **2200** of the device forehead shown in FIGS. 13C & 13D. As shown, a portion of a non-conductive housing component **528** may be molded against the third housing segment **112c** to form a surface **2202** that supports a front cover **106a**. The third housing segment **112c** may also form part of the surface **2202**, or alternatively, the third housing segment **112c** may form the entire surface **2202**. A frame **2204** may be positioned about the periphery of the device, between the surface **2202** and the cover **106a**. In some embodiments, the frame **2204** may be bonded to the surface **2202** and/or cover **106a** by one or more adhesives **2206**, **2208**.

In some embodiments, the frame **2204** may include a plastic outer portion **2204a** that is mechanically coupled to a metallic inner portion **2204b** (e.g., the metallic inner portion **2204b** may be insert-molded into the plastic outer portion **2204a**). In other embodiments, the entire frame **2204** may be plastic or metallic. As shown, the frame **2204** (e.g., the inner portion **2204b** of the frame **2204**) may be attached to a stiffener **2210**, which may be metallic or plastic, for example. The stiffener **2210** may be bonded to the cover **106a**, for example, by an adhesive **2212**.

A bracket, such as the alignment bracket **2214**, may be attached to the stiffener **2210**, and in some cases to the cover **106a**. The alignment bracket **2214** may be metallic or plastic, and in some cases may be welded (e.g., laser welded) or bonded (e.g., adhesively bonded) to the stiffener **2210** and/or cover **106a**. The alignment bracket **2214** may serve as a means for mounting a bio-authentication sensor, camera, speaker, or other components within the device forehead.

The alignment bracket **2214** may be positioned adjacent a device stack **130** including a display, touch sensors, force sensors, or other components.

FIG. 23 shows a sample electrical block diagram of an electronic device **2300**, which electronic device may in some cases take the form of the device **100** described with reference to FIGS. 1A-1C or other devices described herein. The electronic device **2300** may include a display **2302** (e.g., a light-emitting display), a processor **2304**, a power source **2306**, a memory **2308** or storage device, a sensor system **2310**, or an input/output (I/O) mechanism **2312** (e.g., an input/output device, input/output port, or haptic input/output interface). The processor **2304** may control some or all of the operations of the electronic device **2300**. The processor **2304** may communicate, either directly or indirectly, with some or all of the other components of the electronic device **2300**. For example, a system bus or other communication mechanism **2314** can provide communication between the display **2302**, the processor **2304**, the power source **2306**, the memory **2308**, the sensor system **2310**, and the I/O mechanism **2312**.

The processor **2304** may be implemented as any electronic device capable of processing, receiving, or transmit-

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ting data or instructions, whether such data or instructions is in the form of software or firmware or otherwise encoded. For example, the processor **2304** may include a microprocessor, a central processing unit (CPU), an application-specific integrated circuit (ASIC), a digital signal processor (DSP), a controller, or a combination of such devices. As described herein, the term “processor” is meant to encompass a single processor or processing unit, multiple processors, multiple processing units, or other suitably configured computing element or elements. In some embodiments, the processor **2304** may function as the controller described with reference to FIG. 1C.

It should be noted that the components of the electronic device **2300** can be controlled by multiple processors. For example, select components of the electronic device **2300** (e.g., the sensor system **2310**) may be controlled by a first processor and other components of the electronic device **2300** (e.g., the display **2302**) may be controlled by a second processor, where the first and second processors may or may not be in communication with each other.

The power source **2306** can be implemented with any device capable of providing energy to the electronic device **2300**. For example, the power source **2306** may include one or more batteries or rechargeable batteries. Additionally or alternatively, the power source **2306** may include a power connector or power cord that connects the electronic device **2300** to another power source, such as a wall outlet.

The memory **2308** may store electronic data that can be used by the electronic device **2300**. For example, the memory **2308** may store electrical data or content such as, for example, audio and video files, documents and applications, device settings and user preferences, timing signals, control signals, and data structures or databases. The memory **2308** may include any type of memory. By way of example only, the memory **2308** may include random access memory, read-only memory, Flash memory, removable memory, other types of storage elements, or combinations of such memory types.

The electronic device **2300** may also include one or more sensor systems **2310** positioned almost anywhere on the electronic device **2300**. The sensor system(s) **2310** may sense one or more type of parameters, such as but not limited to, force or pressure on the display **2302**, a crown, a button, or a housing of the electronic device **2300**; light; touch; heat; movement; relative motion; biometric data (e.g., biological parameters) of a user; and so on. For example, the sensor system(s) **2310** may include a watch crown sensor system, a heat sensor, a position sensor, a light or optical sensor, an accelerometer, a pressure transducer, a gyroscope, a magnetometer, a bio-authentication sensor, a health monitoring sensor, and so on. Additionally, the one or more sensor systems **2310** may utilize any suitable sensing technology, including, but not limited to, capacitive, ultrasonic, resistive, optical, ultrasound, piezoelectric, and thermal sensing technology.

The I/O mechanism **2312** may transmit or receive data from a user or another electronic device. The I/O mechanism **2312** may include the display **2302**, a touch sensing input surface, one or more buttons (e.g., a graphical user interface “home” button), a crown, one or more cameras, one or more microphones or speakers, one or more ports such as a microphone port, and/or a keyboard. Additionally or alternatively, the I/O mechanism **2312** may transmit electronic signals via a communications interface, such as a wireless, wired, and/or optical communications interface. Examples of wireless and wired communications interfaces include, but are not limited to, cellular and Wi-Fi communications

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interfaces. In some embodiments, the electronic device **2300** may configure one or more of the housing segments **112** described herein to operate as an antenna and/or may configure the electronic device **2300** to communicate in one or more of the wireless frequency bands described with reference to FIG. 19 (or other wireless frequency bands).

The foregoing description, for purposes of explanation, uses specific nomenclature to provide a thorough understanding of the described embodiments. However, it will be apparent to one skilled in the art, after reading this description, that the specific details are not required in order to practice the described embodiments. Thus, the foregoing descriptions of the specific embodiments described herein are presented for purposes of illustration and description. They are not targeted to be exhaustive or to limit the embodiments to the precise forms disclosed. It will be apparent to one of ordinary skill in the art, after reading this description, that many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An electronic device, comprising:

a set of housing segments comprising a first housing segment defining a first corner of the electronic device and a second housing segment defining a second corner of the electronic device; and

wireless communication circuitry configured to, operate in a 2x2 multiple-input multiple-output (MIMO) mode using a first subset of the set of housing segments as antennas, the first subset including the first housing segment and the second housing segment; and

operate in a 4x4 MIMO mode using a second subset of the set of housing segments as antennas, the second subset of the set of housing segments different from the first subset of the set of housing segments and including the first housing segment and the second housing segment.

2. The electronic device of claim 1, wherein:

the second subset of the set of housing segments comprises:

a third housing segment defining a third corner of the electronic device; and

a fourth housing segment defining a fourth corner of the electronic device.

3. The electronic device of claim 1, wherein the wireless communication circuitry is configured to operate in an additional MIMO mode using a third subset of the set of housing segments as antennas.

4. The electronic device of claim 3, wherein the third subset of the set of housing segments comprises:

a third housing segment defining a portion of a first side of the electronic device; and

a fourth housing segment defining a portion of a second side of the electronic device.

5. The electronic device of claim 4, wherein:

the first housing segment is electrically connected to the third housing segment, via a first switchable connection, and the second housing segment is electrically connected to the fourth housing segment, via a second switchable connection, when the wireless communication circuitry operates in the additional MIMO mode; and

the first housing segment is electrically disconnected, via the first switchable connection, from the third housing segment, and the second housing segment is electrically disconnected, via the second switchable connection,

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tion, from the fourth housing segment, when the wireless communication circuitry operates in the 4×4 MIMO mode.

6. The electronic device of claim 4, further comprising: a set of antennas housed within the electronic device; wherein, the wireless communication circuitry, when operated in the additional MIMO mode, further uses the set of antennas.

7. The electronic device of claim 1, further comprising: a set of interior antennas; wherein, the set of housing segments is disposed around an interior volume of the electronic device; the set of interior antennas is disposed within the interior volume; and the wireless communication circuitry is configured to operate in an additional MIMO mode using the set of interior antennas.

8. An electronic device, comprising: a display; a set of housing segments positioned around the display and comprising a first housing segment defining a first corner of the electronic device and a second housing segment defining a second corner of the electronic device; a set of antennas positioned under the display and disposed in an interior volume bounded by the set of housing segments; and wireless communication circuitry operable to, communicate in a 2×2 multiple-input multiple-output (MIMO) mode using a first antenna set including at least a first subset of the set of housing segments as antennas, the first subset including the first housing segment and the second housing segment; and communicate in a 4×4 MIMO mode using a second antenna set different from the first antenna set, the second antenna set including at least the set of antennas positioned under the display and a second subset of the set of housing segments, the second subset of the set of housing segments including the first housing segment and the second housing segment.

9. The electronic device of claim 8, wherein the set of antennas positioned under the display comprises: a first antenna positioned under the display; and a second antenna positioned under the display.

10. The electronic device of claim 8, wherein the second corner is diagonally opposite the first corner.

11. An electronic device, comprising: a first antenna defined by a first housing segment that is positioned at a first corner of the electronic device; a second antenna defined by a second housing segment that is positioned at a second corner of the electronic device;

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a third antenna defined by a third housing segment that is positioned at a third corner of the electronic device, diagonally opposite the first corner;

a fourth antenna defined by a fourth housing segment that is positioned at a fourth corner of the electronic device; and wireless communication circuitry configured to operate in:

a 2×2 multiple-input multiple-output (MIMO) mode using a first set of antennas, the first set including only the first antenna and the third antenna; and

a 4×4 multiple-input multiple-output (MIMO) mode using a second set of antennas, the second set including the first antenna, the second antenna, the third antenna, and the fourth antenna.

12. The electronic device of claim 11, further comprising: a first conductive housing segment forming a portion of a first side of the electronic device; and a second conductive housing segment forming a portion of a second side of the electronic device.

13. The electronic device of claim 12, wherein the first conductive housing segment is switchably connectable to the first antenna.

14. The electronic device of claim 8, wherein the wireless communication circuitry is further configured to operate in an additional 4×4 MIMO mode using a third set of antennas, the third set of antennas including at least the first housing segment electrically connected to a fifth housing segment via a switchable connection.

15. The electronic device of claim 8, wherein: the first housing segment has a first conductive length; and the second housing segment has a second conductive length that is different than the first conductive length.

16. The electronic device of claim 1, wherein the first housing segment and the second housing segment have a same conductive length.

17. The electronic device of claim 1, wherein at least one of the housing segments in the second subset of the set of housing segments is electrically connected to a third housing segment, via a switchable connection.

18. The electronic device of claim 2, wherein: the first housing segment has a first conductive length; and the second housing segment has a second conductive length that is different than the first conductive length.

19. The electronic device of claim 8, wherein the first housing segment has a same conductive length as the second housing segment.

20. The electronic device of claim 8, wherein: wireless communication in the 2×2 MIMO mode occurs in a first frequency band; and wireless communication in the 4×4 MIMO mode occurs in a second frequency band different from the first frequency band.

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